Gauge-Covariant Gaussian Attention through Lie-Algebra Links

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Abstract

Attention selects sources and transmits information through learned interfaces. We formulate these operations for full Gaussian states expressed in learned local frames, with communication carried by group-valued links that are parametrized by Lie-algebra generators. Gaussian query and key observables determine the source weights through an entropy-regularized categorical objective, and at every positive temperature these weights are also the exact posterior of a source label under a binary compatibility record. The comparison cost is exactly a dot product between query expectation parameters and key natural parameters, offset by a key log-partition bias. Under shared projected key covariance it reduces to L2 attention in whitened coordinates, whose key-norm penalty neither LayerNorm nor high-dimensional concentration removes at the conventional temperature.

A link $\Theta_{ij} = U_i e^{X_{ij}} U_j^{-1}$ is invertible for every generator and obeys the endpoint law of a lattice link variable. For the shared low-rank generator built from the value and output maps, the link splits exactly into the flat reference transport and a selective channel term, whose first-order part is the projected matrix link of multi-head attention. With tiled heads and routing outside the exponent, the routed message resums into transformed receiving maps, and its orders separate a head-routed flat transport, the standard value path, and cross-head mixing. The affine, first-order, and exponential charts on the same stored maps represent the same tiled mean maps up to the real-logarithm domain; only the exponential chart cannot leave the identity component, and invertibility supplies inverse-edge holonomy on causal support. Read as evidence rather than as prediction, the same channels define a receiver energy whose exact minimizer is a Gaussian power posterior in closed form. A four-layer model trained on WikiText-103 with a GPT-2 byte-pair vocabulary records token-level test perplexity 31.04 at context 128 in a single-seed feasibility run, and matched single-seed two-layer runs of the three charts differ by at most 0.004 nats per token in test cross-entropy. [**Data pending:** report the final-protocol result with matched baselines and seed variability (Table 8).]

Keywords: gauge covariance, Gaussian attention, Lie-algebra links, variational routing, lattice gauge theory, transformers

1 Introduction

Attention separates source selection from content transmission. Query and key representations determine which inputs are consulted, while value and output maps determine what is communicated (Vaswani et al., 2017). When a representation also carries uncertainty and a local coordinate frame, these operations require compatible transformations of both its mean and covariance. This paper develops a mathematical account of such communication using full Gaussian states, learned comparison interfaces, and matrix-valued links between token positions.

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The starting point is selective use of a framed state. A sending map exposes a lower-dimensional observable, and a receiving map places that observable in the recipient's state space. Distinct query and key maps define the measurements used to choose sources. Composed with the local frames, the content maps give the matrix link $L_{ij}^{(h)} = U_i O_h T_h U_j^{-1}$, directed from source j to receiver i. Its factors specify accessible directions, gains, and correspondences; they need not be orthogonal projections or group elements, and a single head narrower than the state is singular. The flat identification $\Omega_{ij} = U_i U_j^{-1}$ appears as a special case. It does not exhaust the geometry of learned communication.

This paper takes a group-valued link as the primary communication object. The link $\Theta_{ij} = U_i \exp(X_{ij}) U_j^{-1}$ is written through a generator X_{ij} in the Lie algebra $\mathfrak{gl}(K, \mathbb{R})$, expressed in reference coordinates. It is therefore invertible for every generator, it transforms by the endpoint law of a lattice link variable (Wilson, 1974), and its generator plays the role of the algebra-valued potential whose exponential is a lattice link (Creutz, 1983), with the difference that the frames dress the potential and leave the generator invariant under independent site changes of frame. For the low-rank generator assembled from the value and output maps, an exact identity separates the link into the flat reference transport and a selective channel term whose first-order part is the projected matrix link, so the content maps of standard multi-head attention appear as the infinitesimal channels of a group-valued field. With tiled heads and categorical routing outside the exponent, the routed message resums into transformed receiving maps, and the orders of its expansion separate a head-routed flat transport, the standard value path, and cross-head mixing. The affine, first-order, and exponential parametrizations of the same stored maps are charts of one group on a stated domain. They represent the same tiled mean maps up to the real-logarithm domain (Culver, 1966), and only the exponential chart keeps every link in the identity component of $\mathrm{GL}(K)$. Invertibility also supplies reverse edges, and with them triangle holonomies on causal attention graphs, which contain no directed cycles through distinct positions.

Full Gaussian query and key measurements provide a comparison cost. Minimizing its categorical expectation together with a relative-entropy penalty yields Gibbs routing exactly (Niculae and Blondel, 2017). At every positive temperature the same weights are the exact conditional posterior of a source label in a normalized joint that records query-key compatibility as a binary event, the device used for optimality variables in control as inference (Levine, 2018); at unit temperature they are also a coordinate minimization of a normalized mixture KL. The resulting weights determine a predictive mean through the content links; they do not determine those links, a residual connection, a nonlinear layer, or a vocabulary decoder. We therefore formulate the attention operator directly and state its relation to broader variational inference separately (Blei et al., 2017). The same channels also admit an evidential reading, in which the routed messages are likelihood evidence about the receiver rather than a prediction of its state. The resulting receiver energy has an exact Gaussian minimizer, the power posterior of the declared channel likelihoods, and the predictive head sum is its limit under stated conditions.

Several parts of this construction have established precedents. Kernel accounts and modern Hopfield networks describe other mathematical structures underlying attention (Tsai et al., 2019; Katharopoulos et al., 2020; Ramsauer et al., 2021). Structured attention networks and latent alignment integrate probabilistic inference into attention (Kim et al.,

2017; Deng et al., 2018), variational attention treats the attention vector as a latent variable (Bahuleyan et al., 2018), and attention has been interpreted as structural inference and marginalization (Singh and Buckley, 2023a,b). Gaussian-mixture accounts derive softmax routing as mixture responsibilities, including the key-norm term and a prior that cancels it (Movellan and Gabbur, 2020; Gabbur et al., 2021; Nguyen et al., 2022), and L2 self-attention retains the same key-norm term (Kim et al., 2021). Pro-Attention treats the components of each token vector as samples of a token-specific distribution and scores token pairs by the KL divergence between the estimated distributions (Bae et al., 2022), and Gaussian word embeddings are compared by KL divergence in Vilnis and McCallum (2015). Gaussian-KL routing itself is thus not a priority claim here. Our Gaussian observables are multivariate, retain their full reduced covariance, and enter as laws rather than points, so the query covariance contributes a pair-dependent term that point-query mixtures lack.

Geometric deep learning supplies the representation-theoretic setting (Cohen et al., 2019; Bronstein et al., 2021). SE(3)-Transformers combine attention with features organized by irreducible representations, and Gauge Equivariant Transformer constructs gauge-equivariant multi-head attention on meshes (Fuchs et al., 2020; He et al., 2021). In those constructions the frame is geometric data defined only up to the structure group, so shared maps must be intertwiners. Here the frames are learned for each token and position; they act as a local dressing field (Fournel et al., 2014; François, 2019), and the covariance they carry constrains none of the learned maps (Section 4). Transport of distribution parameters also appears in neuronal gauge accounts (Sengupta et al., 2016; Sengupta and Friston, 2017). These precedents make it necessary to distinguish a frame representation from a communication head. For a product of general linear groups, coordinate blocks are irreducible defining representations, whereas learned heads may access overlapping combinations of those blocks.

The links obey the endpoint law of lattice link variables. Conventional lattice connections use group-valued edge maps (Wilson, 1974); linear lattice gauge theory admits more general matrix links with a compact gauge group (Wetterich, 2014), whereas the covariance group here is $\mathrm{GL}(K)$, which is not compact. The exponential links are group valued, but with one shared generator per layer they remain composites of site data and learned maps rather than independent gauge fields. The composition defect and rank loss of individual channels, and the products of routed links around loops, are distinct from the holonomy of the flat reference identification. Because the selective channels factor through an edge space, they are close to the restriction maps of cellular sheaves and sheaf neural networks (Hansen and Ghrist, 2019; Bodnar et al., 2022). Promoting the generators to latent variables with a normalized prior places link dynamics inside a variational free energy; its stationarity condition is a field equation whose source has the form of the delta rule of fast-weight memories (Schmidhuber, 1992; Schlag et al., 2021), and the implemented shared generator is its zero-width limit. That extension is stated in Section 8.4 and is not implemented here. The static construction supplies neither an autonomous lattice action nor a continuum limit.

The contribution is the explicit organization of Gaussian comparison, categorical routing, group-valued communication, and predictive or evidential content within one framed construction, together with exact statements about it. Section 2 states the framed Gaussian states, the comparison and content interfaces, and the attention operator in one place. Section 3 introduces Lie-algebra links, derives the selective channels from a low-rank gen-

erator (Proposition 5), relates the linear and exponential parametrizations as charts of one group (Proposition 6), and derives the aggregation rule, the residual, and the order-by-order structure of the routed message (Propositions 8 to 11). Section 4 proves that the operator factors through frame-invariant reference moments (Theorem 14) and derives the normal form of the evaluated predictor. Section 5 gives the dual-coordinate form of Gaussian-KL routing (Theorem 16) and its reductions to whitened L2 and biased dot-product attention. Section 6 establishes the exact probabilistic status of routing at every temperature and of content aggregation as a predictive mean (Theorem 20). Section 7 derives the channel-likelihood receiver and its closed-form update (Proposition 22). Section 8 develops the rank, composition, and holonomy of the links (Propositions 27 and 32), their relation to lattice gauge theory and cellular sheaves, and the latent-generator extension. Section 9 reports language-model realizations, including a matched comparison of the three charts, and their evaluation limits. The appendices give the smooth-bundle construction, the frame-block representation theory, the positional priors, the prior-bank readout, the relation to a normalized latent-source model, the Gaussian-closure and causal-prediction results, the proofs for latent link generators, the numerical identity checks, and the experimental record.

Scope. The results concern the attention operator for full Gaussian states and its group-valued links. The law of the generator is not derived: the implemented links use one learned generator per layer, shared across edges, and per-edge latent generators are stated as an unimplemented extension. The content maps, the residual coefficient, the normalization, the multilayer perceptron (MLP), and the decoder are taken from the transformer rather than derived. The stored covariance is either a static routing state or the moment-matched covariance of a declared predictive kernel; in neither case is it uncertainty propagated through the nonlinear block. No common variational free energy is claimed for the whole predictor, and the construction does not invoke a general Free Energy Principle. The language-model results are single-seed records without a matched conventional baseline.

2 Framed Gaussian States and the Attention Operator

This section states the objects and the operator in one place; Sections 4 to 8 prove their properties. A token in a language model is one finite-dimensional realization of an agent; attention-based communication among learned agents has a precedent in targeted multi-agent communication (Das et al., 2019). Appendix A gives the smooth-bundle construction behind the frames, with its geometric picture in Figure 5.

2.1 Framed Gaussian states

Each token position $i \in \{0, \dots, N-1\}$ is an agent whose Gaussian state is expressed in local coordinates on a vector space E_i, identified with the reference space $\mathsf{V} = \mathbb{R}^K$ by an invertible frame $U_i : \mathsf{V} \to E_i$. In the bundle construction of Appendix A, the local section u_i of agent i and one reference trivialization σ_0 are related by $u_i = \sigma_0 \cdot U_i^{-1}$, and the token bundle is pulled back along a constant map, so no connection, curvature, or topology of the contextual base enters the token formulas. The state and its local prior, expressed in the

same local coordinates, are

$$q_i = \mathcal{N}(\mu_i, \Sigma_i), \qquad \Sigma_i \succ 0, \tag{1}$$

$$p_i = \mathcal{N}(\mu_{p,i}, \Sigma_{p,i}), \qquad \Sigma_{p,i} \succ 0. \tag{2}$$

Their reference-coordinate representation, in σ_0-coordinates, is obtained through the frame:

$$\widetilde{q}_i = (U_i^{-1})_* q_i = \mathcal{N}(\xi_i, C_i), \qquad \xi_i = U_i^{-1}\mu_i, \quad C_i = U_i^{-1}\Sigma_i U_i^{-\top}. \tag{3}$$

The covariance is an unrestricted positive-definite matrix. The frame determines this coordinate identification; the additional communication interfaces determine which directions another agent receives.

A rechoice $u_i \mapsto u_i \cdot g_i^{-1}$ of the section of agent i, with σ_0 held fixed, changes the coordinates as in Eq. (4) below, by the quotient convention of Eq. (109). These rechoices are independent passive coordinate changes at the token sites. A change of σ_0 is the common reference-basis change $U_i \mapsto U_i a$ of Section 2.2. Under such a passive site change, the same construction is represented by

$$(\mu_i, \Sigma_i, U_i) \mapsto (g_i\mu_i, g_i\Sigma_i g_i^{\top}, g_i U_i), \qquad g_i \in \mathrm{GL}(K). \tag{4}$$

The reference moments (ξ_i, C_i) are unchanged. Reference-space interface parameters are held fixed under this passive action; covectors transform contragrediently. The link endpoint law follows by composing the sending and receiving maps in Section 2.3.

A useful frame parametrization is $U_i = \exp(\phi_{x_i})\exp(Y_i)$, combining the frame coefficient ϕ_{x_i} of the token type (Eq. (23)) with a positional factor. Its determinants are positive. The passive law above is defined for arbitrary invertible frames; a single real exponential is not a global chart of $\mathrm{GL}^+(K)$, and a product of real exponentials need not have a real logarithm (Culver, 1966). These chart restrictions do not alter the link's endpoint law.

2.2 Communication layers, heads, and comparison spaces

A layer is one communication update applied to the collection of token states. The index ℓ counts successive updates: $q_i^{(\ell)} = \mathcal{N}(\mu_i^{(\ell)}, \Sigma_i^{(\ell)})$ is the state entering layer ℓ, and $q_i^{(\ell+1)}$ is its output. The reference moments $\xi_i^{(\ell)}, C_i^{(\ell)}$ are defined by Eq. (3) for that input state.

Each layer contains H communication heads, indexed by $h = 1, \ldots, H$. A head is one channel with its own source comparison and content maps; its weighted messages are combined with those of the other heads. This definition does not require the heads to partition the state or coincide with irreducible frame blocks. For each head choose two real vector spaces,

$$H_h^{\mathrm{cmp}} = \mathbb{R}^{r_h}, \qquad H_h^{\mathrm{val}} = \mathbb{R}^{s_h}. \tag{5}$$

The superscript cmp denotes the comparison space, in which source compatibility is measured; val denotes the content or value space, through which the transmitted message passes. The positive integers r_h and s_h are their dimensions. These spaces are shared across token positions and need not have the same dimension.

The layer's interfaces have the declared types

$$S_{Q,h}^{(\ell)}, S_{K,h}^{(\ell)} : \mathsf{V} \longrightarrow H_h^{\mathrm{cmp}}, \tag{6}$$

$$T_h^{(\ell)} : \mathsf{V} \longrightarrow H_h^{\mathrm{val}}, \qquad O_h^{(\ell)} : H_h^{\mathrm{val}} \longrightarrow \mathsf{V}. \tag{7}$$

The query map S_Q measures the receiver and the key map S_K measures a candidate sender; both have full row rank, hence $r_h \leq K$. The value map T extracts content, and the output map O places it in reference state coordinates. The operator applies an effective receiving map $\widetilde{O}_h^{(\ell)} : H_h^{\mathrm{val}} \to \mathsf{V}$, which equals the stored map $O_h^{(\ell)}$ in the affine chart and is evaluated from the stored pair (T, O) through a Lie-algebra generator in the first-order and exponential charts of Section 3.6. Their composition with local frames is developed in Section 2.3. Superscripts (ℓ, h) identify a layer and head, and layer indices are suppressed when one layer is fixed. Matrices act on column vectors, and ij denotes a map from sender j to receiver i.

The maps are specified in the chosen reference basis. Under a common reference-basis change $U_i \mapsto U_i a$, their representations change as $S_Q \mapsto S_Q a$, $S_K \mapsto S_K a$, $T \mapsto Ta$, and $O \mapsto a^{-1}O$, while $\xi_i \mapsto a^{-1}\xi_i$ and $C_i \mapsto a^{-1}C_i a^{-\top}$. In each chart the effective receiving map then transforms as $\widetilde{O} \mapsto a^{-1}\widetilde{O}$, and a generator transforms by the adjoint action $X \mapsto a^{-1}Xa$ (Section 3). Thus the measured laws and the content link are unchanged. Holding the interface matrices fixed while making this reference-basis change would specify a different communication rule.

The query and key Gaussian pushforwards therefore occupy the same comparison space:

$$q_{Q,i}^{(\ell,h)} = \mathcal{N}\big(S_{Q,h}^{(\ell)}\xi_i^{(\ell)}, S_{Q,h}^{(\ell)}C_i^{(\ell)}S_{Q,h}^{(\ell)\top}\big), \tag{8}$$

$$p_{K,j}^{(\ell,h)} = \mathcal{N}\big(S_{K,h}^{(\ell)}\xi_j^{(\ell)}, S_{K,h}^{(\ell)}C_j^{(\ell)}S_{K,h}^{(\ell)\top}\big), \tag{9}$$

$$D_{ij}^{(\ell,h)} = D_{\mathrm{KL}}\big(q_{Q,i}^{(\ell,h)} \| p_{K,j}^{(\ell,h)}\big). \tag{10}$$

Because $C_i^{(\ell)} \succ 0$ and each comparison map has full row rank, the reduced covariance is positive definite: for $y \neq 0$, $y^\top SCS^\top y = (S^\top y)^\top C(S^\top y) > 0$. These are full reduced covariances, including their off-diagonal entries. Thus the construction uses nonsingular distributions in the measured space, rather than treating a singular ambient projection as a K-dimensional Gaussian density.

When $S_Q = S_K = S$, the score compares a common measurement of the two reference beliefs. When $S_Q \neq S_K$, it compares distinct role observables identified in one comparison space. The latter is an additional modeling choice. The maps are learned interface parameters and are generally rectangular; they are not individually invertible gauge transformations. Full input scope means that every head's maps read all K coordinates, as learned projections do in conventional multi-head attention (Vaswani et al., 2017). It does not mean that the comparison dimension equals K.

2.3 The attention operator

One layer combines invariant source weights with covariant content links: each recipient receives a weighted message from its available sources and adds that message to its mean. Source selection is solved at fixed input states; the mean update is an explicit predictive map.

At fixed input beliefs and role maps, source selection minimizes the headwise objective

$$\mathcal{J}_i^{(\ell,h)}(\beta) = \sum_j \beta_j D_{ij}^{(\ell,h)} + \tau_h \sum_j \beta_j \log \frac{\beta_j}{\pi_{ij}^{(h)}}, \qquad \sum_j \beta_j = 1, \tag{11}$$

on the support of $\pi_i^{(h)}$. Differentiating with a normalization multiplier gives

$$\beta_{ij}^{(\ell,h)} = \mathrm{softmax}_j\left(-D_{ij}^{(\ell,h)}/\tau_h + \log \pi_{ij}^{(h)}\right). \tag{12}$$

This is the unique categorical minimizer for finite costs, nonempty allowed support, and positive temperature. The source prior determines the available graph and any positional preference. The temperature is an independent positive scalar; its relation to conventional attention scaling is considered in Section 5.2. Section 6 derives the optimizer and its probabilistic status, and Appendix C gives the causal, window, ALiBi, and relative-position priors.

Content uses a sending map $T_h^{(\ell)} \in \mathbb{R}^{s_h \times K}$ and the effective receiving map $\widetilde{O}_h^{(\ell)} \in \mathbb{R}^{K \times s_h}$ of Section 2.2. The former extracts transmitted content; the latter places the head's aggregate in the receiver's full state space. The predictive readout and explicit residual are

$$v_j^{(\ell,h)} = T_h^{(\ell)} \xi_j^{(\ell)}, \tag{13}$$

$$z_i^{(\ell,h)} = \sum_j \beta_{ij}^{(\ell,h)} v_j^{(\ell,h)}, \tag{14}$$

$$m_i^{(\ell)} = U_i \sum_{h=1}^{H} \widetilde{O}_h^{(\ell)} z_i^{(\ell,h)}, \tag{15}$$

$$\mu_i^{\mathrm{att},(\ell)} = r\mu_i^{(\ell)} + m_i^{(\ell)}, \qquad r \in \{0,1\}. \tag{16}$$

The sum combines all head contributions in the receiving state. For $r = 1$ the last line defines an attention residual using the pre-update state; for $r = 0$ the predictive message replaces the state. The coefficient is an architectural choice, independent of a belief-gradient step size, and Section 3.5 shows that complement-preserving exponential aggregation with tiled heads has no residual. As a differentiable operator, the layer depends on source states, routing weights, frame factors, and content maps.

Fix a layer and suppress its index. The content interfaces $T_h, \widetilde{O}_h$ compose with the local frames to determine a directed map from the sender's state space to the receiver's. Write $A_{Q,i,h} = S_{Q,h}U_i^{-1}$ and $A_{K,i,h} = S_{K,h}U_i^{-1}$ for the local comparison measurements, $\mathsf{S}_{j,h} = T_h U_j^{-1}$ and $\mathsf{R}_{i,h} = U_i \widetilde{O}_h$ for the content endpoints, and $G_i = U_i^{-\top}U_i^{-1}$ for the local metric induced by the Euclidean reference inner product. Table 1 collects the geometric data and conventions, and Figure 1 shows the comparison and content interfaces of one head.

The content factors define an ambient matrix link without requiring either factor to be square:

$$L_{ij}^{(h)} = \mathsf{R}_{i,h}\mathsf{S}_{j,h} = U_i \widetilde{O}_h T_h U_j^{-1} : E_j \longrightarrow E_i. \tag{17}$$

In the affine chart this is the projected matrix link $U_i O_h T_h U_j^{-1}$; Section 3 derives it as the first-order channel of an exponential link. Under $U_i' = g_i U_i$ and $(\mu_i', \Sigma_i') = (g_i\mu_i, g_i\Sigma_i g_i^\top)$, direct substitution gives

$$\mathsf{S}'_{j,h} = \mathsf{S}_{j,h} g_j^{-1}, \qquad \mathsf{R}'_{i,h} = g_i \mathsf{R}_{i,h}, \qquad L_{ij}^{(h)\prime} = g_i L_{ij}^{(h)} g_j^{-1}. \tag{18}$$

Object	Meaning, domain, and role
N, K, H, B	Number of positions, state dimension, communication heads, and frame blocks
q_i, p_i	Full Gaussian state and local prior on E_i
ξ_i, C_i	Reference moments $U_i^{-1}\mu_i$ and $U_i^{-1}\Sigma_i U_i^{-\top}$
$\pi_{ij}, \beta_{ij}, \tau_h$	Source prior, routing probability, and positive headwise temperature
U_i, $G_i = U_i^{-\top}U_i^{-1}$	Frame $\mathsf{V} \to E_i$ and local metric induced by the Euclidean reference metric
$S_{Q,h}, S_{K,h}$	Full-row-rank maps $\mathsf{V} \to H_h^{\mathrm{cmp}} = \mathbb{R}^{r_h}$ defining comparison observables
T_h, O_h	Stored sending map $\mathsf{V} \to H_h^{\mathrm{val}} = \mathbb{R}^{s_h}$ and stored receiving factor $H_h^{\mathrm{val}} \to \mathsf{V}$
$\widetilde{O}_h$	Effective receiving map of a chart: O_h (affine), or a block of Eq. (43)
$A_{Q,i,h}, A_{K,i,h}$	Local measurements $S_{Q,h}U_i^{-1}, S_{K,h}U_i^{-1} : E_i \to H_h^{\mathrm{cmp}}$
$\mathsf{S}_{j,h}, \mathsf{R}_{i,h}$	Content endpoints $T_h U_j^{-1} : E_j \to H_h^{\mathrm{val}}$ and $U_i\widetilde{O}_h : H_h^{\mathrm{val}} \to E_i$
$L_{ij}^{(h)}$, $\mathcal{M}_{ij}$	Content link $\mathsf{R}_{i,h}\mathsf{S}_{j,h}$ and routed sum in $\mathrm{Hom}(E_j, E_i)$
X_{ij}, Θ_{ij}	Lie-algebra generator in $\mathfrak{gl}(K, \mathbb{R})$, in reference coordinates, and exponential link $U_i e^{X_{ij}} U_j^{-1}$
$\boldsymbol{\beta}_{ij}$	Head-diagonal routing matrix $\bigoplus_h \beta_{ij}^{(h)} I_{s_h}$ acting on the stacked value spaces
Ω_{ij}	Flat reference identification $U_i U_j^{-1}$, the exponential link with zero generator

Table 1: Types and conventions for frame-dressed links. Comparison and content spaces are distinct even when their dimensions agree. Layer indices are suppressed.

This is an endpoint covariance law for the complete geometric data. It does not say that a fixed rectangular matrix commutes with every g_i.

The receiving image is $U_i \operatorname{im} \widetilde{O}_h$; the invisible sending directions are $\ker(T_h U_j^{-1})$. With the metric G_j in Table 1, the orthogonal complement of that kernel is $U_j \operatorname{im} T_h^\top$. Indeed, $\langle U_j x, U_j y\rangle_{G_j} = x^\top y$, reducing this statement to the Euclidean identity $(\ker T_h)^\perp = \operatorname{im} T_h^\top$. The Fisher metric of the Gaussian location family, $\Sigma_j^{-1} = U_j^{-\top} C_j^{-1} U_j^{-1}$, transforms in the same way without a reference inner product; in that metric the complement is $U_j C_j \operatorname{im} T_h^\top$, and the Gaussian lift of Eq. (92) is the minimal-norm right inverse of the query map. Sending, receiving, and comparison subspaces may overlap. Their dimensions do not impose a partition of the frame, and $\widetilde{O}_h T_h$ is generally neither orthogonal nor idempotent.

For invariant scalar routing weights, define

$$\mathcal{M}_{ij} = \sum_h \beta_{ij}^{(h)} L_{ij}^{(h)} = U_i \left(\sum_h \beta_{ij}^{(h)} \widetilde{O}_h T_h \right) U_j^{-1} = U_i \widetilde{O} \boldsymbol{\beta}_{ij} T U_j^{-1}, \tag{19}$$

$$m_i = \sum_j \mathcal{M}_{ij} \mu_j, \tag{20}$$

where $T = [T_1; \cdots; T_H]$ and $\widetilde{O} = [\widetilde{O}_1 \ \cdots \ \widetilde{O}_H]$ stack the heads and $\boldsymbol{\beta}_{ij} = \bigoplus_h \beta_{ij}^{(h)} I_{s_h}$. Equation (18) gives $\mathcal{M}'_{ij} = g_i \mathcal{M}_{ij} g_j^{-1}$ and $m'_i = g_i m_i$. The projected Gaussian routing of Eq. (12) supplies the required scalar invariance: $\xi'_i = \xi_i$ and $C'_i = C_i$, so all comparison laws and their divergences are unchanged. The attention update $r\mu_i + m_i$ consequently has the same vector covariance.

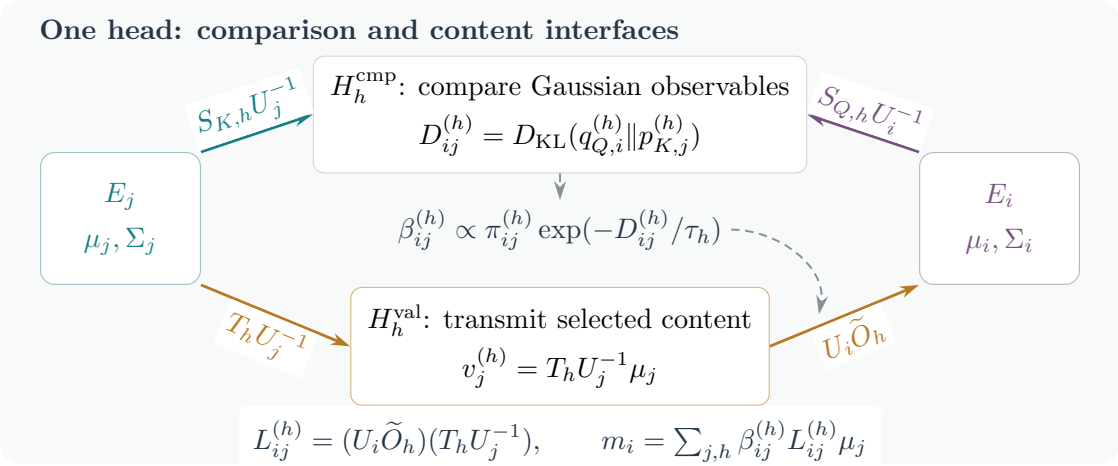


Figure 1: **Separate comparison and content interfaces.** Query and key maps send Gaussian observables into a common comparison space, where their divergence determines the routing weight. The content maps $T_h U_j^{-1}$ and $U_i \widetilde{O}_h$ extract and place the transmitted mean. Their composition is the matrix link $L_{ij}^{(h)}$; dashed arrows show scalar routing control, not an additional state map. The Gibbs weights are derived in Section 6.1.

2.4 Depth, token priors, and prediction

At depth ℓ, choose comparison and content maps $S_Q^{(\ell)}, S_K^{(\ell)}, T^{(\ell)}, O^{(\ell)}$. The frame field may be shared across depths while the communication maps vary. An attention residual can be followed by an affine normalization and a coordinate residual MLP. One such composition is

$$u_i^{(\ell)} = \mathrm{LayerNorm}_{\gamma,b}\big(\mu_i^{\mathrm{att},(\ell)}\big), \tag{21}$$

$$\mu_i^{(\ell+1)} = u_i^{(\ell)} + W_2^{(\ell)}\,\mathrm{GELU}\big(W_1^{(\ell)} u_i^{(\ell)} + b_1^{(\ell)}\big) + b_2^{(\ell)}. \tag{22}$$

Here the parameters γ, b of the layer normalization (Ba et al., 2016) are shared across depths, the MLP parameters may differ, and GELU is the Gaussian error linear unit (Hendrycks and Gimpel, 2016). Two covariance contracts are considered. Covariance passthrough, $\Sigma_i^{(\ell+1)} = \Sigma_i^{(\ell)}$, defines covariance as a routing state rather than the propagated uncertainty of the nonlinear mean map; it remains an input to subsequent comparisons. Moment matching instead returns from attention the covariance of the declared predictive kernel of Theorem 20, which depends on the context through the routing and the source moments, and passes it through the normalization and MLP. Neither contract propagates uncertainty through the nonlinear block. This construction specifies a predictive architecture; it does not identify the normalized mean with a draw from a transformed Gaussian law.

A coordinatewise activation and fixed affine normalization do not in general commute with arbitrary $\mathrm{GL}(K)$ transformations. Consequently, the covariance of Theorem 14 does not extend automatically to this whole block. Corollary 15 gives the normal form of the

evaluated block and its covariant alternatives, and Appendix F.2 makes the Gaussian-closure restriction explicit.

In standard transformers, token embeddings map discrete tokens to continuous vectors via a learned embedding matrix $W_{\text{embed}} \in \mathbb{R}^{V \times K}$. In the present construction, each token type $c \in \{1, \ldots, V\}$ is associated with a learned prior distribution:

$$p_c = \mathcal{N}(\mu_p^c, \Sigma_p^c), \qquad \phi_c \in \mathfrak{gl}(K), \tag{23}$$

comprising a prior mean μ_p^c, prior covariance Σ_p^c, and gauge frame coefficients ϕ_c. At the start of each sequence, agent i's belief is initialized to the prior of its token type: $q_i(0) = p_{\text{token}(i)}$. The collection $\{(\mu_p^c, \Sigma_p^c, \phi_c)\}_{c=1}^V$ constitutes a prior bank: a Gaussian embedding table in the sense of Vilnis and McCallum (2015), extended by frame coefficients. If only the means are retained, their table $W_{\text{embed}} = [\mu_p^1, \ldots, \mu_p^V]^\top$ has the role of an ordinary embedding matrix. Isotropy alone does not remove the covariance or frame parameters.

The bank initializes the input state independently of the choice of output decoder. Positional frame factors can be composed with its token frame before communication. Using Gaussian templates to encode a token therefore does not require the optional Gaussian-bank vocabulary readout of Appendix D.

For input positions $x_0, \ldots, x_{N-1}$, causal source support is $S_i = \{0, \ldots, i\}$. The sequence model has the normalized factorization

$$p_\theta(x_0, \ldots, x_N) = p_\theta(x_0) \prod_{i=0}^{N-1} p_\theta(x_{i+1} \mid x_{\le i}). \tag{24}$$

A proper initial-token law completes the joint. A linear categorical decoder specifies the next-token factors from the final mean,

$$p_\theta(x_{i+1} = v \mid x_{\le i}) = \text{softmax}_v \left(W_{\text{out}} \mu_i^{(L)} + b_{\text{out}} + \kappa_{\text{uni}} \log p_{\text{uni}} \right), \tag{25}$$

where p_{uni} is a positive vocabulary prior. This decoder is itself a Gibbs readout of shared-covariance templates (Section 6.3), and the token-bank decoder of Appendix D is another choice. Minimizing next-token cross-entropy trains this composed conditional predictor through all of its differentiable components. The observed next token supplies the training target; it is not an input to the causal state construction. Algorithm 1 gives the evaluated realization in the experimental methods. In that evaluation each conditional is computed from a window of at most 128 tokens with stride 32; replacing $x_{\le i}$ by that window leaves every factor normalized, so the reported perplexity is the log-loss of the resulting sequence law.

3 Lie-Algebra Links

The content link of Eq. (17) is a composite of site frames and shared maps, and a single head of width $s_h < K$ makes it singular. A lattice link variable is instead a group element attached to an edge, and the algebra-valued potential is its logarithmic coordinate (Wilson, 1974; Creutz, 1983). This section takes the group-valued link as the primary communication object, writes it through a generator in the Lie algebra $\mathfrak{gl}(K, \mathbb{R})$, and derives the selective

content maps of Section 2.3 from that generator. Exponentiation fixes the group domain and the endpoint law. It does not select the generator, the aggregation over sources, or the residual, and each of these is stated below as a separate modeling choice.

3.1 Exponential links and their endpoint law

Definition 1 (Lie-algebra link) *For a generator $X_{ij} \in \mathfrak{gl}(K, \mathbb{R})$ expressed in reference coordinates, the Lie-algebra link, or exponential link, from source j to receiver i is*

$$\Theta_{ij} = U_i \exp(X_{ij}) U_j^{-1} : E_j \longrightarrow E_i, \tag{26}$$

and its reference representative is $U_i^{-1}\Theta_{ij}U_j = \exp(X_{ij})$.

Proposition 2 (Endpoint law and invertibility) *Let $X_{ij} \in \mathfrak{gl}(K, \mathbb{R})$.*

1. *Under the passive site action of Eq. (4), with the reference generator held fixed, $\Theta'_{ij} = g_i \Theta_{ij} g_j^{-1}$.*
2. *Under the common reference change $U_i \mapsto U_i a$, the same local link is represented by the generator $a^{-1}X_{ij}a$, the adjoint action of $\mathrm{GL}(K)$ on its Lie algebra.*
3. *$\det \Theta_{ij} = \det(U_i U_j^{-1}) e^{\operatorname{tr} X_{ij}} \neq 0$. Every exponential link is invertible, including links with singular generators, and $\Theta_{ij}^{-1} = U_j e^{-X_{ij}} U_i^{-1}$.*
4. *A matrix $G \in \mathrm{GL}(K, \mathbb{R})$ equals e^X for some real X exactly when, for each negative eigenvalue of G and each block size, the Jordan blocks of that eigenvalue and size occur an even number of times. Every such G lies in the identity component $\mathrm{GL}^+(K)$, and for $K \geq 2$ not every element of $\mathrm{GL}^+(K)$ is of this form.*

Proof Item 1 follows by substituting $U'_i = g_i U_i$ into Eq. (26). Item 2 follows from $\exp(a^{-1}Xa) = a^{-1}\exp(X)a$, which holds term by term in the exponential series, so that $(U_i a)\exp(a^{-1}X_{ij}a)(U_j a)^{-1} = \Theta_{ij}$. Item 3 is the identity $\det e^X = e^{\operatorname{tr} X}$ (Hall, 2015). The characterization in item 4 is Theorem 1 of Culver (1966), and membership in $\mathrm{GL}^+(K)$ follows from item 3. For the last statement, $\mathrm{diag}(-1, -2) \oplus I_{K-2}$ has positive determinant and two negative eigenvalues, each with a single Jordan block of size one, so it has no real logarithm. ∎

The generator is reference data. The frames carry the entire passive action, as they do for the comparison and content maps of Section 2.2, and X_{ij} changes only under a common change of the reference basis. In lattice gauge theory the link e^{aA} has the algebra-valued potential A as its logarithmic coordinate, and A transforms as a connection under local gauge transformations (Creutz, 1983). Here the frames dress the potential, so the generator is invariant under independent site changes of frame, while the link Θ_{ij} transforms exactly as a lattice link variable. When the reference representative has no eigenvalue on the closed negative real axis, it has a unique real logarithm whose eigenvalues lie in the strip $|\operatorname{Im} z| < \pi$, the principal logarithm (Higham, 2008, Thm. 1.31). The principal logarithm is therefore a chart on the open subset of $\mathrm{GL}^+(K)$ without eigenvalues on the closed negative real axis; it is not a global chart of the group.

3.2 Why the links are group valued

Four structural facts motivate placing the link in the group rather than in the space of all matrices. The first concerns rank. A single head with $s_h < K$ gives $\operatorname{rank} L_{ij}^{(h)} \le s_h$, so for every matrix N the product $L_{ij}^{(h)} N$ is singular, and no group element factors as $L_{ij}^{(h)} N$ through a selective channel. The relation between a selective channel and an invertible link must therefore be additive, through a block, or infinitesimal through the Lie algebra, and the construction of Section 3.3 realizes all three.

The second concerns loops. On the causal support every closed directed walk is a self-loop (Theorem 14), so routed products of causal links contain no loop through distinct positions. An invertible link supplies the reverse edge Θ_{ij}^{-1} without new parameters, and inverse-edge triangles then carry conjugation-invariant holonomies on the causal graph (Section 8.3). A singular link admits closed matrix products only on a separately declared extended graph.

The third concerns the singular boundary. An additive update of an invertible matrix M can reach the singular set in a single step of spectral norm $\sigma_{\min}(M)$: for unit singular vectors u, v belonging to $\sigma_{\min}(M)$, the matrix $M - \sigma_{\min}(M) uv^\top$ annihilates v. The exponential parametrization keeps $\det e^X = e^{\operatorname{tr} X} > 0$ for every finite generator, and $\sigma_{\min}(e^X) \ge e^{-\|X\|_2}$ because $\|e^{-X}\|_2 \le e^{\|X\|_2}$. For an exact Gaussian pushforward, the singular boundary also lies at infinite statistical distance.

Proposition 3 (The singular boundary is at infinite distance) *Let $q_0 = \mathcal{N}(\mu_0, \Sigma_0)$ with $\Sigma_0 \succ 0$, let $g \in \mathrm{GL}(K)$, and let $q_g = \mathcal{N}(g\mu_0, g\Sigma_0 g^\top)$. If $\lambda_{\min}$ is the smallest eigenvalue of $\Sigma_0^{-1/2} g \Sigma_0 g^\top \Sigma_0^{-1/2}$, then*

$$d_{\mathrm{FR}}(q_0, q_g) \ge \tfrac{1}{\sqrt{2}} \big|\log \lambda_{\min}\big|. \tag{27}$$

As g approaches a singular matrix, $\lambda_{\min} \to 0$ and the Fisher–Rao distance diverges.

Proof Let $w = \Sigma_0^{-1/2} e$ for a unit eigenvector e belonging to $\lambda_{\min}$. The scalar observable $w^\top x$ has variance $w^\top \Sigma_0 w = 1$ under q_0 and $w^\top g \Sigma_0 g^\top w = \lambda_{\min}$ under q_g. The univariate Gaussian family with mean m and standard deviation s has Fisher metric $(dm^2 + 2ds^2)/s^2$, so every path between standard deviations 1 and $\lambda_{\min}^{1/2}$ has length at least $\sqrt{2}|\log \lambda_{\min}^{1/2}|$. Marginalization to $w^\top x$ is a Markov morphism and does not increase Fisher–Rao length (Amari and Nagaoka, 2000), which gives Eq. (27). If g tends to a singular matrix $g_\star$, then $g_\star \Sigma_0 g_\star^\top$ is singular, and continuity of eigenvalues gives $\lambda_{\min} \to 0$. ∎

The proposition concerns the exact pushforward of a fixed nondegenerate Gaussian. A positive covariance floor or additive receiving noise changes this geometry, so it is not a claim that a training objective diverges when a content map becomes singular. The fourth fact is that the generator is a linear coordinate on which a Gaussian prior, a natural gradient, and latent inference can be written, and Section 8.4 uses it in this way.

3.3 Low-rank generators and the selective channels

The generator whose first-order term reproduces the content link is built from the stored value and output maps. Its exponential has an exact closed form through the entire function

$$\varphi_1(Z) = \sum_{n\geq 1} \frac{Z^{n-1}}{n!}, \qquad \varphi_1(Z) = (e^Z - I)Z^{-1} \text{ for invertible } Z, \tag{28}$$

the first of the φ-functions of exponential integrators (Hochbruck and Ostermann, 2010).

Lemma 4 (Exponential of a low-rank generator) *For $P \in \mathbb{R}^{K\times s}$ and $Q \in \mathbb{R}^{s\times K}$,*

$$\exp(PQ) = I_K + P\varphi_1(QP)Q. \tag{29}$$

Proof For $n \geq 1$, regrouping the product gives $(PQ)^n = P(QP)^{n-1}Q$. Summing the exponential series term by term gives $I_K + P\big(\sum_{n\geq 1}(QP)^{n-1}/n!\big)Q$, and the inner series converges for every $s \times s$ matrix. ■

Proposition 5 (Flat transport plus a channel term) *For one head with stored maps T_h, O_h and the generator $X_{ij} = O_hT_h$,*

$$\Theta_{ij} = \Omega_{ij} + U_iO_h\varphi_1(T_hO_h)T_hU_j^{-1}. \tag{30}$$

1. $\exp(O_hT_h)$ *acts as the identity on* $\ker T_h$*, so on the subspace* $U_j \ker T_h$ *that the head does not read,* Θ_{ij} *coincides with the flat identification* Ω_{ij}*.*
2. *The image of* $\exp(O_hT_h) - I$ *lies in* $\operatorname{im} O_h$*, so* Θ_{ij} *differs from* Ω_{ij} *only in directions that the head writes.*
3. *For* $X_{ij} = sO_hT_h$*,* $\Theta_{ij} = \Omega_{ij} + sU_iO_hT_hU_j^{-1} + O(s^2)$*, so the projected matrix link is the first-order channel.*
4. $\det \Theta_{ij} = \det(U_iU_j^{-1})e^{\operatorname{tr}(T_hO_h)}$*.*

The same statements hold for the stacked generator $X = OT = \sum_h O_hT_h$, with $O = [O_1 \ \cdots \ O_H] \in \mathbb{R}^{K\times S}$, $T = [T_1; \cdots; T_H] \in \mathbb{R}^{S\times K}$, and $S = \sum_h s_h$.

Proof Equation (30) is Lemma 4 with $P = O_h$ and $Q = T_h$, conjugated by the frames. Items 1 and 2 are read off from the correction term, which has the form $O_h(\cdots)T_h$. Item 3 follows from $\varphi_1(sZ) = I + O(s)$, and item 4 from $\det e^X = e^{\operatorname{tr} X}$ with $\operatorname{tr}(O_hT_h) = \operatorname{tr}(T_hO_h)$. ■

Every entry of Θ_{ij} is determined by the frames and the stored maps. The link is the identity transport on the unread complement and carries the channel term, which equals the projected matrix link to first order and acquires the corrections $\varphi_1(T_hO_h)$ at higher order. Contracting the scaled link $\Theta_{ij}(s) = U_ie^{sO_hT_h}U_j^{-1}$ against one routing row at fixed source states gives

$$\sum_j \beta_{ij}\Theta_{ij}(s)\mu_j = \sum_j \beta_{ij}\Omega_{ij}\mu_j + s\sum_j \beta_{ij}U_iO_hT_hU_j^{-1}\mu_j + O(s^2). \tag{31}$$

The zeroth-order term transports complete source states by the flat identification, which is the message of complete-state alignment (Section 6.1), and the first-order term is the projected predictive message of Eq. (16) in the affine chart. Both message forms are thus orders of one exponential link. This is a statement about the form of the message; it does not recover either objective. With an independent source label for each head, the zeroth-order term becomes a head-routed flat transport (Proposition 11).

3.4 Linear and exponential links as charts of one group

The phrase "first-order channel" could suggest that the matrix link is a weak-field truncation of the exponential link. On a specified logarithm domain the relation is instead an exact change of coordinates. Fix $O \in \mathbb{R}^{K\times s}$ and $T \in \mathbb{R}^{s\times K}$ with TO invertible, so that $\mathbb{R}^K = W \oplus W'$ with $W = \operatorname{im} O$ and $W' = \ker T$, and let $\Pi = O(TO)^{-1}T$ be the projector onto W along W'. For $B \in \mathrm{GL}(s)$ put

$$\Upsilon(B) = (I - \Pi) + OB(TO)^{-1}T, \tag{32}$$

the operator that fixes W' pointwise and acts on W as B in the basis formed by the columns of O.

Proposition 6 (Two charts of one group) *1. $\Upsilon : \mathrm{GL}(s) \to \mathrm{GL}(K)$ is an injective homomorphism with $\det \Upsilon(B) = \det B$. Its image is the factor of the block-diagonal Levi subgroup $\mathrm{GL}(W) \times \mathrm{GL}(W')$ that acts trivially on W'.*

2. *Linear links: for $C \in \mathbb{R}^{s\times s}$, $I + OCT = \Upsilon(I_s + CTO)$, so $\det(I + OCT) = \det(I_s + CTO)$.*

3. *Exponential links: $\exp(OT) = \Upsilon(e^{TO})$.*

4. *Exact conversion: let $O_{\mathrm{lin}} \in \mathbb{R}^{K\times s}$ be such that $I_s + TO_{\mathrm{lin}}$ has no eigenvalue on $(-\infty, 0]$, put $Z = \log(I_s + TO_{\mathrm{lin}})$ with the principal logarithm, and put $O = O_{\mathrm{lin}}\varphi_1(Z)^{-1}$. Then $\varphi_1(Z)$ is invertible, $TO = Z$, and*

$$\exp(OT) = I_K + O_{\mathrm{lin}}T. \tag{33}$$

Proof For item 1, $\Upsilon(B)Oy = OBy$ because $(I - \Pi)O = 0$ and $(TO)^{-1}TO = I$, and $\Upsilon(B)w' = w'$ for $w' \in W'$ because $\Pi w' = 0$. An operator on $W \oplus W'$ is determined by its action on the two summands, so $\Upsilon(B)\Upsilon(B') = \Upsilon(BB')$, $\det \Upsilon(B) = \det B$, and $\Upsilon(B) = I$ only for $B = I$. For item 2, $(I + OCT)Oy = O(I + CTO)y$ and $(I + OCT)w' = w'$. For item 3, $(OT)^n O = O(TO)^n$ gives $\exp(OT)Oy = Oe^{TO}y$, and $\exp(OT)w' = w'$. For item 4, the eigenvalues of Z have imaginary parts in $(-\pi, \pi)$, while the eigenvalues $(e^z - 1)/z$ of $\varphi_1(Z)$ vanish only at $z \in 2\pi i\mathbb{Z} \setminus \{0\}$, so $\varphi_1(Z)$ is invertible. Since $TO_{\mathrm{lin}} = e^Z - I = Z\varphi_1(Z)$, with Z and $\varphi_1(Z)$ commuting, $TO = TO_{\mathrm{lin}}\varphi_1(Z)^{-1} = Z$. Lemma 4 then gives $\exp(OT) = I + O\varphi_1(TO)T = I + O_{\mathrm{lin}}\varphi_1(Z)^{-1}\varphi_1(Z)T = I + O_{\mathrm{lin}}T$. ■

Item 4 says that the flat identification plus a linear channel term, $\Omega_{ij} + U_i O_{\mathrm{lin}} T U_j^{-1}$, is exactly an exponential link for every O_{lin} in an open set containing $O_{\mathrm{lin}} = 0$. On their

common domain the linear and logarithmic parametrizations describe the same group elements, related by $I_s + TO_{\text{lin}} = e^{TO}$, and the expansion of Proposition 5 states that the two charts agree to first order at the identity. Using one numerical generator in both $I + X$ and e^X is a first-order approximation only. The choice of chart changes four things. First, the linear parameter space contains the singular matrices, and a step in (O_{lin}, T) can carry $\det(I_K + O_{\text{lin}}T) = \det(I_s + TO_{\text{lin}})$ through zero, whereas the logarithmic chart cannot reach that boundary; by Proposition 2 it also excludes the matrices without a real logarithm, including every matrix of negative determinant. Second, the two charts carry different optimization geometries, since an optimizer acting on the factors induces different changes of the composite matrix in each. Third, they differ at order one for finite generators: at $X = I$, $e^X = eI$ while $I + X = 2I$. Fourth, these statements concern an unrouted operator with one generator; routed head sums and edge-dependent generators may leave the one-generator family. When the heads tile the state, O_{lin} is the stored factor of the first-order chart of Section 3.6, whose effective receiving map is $T^{-1} + O_{\text{lin}}$.

3.5 Routing outside the exponent: aggregation and the residual

The group law does not choose an aggregation over sources. A link acts on one source state, whereas a predictive mixture also needs a normalized law over source choices. Placing the routing matrix inside the exponent, $X_{ij} = O\boldsymbol{\beta}_{ij}T$, changes the single-edge operator and leaves the aggregation over j unspecified, since neither the sum nor the product of the routed exponentials specifies how source vectors are selected. The construction below keeps categorical routing outside the exponent and uses the routing-independent stacked generator $X = OT$, so that $\beta_{ij}^{(h)}$ retains its meaning as the probability that head h selects source j (Theorem 20). It assumes that the stacked transfer $TO \in \mathbb{R}^{S\times S}$ is invertible, which allows $S \le K$.

Lemma 7 (Complement split) *With* $\Pi = O(TO)^{-1}T$,

$$\exp(OT) = (I - \Pi) + Oe^{TO}(TO)^{-1}T. \tag{34}$$

Proof By Lemma 4, $\exp(OT) = I + O\varphi_1(TO)T = (I - \Pi) + O[(TO)^{-1} + \varphi_1(TO)]T$, and $(TO)^{-1} + \varphi_1(TO) = (TO)^{-1} + (e^{TO} - I)(TO)^{-1} = e^{TO}(TO)^{-1}$ because every term is a function of TO. ∎

The first term is the identity on the unread complement $\ker T$ and annihilates $\operatorname{im} O$. The second reads the S sent coordinates $T\xi$, mixes them through the $S \times S$ matrix $e^{TO}(TO)^{-1}$, and writes them into $\operatorname{im} O$. Write $\widetilde{O} = Oe^{TO}(TO)^{-1}$, and let $\widetilde{O}_h$ be its block column receiving head h. Following the predictive kernel of Theorem 20, each head draws its own source label J^h with probability $\beta_{ij}^{(h)}$, independently across heads, and the sent block $T_h\xi$ is read from the source selected by its head. The complement term needs a carrier of its own, a label J^0. The transported reference state and its expectation over the independent

labels are

$$\xi_i' = (I - \Pi)\xi_{J^0} + \sum_h \widetilde{O}_h T_h \xi_{J^h}, \tag{35}$$

$$\mathbb{E}[\xi_i'] = \mathbb{E}[(I - \Pi)\xi_{J^0}] + \sum_h \sum_j \beta_{ij}^{(h)} \widetilde{O}_h T_h \xi_j. \tag{36}$$

Each sent block travels with the label of its own head, so the cross-head mixing in $e^{TO}(TO)^{-1}$ does not couple the head routings. The weights depend on the states through the comparison costs, but they are evaluated at the layer's input states and held fixed while the message is formed, so Eq. (36) is the expectation conditional on those states.

Proposition 8 (Complement persistence) *Assume that only sent content may cross between agents and that the update retains one unchanged complement taken from an existing agent state, as in Eq.* (35)*. Then the carrier is the receiver, $J^0 = i$, and*

$$\mu_i^+ = U_i(I - \Pi)U_i^{-1}\mu_i + \sum_h \sum_j \beta_{ij}^{(h)} U_i \widetilde{O}_h T_h U_j^{-1} \mu_j. \tag{37}$$

Proof The complement $\ker T$ holds exactly the directions that no head sends. A carrier $J^0 = j \neq i$ would place unsent coordinates of agent j in the state of agent i, contradicting the premise, so the receiver is the only admissible carrier within the declared unchanged-complement rule. Without that rule, locality would also permit discarding the complement or applying a receiver-local map, so locality alone does not force persistence. Equation (36) with $J^0 = i$, conjugated by the frames, gives Eq. (37). ∎

Corollary 9 (Tiled heads leave no residual) *If the heads tile the reference space, $S = K$ with O and T invertible, then $\Pi = I$, the complement is empty, $\widetilde{O} = e^{OT}T^{-1}$, and*

$$\mu_i^+ = \sum_j U_i \widetilde{O} \boldsymbol{\beta}_{ij} T U_j^{-1} \mu_j = \sum_h \sum_j \beta_{ij}^{(h)} U_i \widetilde{O}_h T_h U_j^{-1} \mu_j. \tag{38}$$

Proof With O and T invertible, $\Pi = OO^{-1}T^{-1}T = I$, and $O(TO)^n O^{-1} = (OT)^n$ gives $Oe^{TO}(TO)^{-1} = Oe^{TO}O^{-1}T^{-1} = e^{OT}T^{-1}$. Block multiplication gives $\sum_h \beta_{ij}^{(h)} \widetilde{O}_h T_h = \widetilde{O}\boldsymbol{\beta}_{ij}T$. ∎

Under complement-preserving aggregation, the receiver therefore keeps its own state exactly on the directions that no head writes and takes the transported message on the directions that heads write. That projected residual $U_i(I - \Pi)U_i^{-1}\mu_i$ differs from the full residual $r\mu_i$ of Eq. (16). When the heads tile the state, as in the implementation, the complement is empty, so the residual $r = 1$ has no counterpart in this aggregation, and the receiver's own state enters only through the self-weights $\beta_{ii}^{(h)}$. A residual of the projected kind requires channels that do not tile, $S < K$; the $K - S$ coordinates that no head reads or writes then form the residual stream. Gaussian fusion reaches a related structure by

a different argument: the channel-likelihood receiver of Section 7 keeps its prior on the directions that no head observes and blends prior and message on the observed directions with a precision-weighted gain (Eq. (81)). The additive residual $\mu_i + m_i$ is neither. It is the residual connection of residual networks (He et al., 2016), added in every direction without normalization.

3.6 The tiled message order by order

In the setting of Corollary 9, with tiled heads, square invertible stacked maps, the shared generator $X = OT$, and routing outside the exponent, the receiving maps sum in closed form, and expanding them in the scale of the generator separates the message into orders with distinct roles. Write $\xi_j = U_j^{-1}\mu_j$ for the reference source states.

Proposition 10 (Resummed receiving map) *Under these assumptions,*

$$\widetilde{O} = e^{OT}T^{-1} = T^{-1} + O\varphi_1(TO), \qquad \mu_i^+ = U_i \sum_j \widetilde{O}\boldsymbol{\beta}_{ij}T\xi_j. \tag{39}$$

The effective transfer is $T\widetilde{O} = e^{TO}$, so $\det(T\widetilde{O}) = e^{\mathrm{tr}(TO)} > 0$.

Proof The first form is Corollary 9. Lemma 4 gives $e^{OT} = I + O\varphi_1(TO)T$, and multiplying on the right by T^{-1} gives the second form. Finally $T\widetilde{O} = Te^{OT}T^{-1} = e^{TO}$, and $\det e^Z = e^{\mathrm{tr}\,Z}$. ∎

Replacing O by sO at fixed T, so that $X = sOT$, gives the series

$$\widetilde{O}(s) = T^{-1} + sO\varphi_1(sTO) = T^{-1} + \sum_{n\geq 1} \frac{s^n}{n!} O(TO)^{n-1}, \tag{40}$$

which converges for every s. The order-n message replaces $\widetilde{O}$ in Eq. (39) by its order-n term at $s = 1$.

Proposition 11 (Zeroth order) *Let Δ_h be the coordinate projector onto the head-h block of $\mathbb{R}^S$, and let $\Pi_h = T^{-1}\Delta_h T$. Then $\Pi_h = (T^{-1})_{:,h}T_h$, $\Pi_h\Pi_g = \delta_{hg}\Pi_h$, $\sum_h \Pi_h = I_K$, $\ker\Pi_h = \ker T_h$, $\mathrm{im}\,\Pi_h = \bigcap_{g\neq h}\ker T_g$, and*

$$T^{-1}\boldsymbol{\beta}_{ij}T = \sum_h \beta_{ij}^{(h)}\Pi_h. \tag{41}$$

If every head uses the same routing, $\beta_{ij}^{(h)} = \beta_{ij}$, as always holds for $H = 1$, the zeroth-order message is $\sum_j \beta_{ij}\Omega_{ij}\mu_j$.

Proof The relations $\Delta_h\Delta_g = \delta_{hg}\Delta_h$ and $\sum_h \Delta_h = I$ survive conjugation by T, and Eq. (41) is $\boldsymbol{\beta}_{ij} = \sum_h \beta_{ij}^{(h)}\Delta_h$ conjugated by T. Writing T^{-1} by block columns gives $\Pi_h = (T^{-1})_{:,h}T_h$. Since $\Delta_h Tx$ has the single nonzero block $T_h x$, $\Pi_h x = 0$ exactly when $T_h x = 0$, and $x \in \mathrm{im}\,\Pi_h = T^{-1}\,\mathrm{im}\,\Delta_h$ exactly when $T_g x = 0$ for every $g \neq h$. A shared

routing has $\boldsymbol{\beta}_{ij} = \beta_{ij} I$, so $U_i T^{-1} \boldsymbol{\beta}_{ij} T \xi_j = \beta_{ij} \Omega_{ij} \mu_j$. ■

The zeroth-order message $U_i \sum_h \sum_j \beta_{ij}^{(h)} \Pi_h \xi_j$ is a head-routed flat transport. Head h carries, unmixed and routed by its own weights, the component $\Pi_h \xi_j$ of each source state in the directions that no other head reads. The projectors are oblique in general, and the sum includes the receiver through $\beta_{ii}^{(h)}$, so this term is not the residual $r\mu_i$. The first-order term $U_i \sum_j O \boldsymbol{\beta}_{ij} T \xi_j$ is the standard multi-head value path with value maps T_h and output maps O_h (Section 5.3). The second-order term,

$$\tfrac{1}{2} O(TO) \boldsymbol{\beta}_{ij} T = \tfrac{1}{2} \sum_{g,h} O_g (T_g O_h) \beta_{ij}^{(h)} T_h, \tag{42}$$

is cross-head mixing. The routed block of head h is written by O_h, read again by every head g through the block $T_g O_h$ of the head-transfer matrix TO, and rewritten by O_g, with the same source and no second routing. The order-n term makes $n-1$ such round trips with weight $1/n!$.

The orders do not enlarge the family of mean maps. For invertible T the map $O \mapsto T^{-1} + O$ is a bijection, so the first-order truncation reaches every receiving map, as the affine chart does. Since OT ranges over all of $\mathfrak{gl}(K)$, the exponential chart reaches a receiving map $\widetilde{O}$ exactly when $\widetilde{O}T$ has a real logarithm X, with $O = XT^{-1}$, and by Proposition 2 this forces $\det(\widetilde{O}T) > 0$. The three parametrizations therefore represent the same tiled shared-generator mean maps up to the real-logarithm domain. They differ in what the zero of the stored factor means, in the reachable component, and in the conditioning and optimization geometry of the trained links, and these are the properties that a comparison between them tests.

A finite truncation is a separate model that approximates the exponential only for small generators. Truncation after order n gives $\widetilde{O}_n T = p_n(X)$ with $p_n(z) = \sum_{k \le n} z^k / k!$, and the remainder obeys $\|e^X - p_n(X)\|_2 \le \sum_{k>n} \|X\|_2^k / k!$, which is small only when $\|X\|_2$ is small compared with n. No finite truncation is structurally group valued. Every p_n with $n \ge 1$ has a complex root, which is an eigenvalue of some real Z when $K \ge 2$, through a rotation-scaling block if it is not real, and $p_n(Z)$ is then singular by spectral mapping. For example, $I + Z + Z^2/2 = \frac{1}{2}[(Z+I)^2 + I]$ vanishes at $Z = \left(\begin{smallmatrix} -1 & -1 \\ 1 & -1 \end{smallmatrix}\right)$. The exponential has no zeros and is therefore invertible for every generator.

The stored factors therefore admit three effective receiving maps, which define the affine, first-order, and exponential charts of the implementation:

$$\widetilde{O} = O, \qquad \widetilde{O} = T^{-1} + O, \qquad \widetilde{O} = T^{-1} + O\varphi_1(TO) = e^{OT} T^{-1}. \tag{43}$$

The affine chart keeps only the first-order channel, so $O = 0$ means zero content. The first-order chart keeps orders zero and one. It has the same function class as the affine chart but is centered at the head-routed flat transport, so a decay of the stored factor pulls toward that transport rather than toward zero content. The exponential chart has the same center, differs from the first-order chart from second order in O on, and is the only chart whose effective transfer stays in $\mathrm{GL}^+(K)$. At the block-projector initialization $T = I$, the content $c_{\mathrm{out}} \boldsymbol{\beta}_{ij}$ at output scale $c_{\mathrm{out}} > 0$ requires $O = c_{\mathrm{out}} I$, $(c_{\mathrm{out}} - 1) I$, and $(\log c_{\mathrm{out}}) I$ in the three charts, since $z\varphi_1(z) = e^z - 1$ equals $c_{\mathrm{out}} - 1$ at $z = \log c_{\mathrm{out}}$. At $c_{\mathrm{out}} = 1$ both

centered charts start at $O = 0$, whereas reusing $O = I$ in the exponential chart would start the content at $e\boldsymbol{\beta}_{ij}$. The factor $\varphi_1(TO)$ needs no inverse of TO: for every square A,

$$\exp \begin{pmatrix} A & I \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} e^A & \varphi_1(A) \\ 0 & I \end{pmatrix}, \tag{44}$$

because $\left(\begin{smallmatrix} A & I \\ 0 & 0 \end{smallmatrix}\right)^n = \left(\begin{smallmatrix} A^n & A^{n-1} \\ 0 & 0 \end{smallmatrix}\right)$ for $n \geq 1$. This is the two-block case of the block-triangular exponential formula of Van Loan (1978), so one exponential of size $2S$ returns e^{TO} and $\varphi_1(TO)$.

A chart changes only the effective receiving map. Every statement of Sections 4 to 7 that involves the content maps uses them only through $\widetilde{O}_h$ and T_h, with $M_h = \widetilde{O}_h T_h$, and therefore holds in each chart. In particular, the moment-matched covariance of Theorem 20 holds with that substitution, and the channel-likelihood receiver uses $\widetilde{O}_h^\top$ as its measurement.

Remark 12 (Routed exponents and self-gates) *The routed generator $X_{ij} = O\boldsymbol{\beta}_{ij}T$ gives $\exp(O\boldsymbol{\beta}_{ij}T) = I + O\varphi_1(\boldsymbol{\beta}_{ij}TO)\boldsymbol{\beta}_{ij}T$ by Lemma 4. Its first-order term is the routed head sum, and heads interact from second order on through the blocks of TO. Because the weights depend on the states, $\varphi_1(\boldsymbol{\beta}_{ij}TO)$ is then a matrix function of the states. On the self-link this is a single-source transformation and needs no aggregation rule; for tiled heads with $TO = I$ it equals $\sum_h e^{\beta_{ii}^{(h)}} O_h T_h$, a gate in $(1, e]$ on each head block whose argument is the self-routing weight. A general state-dependent self-generator $X(\xi) = \mathrm{diag}(a(\xi))$ gives the coordinatewise gate $\xi_k \mapsto \xi_k e^{a_k(\xi)}$, and every positive gate has this form. The SiLU $x\,\mathrm{sgm}(x)$, with* sgm *the logistic function, and the GELU $x\Phi_{\mathcal{N}}(x)$, with $\Phi_{\mathcal{N}}$ the standard normal distribution function, are exponential self-links with generators* $\log \mathrm{sgm}$ *and* $\log \Phi_{\mathcal{N}}$ *(Hendrycks and Gimpel, 2016). This rewriting places the MLP in the same group-valued language as the links, but it does not derive the generator.*

4 Covariance and the Reference-Coordinate Normal Form

The operator of Section 2.3 is written with frames, but every routing cost and message it computes factors through the reference moments. This section makes that statement exact, derives its consequences, and states what it does and does not constrain.

Lemma 13 (Invariance of divergences under invertible maps) *For probability laws P and Q on $\mathbb{R}^K$ with densities and any $A \in \mathrm{GL}(K)$,*

$$D_{\mathrm{KL}}(A_* P \| A_* Q) = D_{\mathrm{KL}}(P \| Q). \tag{45}$$

Proof Both densities acquire the same Jacobian factor, which cancels in their ratio (Amari, 2016). The same argument applies to f-divergences, and the data-processing inequality applied to A and A^{-1} gives equality for every Rényi order (van Erven and Harremøes, 2014, Thm. 9). ∎

This common invertible change of coordinates is distinct from a rank-reducing measurement or a change to the learned interfaces.

Theorem 14 (Reference-moment factorization) *Fix a layer, its comparison maps $S_{Q,h}$, $S_{K,h}$, its stored content maps T_h, O_h with effective receiving maps $\widetilde{O}_h$ in one chart of Eq. (43), its source priors, its temperatures, and its residual coefficient r, and write $M_h = \widetilde{O}_h T_h$.*

1. *Every routing cost $D_{ij}^{(h)}$ and weight $\beta_{ij}^{(h)}$ depends on the states only through the reference moments (ξ_j, C_j), and $U_i^{-1}\mu_i^{\mathrm{att}} = r\xi_i + \sum_{j,h} \beta_{ij}^{(h)} M_h \xi_j$.*

2. *Under the passive site action of Eq. (4), the routing weights are unchanged, and the routed contributions $\mathcal{M}_{ij}^{(h)} = \beta_{ij}^{(h)} L_{ij}^{(h)}$ and messages transform as*

$$\mathcal{M}_{ij}^{(h)\prime} = g_i \mathcal{M}_{ij}^{(h)} g_j^{-1}, \qquad m_i' = g_i m_i. \tag{46}$$

3. *The operator is unchanged by the common reference change $(U_i, S_{Q,h}, S_{K,h}, T_h, O_h) \mapsto (U_i a, S_{Q,h}a, S_{K,h}a, T_h a, a^{-1}O_h)$ with $a \in \mathrm{GL}(K)$, under which the stored generator transforms as $OT \mapsto a^{-1}OTa$, by $(S_{Q,h}, S_{K,h}) \mapsto (AS_{Q,h}, AS_{K,h})$ with $A \in \mathrm{GL}(r_h)$, and by the headwise change $(T_h, O_h) \mapsto (B_h T_h, O_h B_h^{-1})$ with $B_h \in \mathrm{GL}(s_h)$, in each chart. The common reference change also leaves every operation in local coordinates unchanged.*

4. *For a closed walk $\gamma = (i_0, \dots, i_r = i_0)$, the routed loop product $\mathcal{H}_\gamma = \mathcal{M}_{i_0 i_1} \cdots \mathcal{M}_{i_{r-1} i_0}$ satisfies $\mathcal{H}_\gamma' = g_{i_0} \mathcal{H}_\gamma g_{i_0}^{-1}$, and its trace is a polynomial in the routing weights with loop-independent coefficients $\mathrm{Tr}(M_{h_1} \cdots M_{h_r})$. On the causal support $S_i = \{0, \dots, i\}$, every closed walk is a self-loop.*

Proof (1) The comparison laws of Eqs. (8) and (9) are built from $S_{Q,h}\xi_i$, $S_{Q,h}C_i S_{Q,h}^\top$, and the corresponding key moments, so the costs and the Gibbs weights of Eq. (12) depend on the states only through the reference moments. Applying U_i^{-1} to Eq. (16) gives the stated reference form. (2) Substituting the transformed variables into Eq. (3) gives $\xi_i' = \xi_i$ and $C_i' = C_i$, so by (1) the weights are unchanged. The content links obey Eq. (18), which leaves only the external factors g_i and g_j^{-1}, and the message transforms by g_i. (3) Under the common reference change, $\xi_i \mapsto a^{-1}\xi_i$ and $C_i \mapsto a^{-1}C_i a^{-\top}$, so every projected mean $S_{Q,h}\xi_i$, every projected covariance $S_{Q,h}C_i S_{Q,h}^\top$, and the corresponding key moments are unchanged. In each chart the effective receiving map transforms as $\widetilde{O} \mapsto a^{-1}\widetilde{O}$: trivially in the affine chart, as $(Ta)^{-1} + a^{-1}O = a^{-1}(T^{-1} + O)$ in the first-order chart, and as $e^{a^{-1}OTa}(Ta)^{-1} = a^{-1}e^{OT}T^{-1}$ in the exponential chart. Hence $M_h \mapsto a^{-1}M_h a$, and $U_i M_h \xi_j$ is unchanged. The local coordinates $\mu_i = U_i\xi_i$ are unchanged, so the operations applied in local coordinates are unaffected. Under A, both comparison laws are pushed forward by one invertible map, so Lemma 13 applied in H_h^{cmp} leaves every $D_{ij}^{(h)}$ unchanged. Under the headwise change, the stacked factors become (BT, OB^{-1}) with $B = \bigoplus_h B_h$, so OT is unchanged, $(BT)^{-1} + OB^{-1} = (T^{-1} + O)B^{-1}$, and $e^{OT}(BT)^{-1} = e^{OT}T^{-1}B^{-1}$; in each chart $\widetilde{O}_h \mapsto \widetilde{O}_h B_h^{-1}$, so every M_h is unchanged, and the messages depend on the content maps only through the M_h. (4) Intermediate factors $g_{i_k}^{-1}g_{i_k}$ cancel in the loop product, and cyclicity of the trace makes its trace invariant. Expanding $U_{i_0}^{-1}\mathcal{H}_\gamma U_{i_0} = M_{i_0 i_1} \cdots M_{i_{r-1} i_0}$ with $M_{ij} = \sum_h \beta_{ij}^{(h)} M_h$ gives the polynomial. Each factor $\mathcal{M}_{i_k i_{k+1}}$ requires $i_{k+1} \le i_k$ on

the causal support, so the indices are nonincreasing along the walk and a closed walk is constant. ∎

Tied comparison maps enter only through their row space. The cogredient redundancy of item (3) differs from the contragredient redundancy $(W_Q, W_K) \mapsto (W_Q A, W_K A^{-\top})$ of dot-product scores; the two share only $A \in \mathrm{O}(r_h)$, the residual freedom of whitening in Proposition 17. In particular, a common rescaling of $S_{Q,h}$ and $S_{K,h}$ does not change the routing temperature. Only H_h^{cmp} and H_h^{val} as abstract vector spaces are determined by the model; their bases are internal parameter redundancies. Gradient flow on a loss that depends on these maps only through the costs and messages therefore conserves $S_{Q,h}S_{Q,h}^\top + S_{K,h}S_{K,h}^\top$ and, in each chart, $T_h T_h^\top - O_h^\top O_h$ (Du et al., 2018; Kunin et al., 2021), whereas adaptive optimizers without weight decay can drift along these directions. The second law requires invariance of the entire loss under $\mathrm{GL}(s_h)$; the channel floor of the receiver of Section 7 reduces that invariance to the orthogonal group.

What the covariance does and does not constrain. Theorem 14 uses only the fact that routing and content depend on the states through the reference moments (ξ_i, C_i). Choosing $g_i = U_i^{-1}$ therefore removes the frames, and the operator becomes Eq. (59) with every $U_i = I$. Any operator with this property is covariant in the same sense, including dot-product attention on dressed means and any coordinate map applied as $U_i G(U_i^{-1}\cdot)$. In the terminology of the dressing-field method (Fournel et al., 2014), U_i is a local dressing field, and the local $\mathrm{GL}(K)$ covariance of the attention operator is artificial in the sense of François (2019, Sec. 2.2). It is the kind of covariance that, as Kretschmann objected in the case of general relativity, a formulation can acquire by introducing auxiliary fields (Norton, 1993). Equivalently, the learned frames fix a global trivialization of the token bundle, an $\{e\}$-structure in the sense of Weiler et al. (2021, Sec. 2.1.1), and the covariance places no constraint on S_Q, S_K, T, O, or a link generator. Gauge-equivariant convolution and attention on manifolds instead treat the frame as geometric data defined only up to the structure group, so their shared maps must be intertwiners (Cohen et al., 2019, Sec. 2.4; He et al., 2021, Secs. 4.3 and 4.4). For $G = \prod_b \mathrm{GL}(d_b)$ that requirement would confine each $\widetilde{O}_h T_h$ to $\mathrm{End}_G(\mathsf{V}) = \bigoplus_b \mathbb{R} I_{d_b}$, so every content link would reduce to $\sum_b c_b \Omega_{ij}^{(b)}$, and full-row-rank comparison maps would reduce routing to blockwise frame-only divergences. The low rank, block mixing, and composition defect of the links studied in Section 8 are available because the frames are learned.

The evaluated architecture applies its normalization, MLP, and decoder in local coordinates, and there the learned frames act beyond the attention operator.

Corollary 15 (Normal form of the evaluated predictor) *In reference coordinates the predictor of Section 2.4 reads*

$$\xi_i^{(0)} = U_i^{-1}\mu_{p,x_i}, \qquad C_i^{(0)} = U_i^{-1}\Sigma_{p,x_i}U_i^{-\top}, \qquad \xi_i^{(\ell+1)} = U_i^{-1}\Phi^{(\ell)}\big(U_i \xi_i^{\mathrm{att},(\ell)}\big), \tag{47}$$

with $\Phi^{(\ell)} = (\mathrm{id} + \mathrm{MLP}^{(\ell)}) \circ \mathrm{LN}$*, with the attention step frame-free given the routing weights, and with decoder logits* $W_{\mathrm{out}} U_i \xi_i^{(L)}$ *plus the class biases. Under covariance passthrough* $C_i^{(\ell)} = C_i^{(0)}$ *at every depth; under moment matching* $C_i^{(\ell+1)}$ *is the reference covariance of Eq.* (69)*, a frame-free function of the routing weights and the reference moments of layer* ℓ*.*

Proof Apply U_i^{-1} to Eqs. (22) and (25), and use item (1) of Theorem 14 for the attention step. The frames are depth independent, so passthrough keeps $C_i^{(0)}$, and Theorem 20 gives the moment-matched covariance in reference coordinates. ■

The frames therefore act through four channels: a multiplicative token-and-position embedding; a routing covariance, which under passthrough is static per token type and position because covariance passes through unchanged and the frames are fixed through the stack, and which under moment matching starts from that value and then depends on the context; a token- and position-specific conjugation of the shared normalization and MLP; and a token- and position-specific decoder $W_{\text{out}}U_i$. Figure 2 depicts these roles. Covariant alternatives to the last two channels exist. Conjugating the coordinate block by the frame, $\mu_i \mapsto U_i G(U_i^{-1}\mu_i)$, or scaling each frame block by a function of the invariant $\mu_b^\top \Sigma_{bb}^{-1}\mu_b$, gives a block that is covariant under the passive site action. With such a block and a decoder acting on ξ_i, the whole predictor depends on the frames only through $\xi_i^{(0)}$ and C_i.

5 Gaussian-KL Routing and Its Transformer Reductions

The correspondence with standard attention starts from the learned comparison and content maps, rather than replacing them by the flat identification Ω_{ij}. We establish an exact dual-coordinate form of the routing cost (Theorem 16), its reduction under stated covariance and bias conditions, and an exact identity for the resulting value/output aggregation. These are conditional equalities of attention operators; Table 2 summarizes them by status.

5.1 The full Gaussian score and its query–key cross term

Fix a layer, a head, and its nonempty allowed source sets; suppress the layer and head indices. Write $r = r_h$ and define the projected moments

$$a_i = S_Q\xi_i, \quad b_j = S_K\xi_j, \quad V_{Q,i} = S_Q C_i S_Q^\top, \quad V_{K,j} = S_K C_j S_K^\top. \tag{48}$$

The state covariances are positive definite and both comparison maps have full row rank, so $V_{Q,i}, V_{K,j} \succ 0$. The Gaussian KL formula gives

$$\begin{aligned} D_{ij} = \frac{1}{2}\big[\,&\operatorname{Tr}(P_j V_{Q,i}) + a_i^\top P_j a_i - 2a_i^\top P_j b_j + b_j^\top P_j b_j - r \\ &+ \log\det V_{K,j} - \log\det V_{Q,i}\big], \qquad P_j = V_{K,j}^{-1}. \end{aligned} \tag{49}$$

Its mean compatibility has the dot-product factorization $a_i^\top(P_j b_j)$; the KL contains the negative of this term. In ambient local coordinates it is $\mu_i^\top B_{ij}\mu_j$ with the comparison tensor of Eq. (90). Thus independent learned S_Q, S_K already supply distinct query and key observables. Neither is required to be a gauge-group element. The other terms retain covariance information: when P_j varies with the source, $a_i^\top P_j a_i$ and $\operatorname{Tr}(P_j V_{Q,i})$ are generally pair dependent and cannot be discarded as row constants.

All of these terms are organized by one exact identity.

Theorem 16 (Dual-coordinate form of Gaussian-KL routing) *Write the query law in expectation coordinates and the key law in natural coordinates,*

$$\eta_{Q,i} = \big(a_i, V_{Q,i} + a_i a_i^\top\big), \qquad \theta_{K,j} = \big(P_j b_j, -\tfrac{1}{2}P_j\big), \qquad \langle \eta, \theta\rangle = \eta_1^\top\theta_1 + \operatorname{Tr}(\eta_2\theta_2), \tag{50}$$

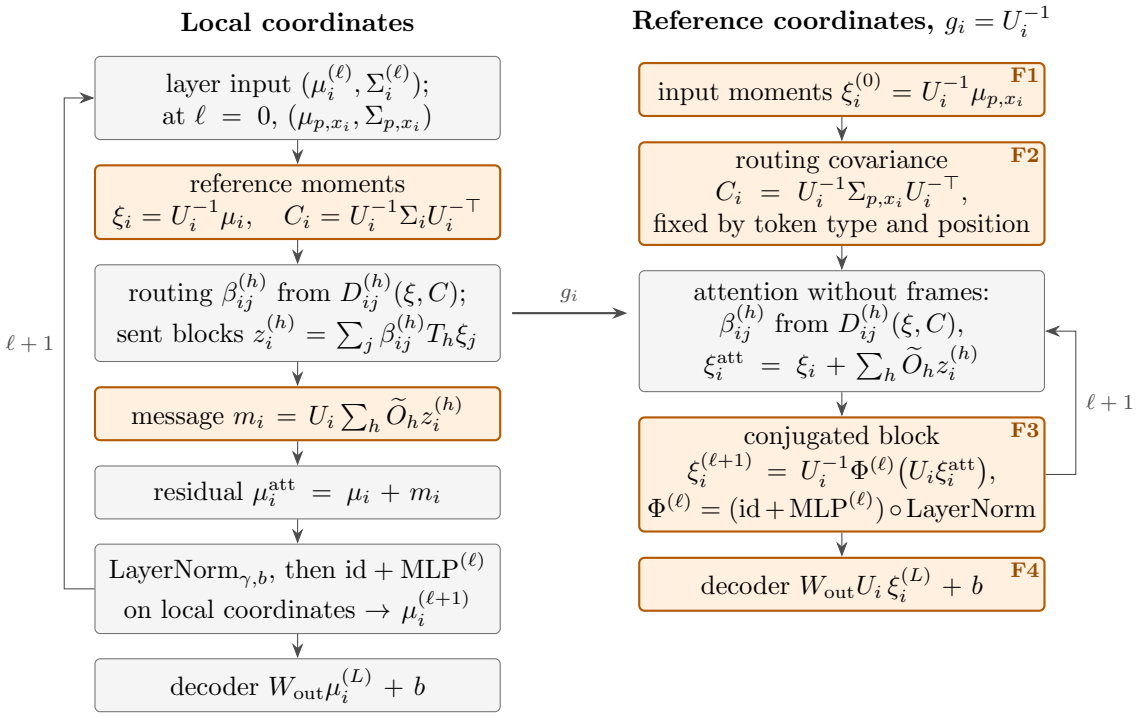


Figure 2: Reference-coordinate form of the evaluated architecture. Shaded boxes are the places where the frames U_i enter; $z_i^{(h)} = \sum_j \beta_{ij}^{(h)} T_h \xi_j$ is the routed sent block of head h in reference coordinates and $\widetilde{O}_h$ the effective receiving map of the chart. Left: one evaluated block in local coordinates, where the frame enters at the reference moments and at the message m_i. Right: the same computation after the site transformation $g_i = U_i^{-1}$. Every routing cost and message factors through (ξ_i, C_i), so attention acts without frames, as Eq. (46) implies, and the frames remain only in the input moments (F1), the routing covariance (F2), the conjugated normalization and MLP (F3), and the decoder (F4). Each tag names the row of Table 5 that removes that role. Under moment matching, C_i is fixed only at the input and is then updated frame-free from the routing weights.

and let $A(\theta_{K,j}) = \frac{1}{2} b_j^\top P_j b_j + \frac{1}{2} \log\det V_{K,j} + \frac{r}{2} \log 2\pi$ *be the Gaussian log-partition, with conjugate* $A^*(\eta_{Q,i}) = -H(q_{Q,i})$*. Then*

$$D_{ij} = A(\theta_{K,j}) + A^*(\eta_{Q,i}) - \langle \eta_{Q,i}, \theta_{K,j} \rangle, \tag{51}$$

and the categorical optimizer is exactly $\beta_{ij} = \mathrm{softmax}_j\big(\langle \eta_{Q,i}, \theta_{K,j} \rangle/\tau - A(\theta_{K,j})/\tau + \log \pi_{ij}\big)$*: a dot product in* $\mathbb{R}^{r+r(r+1)/2}$ *between query expectation parameters and key natural parameters, with the key log-partition as a source bias. No covariance hypothesis is needed.*

Proof With $-H(q_{Q,i}) = -\frac{1}{2} \log\det(2\pi e V_{Q,i})$, expanding the right side of Eq. (51) reproduces Eq. (49) term by term. This is the mixed form of the divergence between members of an exponential family (Wainwright and Jordan, 2008, Sec. 5.2, Eq. (5.11)). The term $A^*(\eta_{Q,i})$ is constant on row i and cancels in the softmax. ■

Proposition 17 is the case in which the second-order key coordinate $-\frac{1}{2} P_j$ is shared, so that only the mean block $a_i^\top V_*^{-1} b_j$ of the pairing varies with the source.

Proposition 17 (Projected Gaussian reduction to biased dot-product routing) *Suppose all keys used by a head have the same projected covariance* $V_{K,j} = V_* \succ 0$*. Let* $\tau > 0$*, let each prior row be a probability distribution with* $\pi_{ij} > 0$ *on its allowed support, and let costs be the uncapped Gaussian divergences. Define* $\mathsf{q}_i = V_*^{-1/2} a_i$ *and* $\mathsf{k}_j = V_*^{-1/2} b_j$*, using the symmetric positive-definite inverse square root. Then the exact categorical optimizer is*

$$\beta_{ij} = \mathrm{softmax}_j \left(\frac{\mathsf{q}_i^\top \mathsf{k}_j}{\tau} - \frac{\|\mathsf{k}_j\|^2}{2\tau} + \log \pi_{ij} \right). \tag{52}$$

The query covariances may differ across receivers. If, more strongly, $V_{Q,i} = V_{K,j} = \sigma_h^2 I_r$*, then* $D_{ij} = \|a_i - b_j\|^2/(2\sigma_h^2)$*.*

Proof Set $P_j = V_*^{-1}$ in Eq. (49). The trace, query quadratic, dimension, and determinant terms are independent of j, except for $b_j^\top V_*^{-1} b_j$ and the cross term. Thus

$$D_{ij} = c_i + \tfrac{1}{2} \|\mathsf{k}_j\|^2 - \mathsf{q}_i^\top \mathsf{k}_j, \tag{53}$$

for a row constant c_i. Substitution into the categorical solution $\mathrm{softmax}_j(-D_{ij}/\tau + \log \pi_{ij})$ removes c_i. Under the stronger isotropic assumption the trace and log-determinant terms cancel and the remaining quadratic is the stated squared distance. ■

Since $D_{ij} = \frac{1}{2} \|\mathsf{q}_i - \mathsf{k}_j\|^2$ up to a row constant, routing with shared projected key covariance is L2 attention in whitened coordinates; under the stronger isotropic hypothesis it is the L2 self-attention of Kim et al. (2021). Dot-product attention is recovered only when projected key norms are constant or compensated (Section 5.2).

Remark 18 (Relation to Gaussian-mixture attention) *Because the entropy* $H(q_{Q,i})$ *is constant on row* i *and* $D_{\mathrm{KL}}(q\|p) = -\mathbb{E}_q \log p - H(q)$*,*

$$\beta_{ij} = \mathrm{softmax}_j\big(\log \pi_{ij} + \mathbb{E}_{q_{Q,i}}[\log p_{K,j}]/\tau\big), \tag{54}$$

the mean-field responsibility of a Gaussian mixture under an uncertain query. For a point query at a_i *it is the mixture responsibility of Movellan and Gabbur (2020) and Nguyen et al.*

(2022), and Eq. (55) *below is the prior with which Movellan and Gabbur (2020, Eqs. 16 to 21) recover dot-product attention. The query covariance adds* $-\operatorname{Tr}(P_j V_{Q,i})/(2\tau)$*, which depends on the pair whenever* P_j *varies with the source.*

The hypotheses of Proposition 17 concern the projected distributions, not merely isotropic ambient states. Even if $C_i = \sigma^2 I_K$, a learned map has covariance $\sigma^2 S_K S_K^\top$, which need not be isotropic. Shared $C_i = C$ across tokens does give a shared key covariance for fixed S_K and therefore satisfies the weaker proposition. It places no equality requirement on S_Q and S_K. Whitening then defines the effective learned maps $\widetilde{S}_Q = V_*^{-1/2} S_Q$ and $\widetilde{S}_K = V_*^{-1/2} S_K$. The simple choice $K = 2r$, $S_Q = [I_r\ 0]$, $S_K = [0\ I_r]$, and $C_i = \sigma^2 I_K$ demonstrates that distinct query and key observables can satisfy even the stronger isotropic assumptions. A trained state with source-dependent projected covariance need not satisfy either specialization.

5.2 Exact bias cancellation, approximation, and temperature

Equation (52) becomes ordinary dot-product attention with the same source prior exactly when the squared key norms are constant on the allowed support. More generally, a desired base prior π^0_{ij} can be reproduced algebraically by choosing

$$\pi_{ij} = \frac{\pi^0_{ij} \exp(\|\mathsf{k}_j\|^2/(2\tau))}{\sum_{k\in S_i} \pi^0_{ik} \exp(\|\mathsf{k}_k\|^2/(2\tau))}. \tag{55}$$

This is an explicit state-dependent compensation. It is not a positional prior fixed independently of the current key states, and it is not assumed to be implemented. A minimal counterexample shows why the condition matters: for $r = 1$, $\tau = 1$, $a_i = 0$, keys $b_1 = 0, b_2 = 1$, shared unit variance, and uniform prior, ordinary dot-product routing gives $(1/2, 1/2)$ while Gaussian routing gives $(1, e^{-1/2})/(1 + e^{-1/2})$.

An approximation can be stated by bounding the remaining bias.

Proposition 19 (Sharp routing perturbation bound) *For any row constant* c_i*, under the hypotheses of Proposition 17, define*

$$\varepsilon_i = \max_{j\in S_i} \frac{|\|\mathsf{k}_j\|^2 - c_i|}{2\tau}, \qquad p_{ij} = \operatorname{softmax}_j\left(\mathsf{q}_i^\top \mathsf{k}_j/\tau + \log \pi_{ij}\right). \tag{56}$$

Then

$$\sum_j |\beta_{ij} - p_{ij}| \le 2\tanh(\varepsilon_i/2) \le \min(\varepsilon_i, 2), \tag{57}$$

with equality for two sources with $p_{i1} = 1/(1 + e^{\varepsilon_i})$ *and key-norm offsets of* $\pm\varepsilon_i$*. The best row constant is the midrange, which gives* $\varepsilon_i = (\max_j \|\mathsf{k}_j\|^2 - \min_j \|\mathsf{k}_j\|^2)/(4\tau)$*.*

Proof By Eq. (52), $\beta_{ij} = p_{ij} e^{\delta_{ij}} / \sum_k p_{ik} e^{\delta_{ik}}$ with $\delta_{ij} = -(\|\mathsf{k}_j\|^2 - c_i)/(2\tau)$, so $|\delta_{ij}| \le \varepsilon_i$. For $A \subseteq S_i$, the derivative of $\beta_i(A)$ in δ_{ik} is positive for $k \in A$ and negative otherwise, so $\beta_i(A) - p_i(A)$ is largest at $\delta_{ij} = \varepsilon_i$ on A and $\delta_{ij} = -\varepsilon_i$ off A, where it equals $x(1-x)(e^{2\varepsilon_i} - 1)/[1 + x(e^{2\varepsilon_i} - 1)]$ with $x = p_i(A)$. The maximum over x is attained at $x = 1/(1 + e^{\varepsilon_i})$ and equals $\tanh(\varepsilon_i/2)$. Since $\sum_j |\beta_{ij} - p_{ij}| = 2\max_A[\beta_i(A) - p_i(A)]$, the bound follows,

and $2\tanh(\varepsilon_i/2) \le \min(\varepsilon_i, 2)$ because tanh is bounded by one and by its argument. The midrange minimizes ε_i over c_i. ∎

The ratio bound $e^{-2\varepsilon_i} \le \beta_{ij}/p_{ij} \le e^{2\varepsilon_i}$ remains available as a multiplicative statement. Therefore uniform control $\varepsilon_i \to 0$ suffices for rowwise convergence, including when the number of keys varies. For the conventional scale $\tau = \sqrt{r}$, this requires maximum squared-norm dispersion $o(\sqrt{r})$ on the row support. High-dimensional relative concentration alone is insufficient: for independent standard Gaussian vectors, the squared norm has variance $2r$ and the dot product of independent vectors has variance r. Both residuals retain order-one fluctuations after division by $\sqrt{r}$. Coordinate LayerNorm followed by an arbitrary learned key map does not impose the required projected-key norm constancy.

Standard attention uses the comparison dimension r_h in its scale $\sqrt{r_h}$ (Vaswani et al., 2017, Sec. 3.2.1). After whitening, the cross term has this conventional scale when $\tau_h = \sqrt{r_h}$. For arbitrary positive τ_h, rescaling both effective query and key by $(\sqrt{r_h}/\tau_h)^{1/2}$ writes the same cross term with denominator $\sqrt{r_h}$. Covariance scale has already entered through whitening and must not be counted a second time as temperature. Numerical modification of the KL cost lies outside the uncapped score identity unless it leaves every relevant cost unchanged.

Geometric dilation must also be separated from statistical sharpness. A passive rescaling of one frame block (Appendix B) transforms its mean, covariance, and frame together as $(\mu, \Sigma, U) \mapsto (c\mu, c^2\Sigma, cU)$, $c \neq 0$. The reference moments and all routing costs are unchanged. At fixed projected means, changing a shared isotropic comparison covariance $\sigma_h^2 I$ instead changes the statistical model and gives $D_{ij}^{(h)} = \|S_{Q,h}\xi_i - S_{K,h}\xi_j\|^2/(2\sigma_h^2)$. The logits then depend on σ_h^2 and τ_h only through $\sigma_h^2\tau_h$, so the two are not separately identifiable from routing. For general projected covariances, trace and determinant terms prevent reducing every covariance change to one scalar temperature.

5.3 Value, output, and residual links

For any routing weights, independently of the dot-product specialization, predictive content has the exact reference-coordinate form

$$v_j^{(h)} = T_h\xi_j, \qquad z_i^{(h)} = \sum_j \beta_{ij}^{(h)} v_j^{(h)}, \tag{58}$$

$$U_i^{-1}\mu_i^{\mathrm{att}} = r\xi_i + \sum_h \widetilde{O}_h z_i^{(h)} = r\xi_i + \sum_{j,h} \beta_{ij}^{(h)} \widetilde{O}_h T_h \xi_j. \tag{59}$$

Concatenate the $z_i^{(h)}$. Then $\sum_h \widetilde{O}_h z_i^{(h)} = \widetilde{O}\,\mathrm{concat}_h(z_i^{(h)})$, the usual multi-head output projection (Vaswani et al., 2017, Sec. 3.2.2). With column-vector conventions, the dictionary is $W_V^{(h)} = T_h$ and output blocks $W_O^{(h)} = \widetilde{O}_h$; the matrices in the original row-vector transformer convention are their transposes. In the affine chart $W_O = O$ is a free parameter. In the exponential chart with tiled heads, $W_O = e^{OT}W_V^{-1}$ is the same output projection reparametrized through a generator and restricted to the matrices for which $W_O W_V$ has a real logarithm (Section 3.6). In local coordinates the same operation is $\sum_{j,h} \beta_{ij}^{(h)} L_{ij}^{(h)} \mu_j$. Thus the content link is the framed value/output composition, not the query–key compatibility tensor.

Each head can read all K coordinates and write into all K output coordinates. The latent channel dimension limits the rank of an individual contribution, not its input or output support. Communication heads need not coincide with a partition into gauge-frame blocks; full-width maps can mix those blocks even when every U_i is block diagonal. Equation (59), together with Proposition 17 and its bias conditions, supplies the conditional attention-layer correspondence. The explicit residual $r = 1$ is part of the architecture of Section 9.2; Theorem 20 reads it as the persistence term of the predictive kernel of Eq. (68), Section 9.2 reports that it is numerically inert in that trained model, and the chart comparison of Section 9.3 uses $r = 0$.

5.4 Restricted rotary correspondence and the remaining stack

Rotary position embedding (RoPE) applies position-dependent orthogonal rotations after query and key projection (Su et al., 2024). For an even comparison dimension, let R_i be a block-diagonal matrix of planar rotations with angles proportional to position, so $R_i^\top R_j = R_{j-i}$. A declared comparison-space construction $a_i \mapsto R_i a_i$, $b_j \mapsto R_j b_j$ has cross term

$$(R_i a_i)^\top (R_j b_j) = a_i^\top R_{j-i} b_j = \xi_i^\top S_Q^\top R_{j-i} S_K \xi_j. \tag{60}$$

Rotating both projected laws gives, by Eq. (45), $D_{\mathrm{KL}}\big((R_i)_* q_{Q,i} \| (R_j)_* p_{K,j}\big) = D_{\mathrm{KL}}\big(q_{Q,i} \| (R_{j-i})_* p_{K,j}\big)$, a relative-position score for arbitrary projected covariances. The dot-product form with an unchanged key-norm bias requires only that the shared key covariance commute with every R_δ; with distinct rotation frequencies this means $V_* = \bigoplus_k \lambda_k I_2$ over the rotation planes. Query covariance is unrestricted. If the rotations act on anisotropic projected laws rather than on means alone, the trace and query-quadratic terms of Eq. (49) also become functions of $i - j$ and contribute relative-position biases.

This identity does not automatically identify position rotations inside an ambient frame with ordinary post-projection RoPE. Moving an ambient positional map through S_Q or S_K requires the corresponding intertwining identities, such as $S_Q e^{-Y_i} = R_i S_Q$ and $S_K e^{-Y_i} = R_i S_K$ for the positional factor of $U_i = e^{\phi_{x_i}} e^{Y_i}$; imposing the identity on the complete U_i^{-1} would require every token factor to act trivially on the comparison row spaces. Arbitrary learned maps need not satisfy them. In addition, ordinary RoPE leaves value aggregation unrotated. The separated comparison and content maps allow this design to be specified, but one shared ambient positional frame acting through every U_i^{-1} does not establish it. The relative rotation is a flat group-valued factor; the full projected compatibility remains a generally low-rank bilinear tensor, and the content link remains a separate operator.

Covariance may be retained as a routing state in a predictive architecture. Passing it unchanged through the mean residual does not compute the covariance of a Gaussian mixture or propagate uncertainty through normalization and the MLP. For example, a declared mixture of content laws with means v_j and covariances W_j would have covariance $\sum_j \beta_j [W_j + (v_j - \bar{v})(v_j - \bar{v})^\top]$, where $\bar{v} = \sum_j \beta_j v_j$. The predictive mean identity does not require this additional propagation rule; Eq. (69) gives its multi-head form for the predictive kernel.

Correspondence	Exact scope or remaining condition
Derived under stated conditions	
Gaussian categorical routing	Exact optimizer for fixed costs, allowed support, and positive temperature; exact label posterior under a binary compatibility record
Query/key dot product	Exact dot product in dual coordinates, Eq. (51); reduces to the r_h-dimensional mean block under shared key covariance
Ordinary attention weights	Shared key covariance, matched scale, and constant or compensated key-norm bias; otherwise L2 attention in whitened coordinates
RoPE score	Relative dependence exact for arbitrary projected covariances when both laws rotate; dot-product form requires a shared key covariance that commutes with the rotations
Identical by construction	
Value/output aggregation	T_h and $\widetilde{O}_h$ are the value and output maps; framed content-link identity at any fixed routing
Exponential content link	Output projection $e^{OT}T^{-1}$ of tiled heads: a reparametrization of W_O on the real-logarithm domain, with orders separating flat transport, value path, and cross-head mixing
Multi-head organization	Full-width maps and output concatenation identity; no frame-block partition is required
Not derived (architectural choices)	
Predictive residual	Coefficient $r \in \{0, 1\}$; $r = 1$ is read as the persistence term of a predictive kernel and was numerically inert after training in the $K = 256$ run; tiled exponential aggregation has no residual
Covariance, normalization, MLP, decoder	Separate processing choices; the linear decoder is a Gibbs readout of shared-covariance templates

Table 2: Transformer correspondences for the projection-based architecture, grouped by status. The equalities concern the specified operators, not an identity between unrestricted model families.

6 Probabilistic Status of Routing and Content

The Gibbs row of Eq. (12) and the content update of Eq. (16) were stated as operations. This section derives the row from a declared local source model, shows that it is an exact posterior at every temperature, and shows that the content update with its residual is an exact predictive mean.

6.1 Variational source selection

Fix a layer, head, and receiver, suppressing those indices. Let $q_{Q,i}(y)$ be the projected query law and $p_{K,j}(y)$ the candidate key laws. On a nonempty allowed support S_i, take positive probabilities π_{ij} summing to one. The normalized local source model and factorized

recognition law are

$$\mathbb{P}_i(y, J = j) = \pi_{ij} p_{K,j}(y), \tag{61}$$

$$\mathbb{Q}_{i,\beta}(y, J = j) = \beta_{ij} q_{Q,i}(y). \tag{62}$$

Here the Gaussian states are fixed values at the given context: they are computed from the prefix before routing and enter the local joint as data, so $p_{K,j}$ reads a state value rather than a recognition law being optimized, and β_i is the only recognition variable. The receiver enters only through the query law, clamped as the recognition marginal of y. Hold the query law fixed during categorical minimization. Expanding the density ratio gives

$$D_{\mathrm{KL}}(\mathbb{Q}_{i,\beta} \| \mathbb{P}_i) = \sum_{j \in S_i} \beta_{ij} D_{ij} + \sum_{j \in S_i} \beta_{ij} \log \frac{\beta_{ij}}{\pi_{ij}}, \tag{63}$$

$$D_{ij} = D_{\mathrm{KL}}(q_{Q,i} \| p_{K,j}). \tag{64}$$

The categorical entropy belongs to this variational objective. The direction of D_{ij} is fixed by the declared joint and the exclusive divergence: the key laws are the generative mixture components and the query law is the recognition factor. The inclusive divergence gives $D_{\mathrm{KL}}(\mathbb{P}_i \| \mathbb{Q}_{i,\beta}) = D_{\mathrm{KL}}(\pi_i \| \beta_i) + \sum_j \pi_{ij} D_{\mathrm{KL}}(p_{K,j} \| q_{Q,i})$, whose minimizer is $\beta_i = \pi_i$ for all costs. The reverse cost $D_{\mathrm{KL}}(p_{K,j} \| q_{Q,i})$ follows from the joint $\pi_{ij} q_{Q,i}(y)$ with recognition $\beta_{ij} p_{K,j}(y)$; its reduction to a dot product requires a shared query covariance and retains a source-dependent bias. Symmetric divergences require both projected covariances to be shared.

At a declared temperature $\tau > 0$, use

$$\mathcal{A}_i(\beta) = \sum_{j \in S_i} \beta_{ij} D_{ij} + \tau D_{\mathrm{KL}}(\beta_i \| \pi_i). \tag{65}$$

The Hessian $\tau \operatorname{diag}(1/\beta_{ij})$ is positive definite on nonzero simplex tangent directions. The Lagrange condition is $D_{ij} + \tau(\log(\beta_{ij}/\pi_{ij}) + 1) + \lambda_i = 0$. For finite costs the unique minimizer is interior on S_i:

$$\beta^*_{ij} = \frac{\pi_{ij} \exp(-D_{ij}/\tau)}{\sum_{k \in S_i} \pi_{ik} \exp(-D_{ik}/\tau)}, \tag{66}$$

$$\beta^*_{ij} = \operatorname{softmax}_{j \in S_i}(-D_{ij}/\tau) \quad \text{for uniform } \pi_i. \tag{67}$$

Substitution gives $-\tau \log Z_i$, with $Z_i = \sum_{j \in S_i} \pi_{ij} \exp(-D_{ij}/\tau)$, by the usual entropy-regularized variational calculation (Boyd and Vandenberghe, 2004; Wainwright and Jordan, 2008; Niculae and Blondel, 2017). At $\tau = 1$ this is the KL of the declared local joint; general temperature changes that objective. The categorical minimizer needs no gradient-based update of q_i. Local mixture normalization does not supply a population model or a language-model observation likelihood.

Complete-state alignment is a special comparison: for $S_Q = S_K = I_K$, common push-forward by U_i gives $D_{ij} = D_{\mathrm{KL}}(q_i \| (\Omega_{ij})_* q_j)$. Independent lower-dimensional role measurements therefore extend the comparison beyond this frame-only case. For tied maps, $D_{ij} \le D_{\mathrm{KL}}(q_i \| (\Omega_{ij})_* q_j)$ by the data-processing inequality (van Erven and Harremøes, 2014, Thm. 9), so a shared comparison coarse-grains complete-state alignment; for $S_Q \neq S_K$ no such bound holds, and identical states can have $D_{ij} > 0$.

6.2 Exact posterior and predictive mean

Theorem 20 (Exact probabilistic status of routing and content) *Fix a layer and its input states, and let $D_{ij}^{(h)} \geq 0$ be the routing costs.*

1. *For every $\tau_h > 0$, give each receiver and head a source label $J_i^{(h)}$ with prior $\pi_i^{(h)}$ and a binary compatibility record $o_i^{(h)}$ with $P(o_i^{(h)} = 1 \mid J_i^{(h)} = j) = \exp(-D_{ij}^{(h)}/\tau_h)$. The joint is normalized on any source graph, and $P(J_i^{(h)} = j \mid o_i^{(h)} = 1) = \beta_{ij}^{(h)}$ of Eq. (12). The objective of Eq. (11) is τ_h times the label free energy $D_{\mathrm{KL}}(\beta \| \pi_i^{(h)}) - \sum_j \beta_j \log P(o_i^{(h)} = 1 \mid J_i^{(h)} = j)$, whose minimum is the surprisal $-\log P(o_i^{(h)} = 1)$.*

2. *Let the head labels be independent with laws $\beta_i^{(h)}$, and take the directed kernel*

$$k_i' \mid k, J_i \sim \mathcal{N}\Big(rk_i + \sum_h L_{iJ_i^{(h)}}^{(h)} k_{J_i^{(h)}}, Q_i\Big), \qquad r \in \{0,1\}, \quad Q_i \succeq 0, \tag{68}$$

which is normalized for any link rank, as a degenerate Gaussian when Q_i is singular. At state values $k_j = \mu_j$ its predictive mean is μ_i^{att} of Eq. (16). With independent source states $k_j \sim \mathcal{N}(\mu_j, \Sigma_j)$, the self edge included, its moment-matched covariance in reference coordinates is

$$\begin{aligned} C_i^{\mathrm{att}} = \widetilde{Q}_i + r\Big[C_i + \sum_h \beta_{ii}^{(h)}\big(M_h C_i + C_i M_h^\top\big)\Big] + \sum_j \sum_{h,h'} c_{ij}^{hh'} M_h C_j M_{h'}^\top \\ + \sum_h \Big[\sum_j \beta_{ij}^{(h)} n_{ij}^{(h)} n_{ij}^{(h)\top} - \bar{n}_i^{(h)} \bar{n}_i^{(h)\top}\Big], \end{aligned} \tag{69}$$

where $\widetilde{Q}_i = U_i^{-1} Q_i U_i^{-\top}$, $M_h = \widetilde{O}_h T_h$, $c_{ij}^{hh} = \beta_{ij}^{(h)}$, $c_{ij}^{hh'} = \beta_{ij}^{(h)} \beta_{ij}^{(h')}$ for $h \neq h'$, $n_{ij}^{(h)} = M_h \xi_j$, and $\bar{n}_i^{(h)} = \sum_j \beta_{ij}^{(h)} n_{ij}^{(h)}$. The local covariance $U_i C_i^{\mathrm{att}} U_i^\top$ obeys the passive law.

Proof (1) Because $D_{ij}^{(h)} \geq 0$, each record kernel takes values in $(0, 1]$ and is normalized. Records attached to different receivers and heads are conditionally independent children of the states and labels, so the joint is normalized. Bayes' rule gives $P(J = j \mid o = 1) = \pi_{ij} e^{-D_{ij}/\tau} / \sum_k \pi_{ik} e^{-D_{ik}/\tau}$, which is Eq. (12). Since $-\log P(o = 1 \mid J = j) = D_{ij}/\tau$, the label free energy equals τ^{-1} times the objective, and its minimum is $-\log \sum_j \pi_{ij} e^{-D_{ij}/\tau} = -\log P(o = 1)$. (2) Write $X = rk_i + \sum_h L_{iJ^{(h)}}^{(h)} k_{J^{(h)}}$. Conditioning on the labels gives $\mathbb{E}[X \mid J] = r\mu_i + \sum_h L_{iJ^{(h)}}^{(h)} \mu_{J^{(h)}}$, whose average over the independent labels is μ_i^{att}. By the law of total covariance, $\operatorname{Cov} X = \mathbb{E} \operatorname{Cov}(X \mid J) + \operatorname{Cov} \mathbb{E}(X \mid J)$. Given the labels, each distinct source s contributes $A_s \Sigma_s A_s^\top$, with $A_i = rI + \sum_h \mathbf{1}[J^{(h)} = i] L_{ii}^{(h)}$ and $A_s = \sum_h \mathbf{1}[J^{(h)} = s] L_{is}^{(h)}$ for $s \neq i$. Averaging the indicator products with $\mathbb{E}\big(\mathbf{1}[J^{(h)} = s]\mathbf{1}[J^{(h')} = s]\big) = c_{is}^{hh'}$ gives the terms in C_i, $\beta_{ii}^{(h)}$, and $c_{ij}^{hh'}$, and independence of the labels across heads gives the spread term. Conjugating by U_i^{-1} removes every frame, since $L_{ij}^{(h)} \Sigma_j L_{ij}^{(h')\top} = U_i M_h C_j M_{h'}^\top U_i^\top$, and adding Q_i completes the kernel covariance. ∎

The record is a declared modeling event of the kind used for optimality variables in control as inference (Toussaint and Storkey, 2006; Kappen et al., 2012; Levine, 2018); it is not an observed datum. Exactness of this kind is available for any nonnegative cost, so item (1) shows that Gibbs routing is consistent with exact inference, not that attention must be inference. Covariance passthrough with $r = 1$ retains only the term C_i of Eq. (69). The derivation of item (2) leaves T_h, $\widetilde{O}_h$, and Q_i free; it holds in every chart of Eq. (43); it declares the persistence coefficient, additivity across heads, and one label for both the compatibility record and the content channel; and it does not cover the normalization and MLP maps or the collapse of the predictive mixture to a point state between layers.

The implemented moment-matched contract evaluates Eq. (69) with the reference floor $\widetilde{Q}_i = \epsilon_Q \operatorname{Tr}(C_i) I/K$, so that $Q_i = \epsilon_Q \operatorname{Tr}(C_i) U_i U_i^\top/K$ transforms as a covariance because $\operatorname{Tr} C_i$ is invariant under the passive action. The output then satisfies $\lambda_{\min}(C_i^{\mathrm{att}}) \geq \epsilon_Q \operatorname{Tr}(C_i)/K$ whatever the routing and the rank of M_h. With $r = 0$ the covariance is that of the message alone plus the floor; with $r = 1$ the self-edge cross terms $\beta_{ii}^{(h)}(M_h C_i + C_i M_h^\top)$ can be indefinite, so no monotonicity in depth holds in general. When only the local diagonal of the covariance is stored, matching marginal variances is the projection that minimizes $D_{\mathrm{KL}}(p\|q)$ over diagonal Gaussian q, and it does not commute with arbitrary $\mathrm{GL}(K)$ congruences, so the passive law applies before that projection.

For fixed $q_{Q,i}$ the optimizer of Eq. (66) minimizes $\mathbb{E}_{y\sim q_{Q,i}} D_{\mathrm{KL}}\big(\beta_i \| p_i^{(\tau)}(\cdot \mid y)\big)$ with $p_i^{(\tau)}(j \mid y) \propto \pi_{ij} p_{K,j}(y)^{1/\tau}$: it is the normalized geometric mean of the pointwise source posteriors under the query law. The posterior-predictive source probability $\mathbb{E}_{y\sim q_{Q,i}} \mathbb{P}_i(J = j \mid y)$ minimizes the reverse expected divergence. It is also the unrestricted minimizer of the KL to $\mathbb{P}_i$ over joints whose y-marginal is clamped to $q_{Q,i}$, and it generally differs from the Gibbs row, as for the readout of Appendix D. Apart from a row constant, the query covariance enters D_{ij} only through the trace term of Eq. (49). That term penalizes sources whose key laws are precise along the uncertain directions of the query, and it drops out when key covariances are shared. A broad query law therefore does not flatten Gibbs routing.

Remark 21 (Noisy-query routing) *A rule under which query uncertainty does flatten routing requires a different local model, in which the projected query mean is data. Under source j, let $y \sim p_{K,j}$ and let the observed query mean be $a_i = y + \varepsilon$ with independent noise $\varepsilon \sim \mathcal{N}(0, V_{Q,i})$. Then $p(a_i \mid J = j) = \mathbb{E}_{y\sim q_{Q,i}}[p_{K,j}(y)] = \mathcal{N}(a_i; b_j, V_{Q,i} + V_{K,j})$, the expected likelihood kernel of the two laws (Jebara et al., 2004, Secs. 2.1 and 3.1), and Bayes' rule gives the exact posterior*

$$P(J = j \mid a_i) = \frac{\pi_{ij} \mathcal{N}(a_i; b_j, V_{Q,i} + V_{K,j})}{\sum_{k\in S_i} \pi_{ik} \mathcal{N}(a_i; b_k, V_{Q,i} + V_{K,k})}. \tag{70}$$

At unit temperature, Gibbs routing normalizes $\pi_{ij} \exp \mathbb{E}_{q_{Q,i}}[\log p_{K,j}]$, whereas Eq. (70) normalizes $\pi_{ij} \mathbb{E}_{q_{Q,i}}[p_{K,j}]$; in the point-query limit both give the mixture responsibility of Remark 18. With shared key covariance V_*, Eq. (70) is Eq. (52) at $\tau = 1$ with V_* replaced by $V_{Q,i} + V_*$, so the query covariance enters as a receiver-dependent whitening that tempers the row. For $r = 1$, keys $b_1 = 0$ and $b_2 = 1$ with unit variance, $a_i = 0$, and a uniform prior, the unit-temperature Gibbs weight on the first source is $(1 + e^{-1/2})^{-1} \approx 0.622$ for every query variance v, whereas Eq. (70) gives $(1 + e^{-1/(2+2v)})^{-1}$, which is 0.562 at $v = 1$ and 0.511 at $v = 10$. The weights depend on the states only through projected moments, and a common

invertible map of the comparison space multiplies every term by the same Jacobian factor, so they share the invariances of Theorem 14. A temperature can be introduced through the binary record of Theorem 20 after a row shift that makes the cost $-\log \mathcal{N}(a_i; b_j, V_{Q,i}+V_{K,j})$ *nonnegative. The rule is a different routing operator rather than a reading of Eq.* (12)*, and the evaluated model does not implement it.*

6.3 Content laws and the shared-latent alternative

For one head, $z_i^{(h)}$ is exactly the mean of a declared mixture of content laws with means $T_h\xi_j$ and weights β_{ij}. A normalized local construction makes the limited probabilistic interpretation explicit. At fixed context, let $p_{K,j}(y)$ be the comparison densities of Eq. (9), and choose normalized content laws $\nu_j(\zeta)$ with those content means. Then

$$\mathbb{P}_i(j, y, \zeta) = \pi_{ij} p_{K,j}(y)\nu_j(\zeta), \tag{71}$$

$$\mathbb{Q}_{i,\beta}(j, y, \zeta) = \beta_j q_{Q,i}(y)\nu_j(\zeta), \tag{72}$$

$$D_{\mathrm{KL}}(\mathbb{Q}_{i,\beta}\|\mathbb{P}_i) = \sum_j \beta_j D_{\mathrm{KL}}(q_{Q,i}\|p_{K,j}) + D_{\mathrm{KL}}(\beta\|\pi_i). \tag{73}$$

The identity follows by cancellation of ν_j in the density ratio and integration of normalized factors. It recovers the uncapped routing objective at $\tau_h = 1$. Since every normalized ν_j cancels, this construction permits a chosen content bank but does not select its means, T_h, $\widetilde{O}_h$, or the residual. At general τ_h, Eq. (11) is the declared entropy-regularized extension, not the KL of this same joint without additional modeling changes; the binary compatibility record of Theorem 20 realizes it exactly at every temperature. The identity determines routing within the declared joint; a predictive architecture requires additional choices.

The independent-content construction also distinguishes two measurements of a distribution from two observations of one latent draw. If $X_j \sim \mathcal{N}(\xi_j, C_j)$, $Y = S_K X_j + \varepsilon$ with independent Gaussian noise $\varepsilon \sim \mathcal{N}(0, R)$, $R \succeq 0$, and $V = TX_j$, then Gaussian conditioning instead gives (Petersen and Pedersen, 2012)

$$\mathbb{E}[V \mid Y = y, J = j] = T\xi_j + TC_jS_K^\top(S_KC_jS_K^\top + R)^{-1}(y - S_K\xi_j). \tag{74}$$

Thus uncorrected values $T\xi_j$ do not generally equal shared-latent conditional means. Overlapping transmitting and comparison subspaces are permitted, but they do not remove this correlation term. The mixture-mean readout in Eq. (16) specifies a different statistical operation from this shared-latent conditioning law. With $R = 0$ and the query law as the recognition of Y, this model gives the same routing objective and the pair-dependent content $T\big(\xi_j + R_{K,j}(S_Q\xi_i - S_K\xi_j)\big)$, where $R_{K,j} = C_jS_K^\top(S_KC_jS_K^\top)^{-1}$ is the key-side analogue of the lift in Eq. (92). The uncorrected value is recovered for every query exactly when $TC_jS_K^\top = 0$.

The same variational reading covers the decoder. The linear decoder of Eq. (25) is exactly the unit-temperature Gibbs readout of shared-covariance Gaussian templates; Appendix D derives this identity and develops the Gaussian-bank readout with class-specific covariances.

7 Receiver Inference over the Same Channels

Section 6 reads content predictively: each head's routed message is averaged, and the predictive mean replaces or increments the receiver's state. The same channels can instead be read as evidence about the receiver. The receiver holds a Gaussian prior, each head observes the receiver through a receiving measurement, and the routed source messages are the observations, so the update is a variational posterior. This section derives that receiver energy and its exact minimizer, states when the predictive head sum is its limit, and records the realization used in Section 9.3. Routing is computed from the comparison laws as before and held fixed during the update.

7.1 The receiver episode and the channel likelihood

A receiver episode snapshots the incoming states $(\bar{\xi}_j, \bar{C}_j)$, the routing weights, and the frames of one layer. It then moves each receiver's state (ξ, C) against an energy $F_i(\xi, C)$ with the snapshot held fixed in value, while the outer supervised graph through the snapshot remains attached. The frames do not change within the episode, so $(\mu, \Sigma) \mapsto (\xi, C)$ is a fixed linear reparametrization. Every energy below is a function of reference moments. Its local gradients are $\nabla_\mu F = U_i^{-\top} \nabla_\xi F$ and $\nabla_\Sigma F = U_i^{-\top} (\nabla_C F) U_i^{-1}$, and the Gaussian Fisher natural gradient $(\Sigma \nabla_\mu F, 2\Sigma \nabla_\Sigma F \Sigma)$ equals the reference natural gradient $(C \nabla_\xi F, 2C \nabla_C F C)$ mapped by U_i, with $U_i^\top$ on the right of the covariance term. A Fisher step taken in local coordinates is therefore the same step taken in reference coordinates and mapped back, because the Fisher metric of the Gaussian family is invariant under affine reparametrization of the sample space (Amari, 1998). We work in reference coordinates and suppress the receiver index.

Head h observes the receiver through a receiving measurement $\mathsf{P}_h \in \mathbb{R}^{s_h \times K}$, and the message of source j, with the source state marginalized, predicts that measurement through a channel with noise $Q_h^{\mathrm{ch}} \succeq 0$:

$$\begin{aligned} \mathsf{P}_h x = T_h x_j + \epsilon, \qquad x_j \sim \mathcal{N}(\bar{\xi}_j, \bar{C}_j), \qquad \epsilon \sim \mathcal{N}(0, Q_h^{\mathrm{ch}}), \\ p(T_h \bar{\xi}_j \mid x) = \mathcal{N}\big(T_h \bar{\xi}_j; \mathsf{P}_h x, R_{hj}\big), \qquad R_{hj} = T_h \bar{C}_j T_h^\top + Q_h^{\mathrm{ch}}. \end{aligned} \tag{75}$$

With the routing weights declared as likelihood powers $w_{hj} = \lambda_\beta \beta_{ij}^{(h)}$, $\lambda_\beta > 0$, and a Gaussian prior $p = \mathcal{N}(\xi_p, C_p)$ with coefficient $\lambda_\alpha > 0$, the receiver energy is

$$F(\xi, C) = \lambda_\alpha D_{\mathrm{KL}}\big(\mathcal{N}(\xi, C) \| \mathcal{N}(\xi_p, C_p)\big) + \sum_{h,j} w_{hj} \mathbb{E}_{x \sim \mathcal{N}(\xi, C)} \big[-\log \mathcal{N}(T_h \bar{\xi}_j; \mathsf{P}_h x, R_{hj}) \big]. \tag{76}$$

Proposition 22 (Closed-form channel receiver) *Let $C_p \succ 0$ and $R_{hj} \succ 0$, and put*

$$\begin{aligned} W_h = \sum_j w_{hj} R_{hj}^{-1}, \qquad \omega_h = \sum_j w_{hj} R_{hj}^{-1} T_h \bar{\xi}_j, \\ \Lambda = \lambda_\alpha C_p^{-1} + \sum_h \mathsf{P}_h^\top W_h \mathsf{P}_h, \qquad \eta = \lambda_\alpha C_p^{-1} \xi_p + \sum_h \mathsf{P}_h^\top \omega_h. \end{aligned} \tag{77}$$

Then

$$F(\xi, C) = \tfrac{1}{2} \operatorname{Tr}(\Lambda C) + \tfrac{1}{2} \xi^\top \Lambda \xi - \eta^\top \xi - \tfrac{\lambda_\alpha}{2} \log\det C + \text{const}, \tag{78}$$

F is strictly convex on $\mathbb{R}^K \times \{C \succ 0\}$, and its unique minimizer is

$$\xi^\star = \Lambda^{-1}\eta, \qquad C^\star = \lambda_\alpha \Lambda^{-1}. \tag{79}$$

Equivalently, $F = \lambda_\alpha D_{\mathrm{KL}}(\mathcal{N}(\xi, C)\|p_{\mathrm{pow}}) + \text{const}$, where $p_{\mathrm{pow}}(x) \propto p(x)\prod_{h,j} p(T_h\bar{\xi}_j \mid x)^{w_{hj}/\lambda_\alpha}$ is the normalized Gaussian power posterior with mean $\xi^\star$ and covariance $C^\star$.

Proof For a Gaussian likelihood, $\mathbb{E}_{\mathcal{N}(\xi,C)}[-\log \mathcal{N}(y; \mathsf{P}x, R)] = \frac{1}{2}(y - \mathsf{P}\xi)^\top R^{-1}(y - \mathsf{P}\xi) + \frac{1}{2}\operatorname{Tr}(\mathsf{P}^\top R^{-1}\mathsf{P}C) + \frac{1}{2}\log\det(2\pi R)$, and the Gaussian divergence is $\frac{1}{2}[\operatorname{Tr}(C_p^{-1}C) + (\xi - \xi_p)^\top C_p^{-1}(\xi - \xi_p) - K + \log\det C_p - \log\det C]$. Collecting terms gives Eq. (78). The expected log-likelihoods contain no entropy of the receiver, which enters once, through the prior divergence. The energy is a strictly convex quadratic in ξ plus a function of C that is linear apart from the strictly convex term $-\frac{\lambda_\alpha}{2}\log\det C$, and its gradients $\Lambda\xi - \eta$ and $\frac{1}{2}(\Lambda - \lambda_\alpha C^{-1})$ vanish exactly at Eq. (79). Finally, $F/\lambda_\alpha = \mathbb{E}_q[\log q - \log p - \sum_{h,j}(w_{hj}/\lambda_\alpha)\log p(T_h\bar{\xi}_j \mid x)]$ up to a constant, which is $D_{\mathrm{KL}}(q\|p_{\mathrm{pow}})$ minus the logarithm of the normalizer of p_{pow}, and the product of the Gaussian prior with Gaussian likelihood powers is Gaussian with precision Λ/λ_α and information vector η/λ_α. ■

Corollary 23 (Properties of the channel receiver) *1. The receiver entropy enters with the single coefficient λ_α, so $C^\star$ is the covariance of the declared power posterior. At $\lambda_\alpha = 1$ and unit weights the update is the conjugate linear-Gaussian update (Bishop, 2006, Sec. 2.3.3), and the Gaussian variational optimum is exact because the target is Gaussian (Opper and Archambeau, 2009).*

2. *With $Q_h^{\mathrm{ch}} = 0$ the energy is finite whenever T_h has full row rank, because then $T_h\bar{C}_j T_h^\top \succ 0$. A positive channel floor guards against loss of rank as T_h is learned; it is not needed for finiteness.*

3. *The likelihoods depend on x only through $\mathsf{P}_h x$. On $V_0 = \bigcap_h \ker \mathsf{P}_h$, the directions that no head observes, the posterior conditional law given the observed component equals the prior conditional law, and if C_p does not couple V_0 to its orthogonal complement, the posterior marginal on V_0 equals the prior marginal. No head imposes a value on V_0.*

4. *At $C = C^\star$, $C\Lambda = \lambda_\alpha I$, so the explicit Fisher mean step $\xi \leftarrow \xi - \varepsilon C(\Lambda\xi - \eta)$ contracts exactly for $0 < \varepsilon < 2/\lambda_\alpha$, and $\varepsilon = 1/\lambda_\alpha$ is the Newton step.*

Proof Item 1 restates Proposition 22. For item 2, $y^\top T_h\bar{C}_j T_h^\top y = (T_h^\top y)^\top \bar{C}_j (T_h^\top y) > 0$ for $y \neq 0$. For item 3, decompose $x = u + v$ with $u \in V_0^\perp$ and $v \in V_0$. Every likelihood depends on x only through $\mathsf{P}_h u$, so $p_{\mathrm{pow}}(v \mid u) = p(v \mid u)$; if the prior factorizes over $V_0^\perp \oplus V_0$, so does p_{pow}, and its marginal on V_0 is the prior marginal. For item 4, the step maps the error $e = \xi - \xi^\star$ to $(I - \varepsilon C\Lambda)e = (1 - \varepsilon\lambda_\alpha)e$ at $C = C^\star$. ■

7.2 The receiving measurement and the predictive limit

The posterior moves the state along the gain $\Lambda^{-1}\mathsf{P}_h^\top W_h = (C^\star/\lambda_\alpha)\mathsf{P}_h^\top W_h$, as in a Kalman update, so the receiving interface of this energy is the measurement P_h rather than a free write-back map. The implementation ties $\mathsf{P}_h = \widetilde{O}_h^\top$, the transposed effective receiving map of the chart in use. This keeps the stored parameters and their block-identity initialization and gives the following limit.

Proposition 24 (Predictive limit) *Assume $\mathsf{P}_h = \widetilde{O}_h^\top$ with $\widetilde{O}_h^\top \widetilde{O}_g = \delta_{hg} I_{s_h}$ and $\sum_h s_h = K$, so that $\sum_h \widetilde{O}_h \widetilde{O}_h^\top = I_K$, assume $R_{hj} = R_h$ for every source, and let $\lambda_\alpha C_p^{-1} \to 0$ and $\lambda_\alpha C_p^{-1}\xi_p \to 0$. Then*

$$\xi^\star \to \sum_h \widetilde{O}_h \sum_j \beta_{ij}^{(h)} T_h \bar{\xi}_j, \tag{80}$$

the residual-free predictive head sum of Eq. (16) *in reference coordinates.*

Proof In the limit, $\Lambda = \sum_h \widetilde{O}_h(\lambda_\beta R_h^{-1})\widetilde{O}_h^\top$ and $\eta = \sum_h \widetilde{O}_h \lambda_\beta R_h^{-1} \sum_j \beta_{ij}^{(h)} T_h \bar{\xi}_j$, using $\sum_j \beta_{ij}^{(h)} = 1$. Both are block diagonal in the orthonormal basis formed by the columns of the $\widetilde{O}_h$, and solving blockwise gives $\widetilde{O}_h^\top \xi^\star = \sum_j \beta_{ij}^{(h)} T_h \bar{\xi}_j$. ∎

Each relaxed hypothesis has a definite meaning. With source-dependent channel covariances, the channel mean of head h becomes the precision-weighted average $(\sum_j \beta_{ij}^{(h)} R_{hj}^{-1})^{-1} \sum_j \beta_{ij}^{(h)} R_{hj}^{-1} T_h \bar{\xi}_j$, so confident senders count for more than their routing weight. Overlapping heads fuse their measurements by least squares. A finite prior blends the prior mean and the messages with a gain set by their precisions. For an isotropic prior precision $\lambda_\alpha C_p^{-1} = aI$ and message evidence of precision ℓ on the range W of the measurements, $\sum_h \mathsf{P}_h^\top W_h \mathsf{P}_h = \ell P_W$ and $\sum_h \mathsf{P}_h^\top \omega_h = \ell P_W m$ with P_W the orthogonal projector onto W, the minimizer is

$$\xi^\star = (I - P_W)\xi_p + P_W \frac{a\xi_p + \ell m}{a + \ell}. \tag{81}$$

The coefficient of the prior state is its share of precision, $a/(a+\ell)$, on observed directions and one on unobserved directions, and the complement-preserving aggregation of Proposition 8 is the limit $\ell/a \to \infty$ when its splitting is orthogonal. The posterior covariance $C^\star = \lambda_\alpha \Lambda^{-1}$ depends on the context but omits the between-source spread of the moment-matched predictive covariance, because geometric pooling of likelihoods differs from marginalizing a categorical predictive mixture. In the exponential chart the tied measurement has the zeroth-order part $(T^{-1})_{:,h}^\top$, which reads through $T^{-\top}$ and agrees with the head-routed flat transport of Proposition 11 only when T is orthogonal; tying the chart into the measurement is an implementation choice, not a consequence of the exponential construction.

7.3 Updates, outer derivatives, and the profiled coupling

Proposition 25 (Natural-parameter flow) *For an energy of the form of Eq.* (78) *with an entropy coefficient $c_{\mathrm{H}} > 0$ in place of λ_α, the Fisher flow $\dot{\xi} = -C(\Lambda\xi - \eta)$, $\dot{C} =$*

$c_{\mathrm{H}}C - C\Lambda C$ has the exact solution

$$J(t) = (1-\rho)J(0) + \rho\Lambda/c_{\mathrm{H}}, \qquad \vartheta(t) = (1-\rho)\vartheta(0) + \rho\eta/c_{\mathrm{H}}, \qquad \rho = 1 - e^{-c_{\mathrm{H}}t}, \tag{82}$$

in the natural parameters $J = C^{-1}$ and $\vartheta = C^{-1}\xi$. Every iterate with $\rho \in [0,1]$ is positive definite, the energy does not increase along the flow, and $\rho = 1$ gives the minimizer $(\Lambda^{-1}\eta, c_{\mathrm{H}}\Lambda^{-1})$.

Proof $\dot{J} = -J\dot{C}J = \Lambda - c_{\mathrm{H}}J$ and $\dot{\vartheta} = \dot{J}\xi + J\dot{\xi} = \eta - c_{\mathrm{H}}\vartheta$ are linear with constant coefficients. A convex combination of positive-definite matrices is positive definite. Along the flow, $\frac{d}{dt}F = -\nabla_\xi F^\top C \nabla_\xi F - 2\,\mathrm{Tr}(C\nabla_C F C \nabla_C F) \le 0$, since $c_{\mathrm{H}}C - C\Lambda C = -2C\nabla_C F C$. ■

With the snapshot fixed in value, the closed form, $\rho = 1$, completes a receiver episode in one solve, and further iterations leave it unchanged when the coupling is constant. Its outer derivatives follow by implicit differentiation, $d\xi^\star = \Lambda^{-1}\big(d\eta - (d\Lambda)\xi^\star\big)$ and $dC^\star = (d\lambda_\alpha)\Lambda^{-1} - \lambda_\alpha\Lambda^{-1}(d\Lambda)\Lambda^{-1}$, which automatic differentiation through a Cholesky solve computes. This removes the step-size restriction of the explicit step, but it does not bound every outer derivative of the network. Partial damping $\rho < 1$ descends the frozen energy in the full family: with a scalar coefficient the energy equals $c_{\mathrm{H}} D_{\mathrm{KL}}(q\|q^\star)$ up to a constant, with $q^\star = \mathcal{N}(\Lambda^{-1}\eta, c_{\mathrm{H}}\Lambda^{-1})$, and along the segment $\theta_\rho = (1-\rho)\theta_0 + \rho\theta^\star$ of natural parameters with log-partition A, $\frac{d}{d\rho}D_{\mathrm{KL}}(q_{\theta_\rho}\|q^\star) = -(1-\rho)\delta^\top\nabla^2 A(\theta_\rho)\delta \le 0$ with $\delta = \theta^\star - \theta_0$.

When only local variances s_k are stored, the energy restricted to the diagonal family separates in the variances as $\frac{1}{2}\sum_k(\Lambda_{\mathrm{loc}})_{kk}s_k - \frac{c_{\mathrm{H}}}{2}\sum_k \log s_k$, with $\Lambda_{\mathrm{loc}} = U_i^{-\top}\Lambda U_i^{-1}$ and $\eta_{\mathrm{loc}} = U_i^{-\top}\eta$, so its exact restricted minimizer is $\mu^\star = \Lambda_{\mathrm{loc}}^{-1}\eta_{\mathrm{loc}}$ and $s_k^\star = c_{\mathrm{H}}/(\Lambda_{\mathrm{loc}})_{kk}$. Interpolating natural parameters does not stay in the diagonal family when Λ_{loc} is dense, and the coordinatewise damping $s_k^{-1} \mapsto (1-\rho)s_k^{-1} + \rho(\Lambda_{\mathrm{loc}})_{kk}/c_{\mathrm{H}}$ carries no descent guarantee. For $\Lambda = \left(\begin{smallmatrix} 1 & 0.99 \\ 0.99 & 1 \end{smallmatrix}\right)$, $\eta = 0$, $c_{\mathrm{H}} = 1$, mean $(5,-5)$, variances $(1/2, 2)$, and $\rho = 1/2$, the frozen energy without constants rises from 1.500 to 2.503, while $\rho = 1$ lowers it to 1.000.

The coefficient of the prior can also be profiled. With $D(q) = D_{\mathrm{KL}}(q\|p)$ and the regularizer $b_0\lambda_\alpha - c_0\log\lambda_\alpha$, $b_0, c_0 > 0$, minimizing over λ_α gives $\lambda_\alpha^\star(D) = c_0/(b_0 + D)$ and the profiled energy $c_0\log(b_0 + D(q)) + G(q) + \mathrm{const}$, where G is the energy at $\lambda_\alpha = 0$. Its gradient equals the gradient of the frozen energy at $\lambda_\alpha = \lambda_\alpha^\star(D(q))$. Because $x \mapsto c_0\log(b_0 + x)$ is concave, freezing λ_α at the current state gives a majorizer that touches the profiled energy there, so the closed form at the frozen coefficient never increases the profiled energy: it is a majorize-minimize step (Hunter and Lange, 2004). On the diagonal family with a diagonal prior the divergence separates over local coordinates, and the per-coordinate coefficients $\lambda_{\alpha,k}^\star = c_0/(b_0 + D_k)$ keep the restricted closed form exact with entropy weights $c_{\mathrm{H},k} = \lambda_{\alpha,k}^\star$. That per-coordinate form is tied to the local basis and is not covariant under a change of local frame.

In the implementation the receiver of each layer receives as its prior the state handed off by the layer below: the prior mean is the incoming mean, and the prior covariance is the token-type covariance of the encoding. In the configuration of Section 9.3, which holds the stored covariance at that encoding value, the incoming state equals the layer prior at the start of every episode. Then $D_k = 0$, the profiled coefficients equal c_0/b_0, and with

$b_0 = c_0 = 1$ and one closed-form step per layer the update is the Kalman update of the incoming mean,

$$\mu^+ = \mu + \big(\Sigma_p^{-1} + \Lambda_{\mathrm{loc}}^{\mathrm{ch}}\big)^{-1}\big(\eta_{\mathrm{loc}}^{\mathrm{ch}} - \Lambda_{\mathrm{loc}}^{\mathrm{ch}}\mu\big), \tag{83}$$

where $\Lambda_{\mathrm{loc}}^{\mathrm{ch}}$ and $\eta_{\mathrm{loc}}^{\mathrm{ch}}$ are the channel terms of Eq. (77) pulled back to local coordinates and Σ_p is the stored diagonal covariance. The routed messages are the measurements, and the gain weighs the stored token-type precision against the channel precision. This is a precision-weighted residual of the kind in Eq. (81), not the additive residual, and the profiled coupling acts only when the receiver iterates or when its incoming state differs from the layer prior.

7.4 Fixed routing and the ambient alternative

Proposition 26 (Routing by the comparison cost alone) *Consider a functional of one routing row that is affine in β apart from the routing entropy $\tau \sum_j \beta_j \log(\beta_j/\pi_j)$, with linear coefficients $D_j + g_j(q)$ that may depend on the receiver state q. Its minimizing weights equal the Gibbs row of Eq.* (12) *for every q if and only if $g_j(q)$ does not depend on j on the support of π.*

Proof The minimizer is $\mathrm{softmax}_j(\log \pi_j - (D_j + g_j)/\tau)$, and a softmax is unchanged by j-independent shifts of its arguments and injective modulo them. ∎

A content term that carries source-specific information therefore changes the routing, and routing by the comparison costs while content moves the receiver is not coordinate descent of one such functional. It is exact inference in a declared modular model, in which comparison supplies a source-label posterior that content does not feed back into, the intention of cut inference (Plummer, 2015). The expected-log-likelihood receiver targets the geometric likelihood pool p_{pow}; a literal categorical cut posterior would instead average normalized conditionals and is generally a mixture. No population evidence bound for the composed stack follows from these local identities.

Replacing each channel likelihood by the divergence $D_{\mathrm{KL}}(\mathcal{N}(\xi, C)\|\mathcal{N}(M_h\bar{\xi}_j, M_h\bar{C}_j M_h^\top + \nu I_K))$, which aligns the receiver with lifted K-dimensional messages, gives an energy of the same quadratic form with entropy coefficient $\lambda_\alpha + \lambda_\beta H$, because each divergence carries its own copy of the receiver entropy and each routing row sums to one. For $s_h < K$ that energy is finite only with $\nu > 0$, since the lifted message law is otherwise singular, and the floor then makes every head an opinion of mean zero and precision $1/\nu$ on the complement of its range. The implementation offers this ambient law as an alternative; the runs reported here use the channel likelihood.

8 Matrix Links: Rank, Composition, Holonomy, and Lattice Structure

Communication uses selected directions of an agent's framed state. The mathematical meaning of using a subset of a frame is that a receiving or sending map accesses a lower-dimensional observable of the state expressed in that frame. These maps also set gains and identify transmitted directions with received directions. They need not select coordinate entries, define orthogonal projections, or select a subset of group elements. The possible

analogy with information coding is an interpretation to develop; no coding theorem or channel-capacity claim is assumed here.

Proposition 27 (Rank and composition of content links) *Fix a layer and write* $M_h = \widetilde{O}_h T_h$.

1. *Each head has rank at most* s_h*, and the routed link of Eq.* (20) *satisfies*

$$\operatorname{rank} \mathcal{M}_{ij} \le \min\left(K, \sum_h s_h\right). \tag{84}$$

2. *For one head, intermediate frame factors cancel:*

$$L^{(h)}_{ij} L^{(h)}_{jk} - L^{(h)}_{ik} = U_i \widetilde{O}_h (T_h \widetilde{O}_h - I_{s_h}) T_h U_k^{-1}, \tag{85}$$

$$L^{(h)}_{i_0 i_1} \cdots L^{(h)}_{i_{a-1} i_a} = U_{i_0} M_h^a U_{i_a}^{-1}. \tag{86}$$

   If $T_h \widetilde{O}_h = I_{s_h}$*, the composition defect vanishes. Conversely, if* $\widetilde{O}_h$ *has full column rank and* T_h *has full row rank, a vanishing defect implies* $T_h \widetilde{O}_h = I_{s_h}$.

3. *For a declared link* $\mathcal{L}_{ij} : E_j \to E_i$ *of any rank and* $Q_i \succ 0$*, the kernel*

$$k_i \mid k_j \sim \mathcal{N}(\mathcal{L}_{ij} k_j, Q_i) \tag{87}$$

   is normalized, and a source $k_j \sim \mathcal{N}(\mu_j, \Sigma_j)$ *has the nondegenerate receiving marginal*

$$k_i \sim \mathcal{N}\left(\mathcal{L}_{ij}\mu_j, Q_i + \mathcal{L}_{ij}\Sigma_j \mathcal{L}_{ij}^\top\right). \tag{88}$$

Proof (1) Each head contributes $U_i M_h U_j^{-1}$, of rank at most s_h, and the rank of a sum is at most the sum of the ranks. (2) Substituting Eq. (17), the factors $U_j^{-1} U_j$ cancel and leave $U_i \widetilde{O}_h T_h \widetilde{O}_h T_h U_k^{-1} - U_i \widetilde{O}_h T_h U_k^{-1}$; the path identity follows by induction. For the converse, remove the invertible frames and multiply the defect by a left inverse of $\widetilde{O}_h$ on the left and a right inverse of T_h on the right. (3) Write $k_i = \mathcal{L}_{ij} k_j + \varepsilon_i$ with independent $\varepsilon_i \sim \mathcal{N}(0, Q_i)$. Means and covariances give the marginal, and $Q_i \succ 0$ makes it nondegenerate. These are affine-Gaussian identities (Bishop, 2006). ■

For example, at $K = 2$, take $\widetilde{O}_1 T_1 = \operatorname{diag}(1, 0)$ and $\widetilde{O}_2 T_2 = \operatorname{diag}(0, 1)$. Positive weights produce the invertible reference operator $\operatorname{diag}(\beta^{(1)}_{ij}, \beta^{(2)}_{ij})$, so rank-deficient heads can sum to an invertible link. Conversely, a single head with $s_h < K$ cannot be a $\mathrm{GL}(K)$ element. The realized matrices also remain in the span of the shared $\widetilde{O}_h T_h$ at a fixed layer; they are not unrestricted independent edge parameters.

The composition defect measures repeated receiving-to-sending processing. If $T_h \widetilde{O}_h = I_{s_h}$, then $L^{(h)}_{ij}$ identifies $U_j \operatorname{im} \widetilde{O}_h$ with $U_i \operatorname{im} \widetilde{O}_h$ and composes flatly on these subspaces; its ambient self-map is a projector. At $K = s_h = 1$, $U_i = 1$, $T_h = 1$, and $\widetilde{O}_h = 2$, the one-edge map is 2 while the two-edge map is 4. Even invertibility therefore does not make the shared processing family a flat reference transport. Paths of the same length and endpoints still

agree for this unweighted, shared-head specialization. Routed mixtures can depend on the intermediate sites through their coefficients. Without the rank assumptions the converse can fail.

Under a passive frame change, $Q_i' = g_i Q_i g_i^\top$ together with the link endpoint law gives the corresponding pushforward conditional and marginal. This channel provides a probabilistic use of the sending and receiving map, but it does not force a particular predictive architecture. In particular, a residual update of the mean with covariance retained as a routing state does not propagate the covariance in Eq. (88); Theorem 20 gives the moment-matched alternative for the full multi-head update. A population model requires a normalized directed assembly of kernels, or a separately normalized cyclic law. Multiplying reciprocal conditionals without that construction does not specify a joint evidence model.

8.1 Comparison maps and induced links

The query and key maps first define the comparison cospan

$$E_i \xrightarrow{A_{Q,i,h}} H_h^{\mathrm{cmp}} \xleftarrow{A_{K,j,h}} E_j. \tag{89}$$

The Gaussian observables of Eqs. (8) and (9) occupy this common comparison space. A cospan compares two observables; it does not by itself choose a map back from the comparison space to E_i.

The distinction can be seen directly in the mean cross term of the Gaussian KL. Suppress h, and put $P_{K,j} = (A_{K,j}\Sigma_j A_{K,j}^\top)^{-1}$. The comparison tensor is

$$B_{ij} = A_{Q,i}^\top P_{K,j} A_{K,j}, \qquad B_{ij}' = g_i^{-\top} B_{ij} g_j^{-1}. \tag{90}$$

It maps E_j to E_i^*, with invariant scalar $\mu_i^\top B_{ij}\mu_j$. A vector link instead maps E_j to E_i. A declared receiver metric can raise the first index: $G_i^{-1}B_{ij}$ then obeys the link law because $G_i' = g_i^{-\top} G_i g_i^{-1}$. The covariance Σ_i, viewed as a map $E_i^* \to E_i$, supplies another such choice; with the Fisher metric $G_i = \Sigma_i^{-1}$ the two choices coincide. Neither choice is an inverse of the rectangular query map.

A more specific lift uses the receiver's Gaussian geometry. Define

$$R_{Q,i} = C_i S_Q^\top (S_Q C_i S_Q^\top)^{-1}, \qquad S_Q R_{Q,i} = I_{r_h}, \tag{91}$$

$$L_{ij}^{\mathrm{cmp}} = U_i R_{Q,i} S_K U_j^{-1}. \tag{92}$$

For every $y \in \mathbb{R}^{r_h}$, $R_{Q,i}y$ is the unique minimizer of $\delta^\top C_i^{-1}\delta$ subject to $S_Q\delta = y$. To check this, write $\delta = R_{Q,i}y + w$ with $S_Q w = 0$. Then $(R_{Q,i}y)^\top C_i^{-1} w = y^\top (S_Q C_i S_Q^\top)^{-1} S_Q w = 0$, so the objective is its value at $R_{Q,i}y$ plus $w^\top C_i^{-1} w$. This also identifies the lift in the Gaussian conditional mean,

$$\mathbb{E}[X_i \mid S_Q X_i = y] = \xi_i + R_{Q,i}(y - S_Q\xi_i), \quad X_i \sim \mathcal{N}(\xi_i, C_i). \tag{93}$$

For the conditional identity, the Gaussian residual $X_i - \xi_i - R_{Q,i}(S_Q X_i - S_Q\xi_i)$ has zero cross covariance with $S_Q X_i$, since $C_i S_Q^\top - R_{Q,i} S_Q C_i S_Q^\top = 0$. Its joint Gaussian characteristic function therefore factors, making the residual independent of the measurement and giving

Eq. (93). Thus L_{ij}^{cmp} is a precise state-dependent comparison link induced by a specified lift. Its covariance follows because $R_{Q,i}$ is unchanged by the passive site action. It transports a comparison displacement into the receiver; the complete conditional mean includes the affine centering term in Eq. (93). This construction gives S_Q and S_K a precise induced-link interpretation. It is distinct from the independently chosen content link $L_{ij}^{(h)}$ built from $T_h, \widetilde{O}_h$; equality would require additional conditions on those maps.

8.2 Flat reference identification, loops, and lattice structure

The separate reference identification is

$$\Omega_{ij} = U_i U_j^{-1}, \qquad k_j \mapsto \Omega_{ij} k_j. \tag{94}$$

It is the exponential link of Definition 1 with zero generator, and it is recovered from a content link when $\widetilde{O}_h T_h = I_K$. When the frames are each represented by one exponential, it equals $e^{\phi_i} e^{-\phi_j}$; the chart restrictions of Section 2.1 remain applicable.

Remark 28 (Flat reference identification) *For any invertible frames and any finite sequence of sites, successive factors cancel, $\Omega_{ij}\Omega_{jk} = U_i U_k^{-1} = \Omega_{ik}$. The ordered product of the Ω maps therefore depends only on its endpoints, and every closed product equals I_K; in particular $\Omega_{ij}\Omega_{jk}\Omega_{ki} = I_K$. The reference identification is pure gauge: it is the coboundary of the frame field.*

The global reference space supplies this trivialization explicitly. A finite discrete base has no overlap or continuum topology obstruction here; the graph may nevertheless support independent connection data with nonidentity cycle products. The identity just proved applies to Ω, not to every matrix communication link.

Theorem 14 records the loop invariants of routed links; the following definition applies to any endpoint-covariant edge matrices. For a declared closed walk $\gamma = (i_0, \dots, i_r = i_0)$ and endpoint-covariant edge matrices $L_{ij} : E_j \to E_i$, define

$$\mathcal{H}_\gamma(L) = L_{i_0 i_1} \cdots L_{i_{r-1} i_0}, \tag{95}$$

$$\mathcal{H}_\gamma(L)' = g_{i_0} \mathcal{H}_\gamma(L) g_{i_0}^{-1}, \tag{96}$$

$$\mathcal{W}_\gamma(L) = \operatorname{Tr} \mathcal{H}_\gamma(L). \tag{97}$$

Intermediate gauge factors cancel, and cyclicity of trace proves that $\mathcal{W}_\gamma$ is invariant even for singular links. It is not a complete invariant of the loop matrix. For routed links, write $M_{ij} = \sum_h \beta_{ij}^{(h)} M_h$. Then

$$\mathcal{H}_\gamma(\mathcal{M}) = U_{i_0} \big(M_{i_0 i_1} \cdots M_{i_{r-1} i_0} \big) U_{i_0}^{-1}. \tag{98}$$

Expanding the product gives the loop-trace polynomial of Theorem 14. A nonidentity communication product can express gain, contraction, or rank loss and does not imply nonzero holonomy of Ω. Causal support has no directed cycles through distinct token positions, so on it the only closed walks are self-loops, whose traces are those of the frame-free operator M_{ii}. For singular links, a loop observable therefore requires a declared extended graph and its additional links, because causal routing weights alone do not provide reverse transports.

Invertible links do provide them, and Section 8.3 constructs the resulting inverse-edge holonomy. Products at different network depths are compositions of different operators and are not automatically loops of one fixed connection.

A group-valued connection instead supplies invertible edge maps Θ_{ij} together with inverse reverse maps $\Theta_{ji} = \Theta_{ij}^{-1}$ (Wilson, 1974). The exponential links of Section 3 are of this kind. With one shared generator, $\Theta_{ij} = U_i e^X U_j^{-1} = F_i \Omega_{ij}$ with $F_i = U_i e^X U_i^{-1}$, so they form a factorized family that is group valued but remains a composite of site data and one learned matrix. A factorized content head decomposes in the same way,

$$L_{ij}^{(h)} = F_{i,h} \Omega_{ij}, \qquad F_{i,h} = U_i \widetilde{O}_h T_h U_i^{-1}. \tag{99}$$

Replacing Ω_{ij} by an independently specified edge field, for example by generators X_{ij} that vary with the edge, would define a connection with the same endpoint covariance and possibly nontrivial holonomy. It is a broader construction than the factorized families above, and Section 8.4 states a latent-variable version. An invertible routed link can be supplied with an inverse reverse edge, but a separately computed reverse attention operator need not be that inverse. General matrix links have a precedent in linear lattice gauge theory (Wetterich, 2014); its compact-group action and continuum conclusions are not consequences of the present algebra. Because the attention links are composites of site data rather than independent edge variables, the closer counterpart is a cellular sheaf, whose restriction maps into an edge stalk play the role of the comparison cospan of Eq. (89) (Hansen and Ghrist, 2019; Bodnar et al., 2022).

Finally, arbitrary invertible coordinate changes must be distinguished from symmetries of a fixed numerical metric or a restricted frame parameterization. Here G_i and all Gaussian tensors transform together. Holding G_i fixed restricts metric-preserving active maps to its orthogonal group, and the Euclidean quantity $\mathrm{Tr}(L_{ij}^\top L_{ij})$ is not invariant under general independent $\mathrm{GL}(K)$ actions. The results above establish covariance of the full-Gaussian communication construction. They do not supply an invariant lattice action or a normalized population model. The link prior of Section 8.4 is a declared action whose curvature term uses the reference metric; it is invariant under passive site changes and under reparametrizations by nonzero multiples of orthogonal matrices, not under all of $\mathrm{GL}(K)$.

8.3 Inverse-edge holonomy of invertible links

Invertible links admit loop observables on the causal graph without adding reverse attention operators. Write $G_{ij} = U_i^{-1} \Theta_{ij} U_j$ for the reference representative of an invertible link Θ_{ij}, $j < i$.

Definition 29 (Triangle holonomy) *For $k < j < i$ put*

$$F_{ijk} = \Theta_{ik}^{-1} \Theta_{ij} \Theta_{jk} = U_k G_{ik}^{-1} G_{ij} G_{jk} U_k^{-1}. \tag{100}$$

The field is flat on the complete causal support when $F_{ijk} = I$ for every strict triple.

Under the passive site action $F_{ijk} \mapsto g_k F_{ijk} g_k^{-1}$, so the conjugacy class of F_{ijk}, and with it the trace, determinant, and spectrum, is invariant. F_{ijk} is the holonomy of the closed loop $k \to j \to i \to k$ based at k, whose last step traverses the long edge backward with its

inverse. Invertibility is what gives a causal field these loop observables. On graphs with missing edges an explicit cycle set must be declared, and the recurrence characterization below need not apply.

Proposition 30 (Flatness and pure gauge) *1. Let Θ_{ij}, $j < i$, be arbitrary $K \times K$ matrices with $\Theta_{ij}\Theta_{jk} = \Theta_{ik}$ for all $k < j < i$. Then $\Theta_{ij} = \Theta_{i,i-1}\Theta_{i-1,i-2}\cdots\Theta_{j+1,j}$, and conversely every choice of adjacent matrices defines a flat field through these ordered products.*

2. If every Θ_{ij} is invertible, the field is flat exactly when $\Theta_{ij} = g_i g_j^{-1}$ for some $g_i \in \mathrm{GL}(K)$, and then $g_i = \Theta_{i0}$ up to a common right factor. For singular matrices this equivalence fails: a shared projector $\Theta_{ij} = P = P^2$ is flat and is not of the form $g_i g_j^{-1}$.

Proof For item 1 with $i - j \geq 2$, flatness with middle index $i - 1$ gives $\Theta_{ij} = \Theta_{i,i-1}\Theta_{i-1,j}$, and induction on $i - j$ gives the ordered product; conversely, ordered products of adjacent matrices are associative. For item 2, set $g_0 = I$ and $g_i = \Theta_{i0}$. By item 1, $g_i g_j^{-1} = \Theta_{i,i-1}\cdots\Theta_{j+1,j} = \Theta_{ij}$, and the converse is the cocycle identity $g_i g_j^{-1} g_j g_k^{-1} = g_i g_k^{-1}$. The projector satisfies $PP = P$ but is singular. ■

Three consequences follow. The flat identification Ω_{ij} is pure gauge for arbitrary frames, including positional factors, and so is every reference field of the form $h_i h_j^{-1}$, such as a rotary link $e^{(p_i - p_j)Y}$, because it is absorbed into the frames. By Eq. (100), F_{ijk} is conjugate to $G_{ik}^{-1} G_{ij} G_{jk}$, so curvature is a property of the reference field: no choice of frames creates or removes it, and a constant reference link $G \neq I$ is never flat. Finally, the triangle holonomies detect failure of the cocycle condition on the complete ordered graph, but their separate traces or spectra need not form a complete set of invariants. Inverse-edge completion thus supplies loop observables without contradicting the absence of directed cycles on causal support.

Lemma 31 (Second-order holonomy) *For exponential links with generators $A = X_{ij}$, $B = X_{jk}$, and $Z = X_{ik}$, $F_{ijk} = U_k e^{-Z} e^A e^B U_k^{-1}$, and on a neighborhood of zero where the logarithm is analytic,*

$$\log\big(e^{-Z} e^A e^B\big) = (A + B - Z) + \tfrac{1}{2}\big([A, B] - [Z, A] - [Z, B]\big) + O(3). \tag{101}$$

The first-order term is the additive cocycle defect of the algebra-valued field. On the complete ordered graph it vanishes for all triples exactly when $X_{ij} = \psi_i - \psi_j$ for some $\psi_i \in \mathfrak{gl}(K)$, and an additive coboundary does not make the exponential links flat when its values fail to commute.

Proof Apply the Baker–Campbell–Hausdorff series twice (Hall, 2015): $\log(e^A e^B) = A + B + \frac{1}{2}[A, B] + \cdots$, and then $\log(e^{-Z} e^C) = C - Z - \frac{1}{2}[Z, C] + \cdots$ with $C = A + B + \frac{1}{2}[A, B]$. For the coboundary statement take $\psi_0 = 0$ and $\psi_i = X_{i0}$, and apply the vanishing defect with $k = 0$. ■

Link field	Triangle holonomy F_{ijk}	Flat when
Positional, $U_i e^{(p_i-p_j)Y} U_j^{-1}$	I	always
Shared generator, $U_i e^X U_j^{-1}$	$U_k e^X U_k^{-1}$	only if $e^X = I$
Routed exponent, $X_{ij} = O\boldsymbol{\beta}_{ij} T$	$U_k O e^{\boldsymbol{\beta}_{ij}+\boldsymbol{\beta}_{jk}-\boldsymbol{\beta}_{ik}} T U_k^{-1}$	$\boldsymbol{\beta}_{ij} + \boldsymbol{\beta}_{jk} = \boldsymbol{\beta}_{ik}$
Head sum, $U_i O\boldsymbol{\beta}_{ij} T U_j^{-1}$	$U_k O\boldsymbol{\beta}_{ik}^{-1}\boldsymbol{\beta}_{ij}\boldsymbol{\beta}_{jk} T U_k^{-1}$	$\boldsymbol{\beta}_{ij}\boldsymbol{\beta}_{jk} = \boldsymbol{\beta}_{ik}$

Table 3: Triangle holonomy of candidate link fields. The routing matrices act headwise. The last two rows assume square invertible stacked maps with $TO = I_K$, and the head sum also requires positive weights on the three edges.

Table 3 collects exact holonomies; the shared-generator row shows that the static exponential link has the same nonflat holonomy e^X on every triangle. The tiled routed links of the implementation fall under the following statement, which does not assume $TO = I$.

Proposition 32 (Triangle spectrum of tiled routed links) *Assume tiled heads with invertible T and an invertible effective receiving map $\widetilde{O}$, and positive routing weights on the edges ij, jk, and ik. Put $\boldsymbol{\rho}_{ijk} = \bigoplus_h \rho_{ijk}^{(h)} I_{s_h}$ with $\rho_{ijk}^{(h)} = \beta_{ij}^{(h)}\beta_{jk}^{(h)}/\beta_{ik}^{(h)}$. The triangle holonomy of the edge operators $\mathcal{M}_{ab} = U_a \widetilde{O}\boldsymbol{\beta}_{ab} T U_b^{-1}$ is similar to $\boldsymbol{\rho}_{ijk} T\widetilde{O}$. In the affine chart $T\widetilde{O} = TO$, in the first-order chart $T\widetilde{O} = I + TO$, and in the exponential chart $T\widetilde{O} = e^{TO}$.*

Proof Removing the external conjugation by U_k gives $T^{-1}\boldsymbol{\beta}_{ik}^{-1}\widetilde{O}^{-1}\widetilde{O}\boldsymbol{\beta}_{ij} T\widetilde{O}\boldsymbol{\beta}_{jk} T = T^{-1}\boldsymbol{\beta}_{ik}^{-1}\boldsymbol{\beta}_{ij}(T\widetilde{O})\boldsymbol{\beta}_{jk} T$. Conjugation by $\boldsymbol{\beta}_{jk} T$ gives $\boldsymbol{\beta}_{jk}\boldsymbol{\beta}_{ik}^{-1}\boldsymbol{\beta}_{ij}(T\widetilde{O}) = \boldsymbol{\rho}_{ijk} T\widetilde{O}$, because head-diagonal matrices commute with one another; they need not commute with $T\widetilde{O}$. The last statement is Eq. (43) and Proposition 10. ∎

The routing factor and the head transfer therefore remain distinct. A vanishing routing defect, $\boldsymbol{\rho}_{ijk} = I$, leaves a holonomy similar to $T\widetilde{O}$, which in the exponential chart is e^{TO} and is generally not the identity. For softmax routing $\beta_{ij} = e^{s_{ij}}/Z_i$ the receiver normalizer cancels between β_{ij} and β_{ik}, and for Gaussian-KL routing

$$\log \rho_{ijk}^{(h)} = -\frac{D_{ij}^{(h)} + D_{jk}^{(h)} - D_{ik}^{(h)}}{\tau_h} + \log \frac{\pi_{ij}^{(h)}\pi_{jk}^{(h)}}{\pi_{ik}^{(h)}} - \log Z_j^{(h)}. \tag{102}$$

The leading term is the triangle defect of the routing costs. The Kullback–Leibler divergence obeys no triangle inequality, and here it compares a query law with a key law under different maps, so the defect can take either sign. With $T\widetilde{O} = I$, flatness on every strict triangle forces $\beta_{ij} = c_i/c_j$ for $0 < j < i$ with $c_i = \beta_{i0}$, as the case $k = 0$ shows; this is a thin set, and uniform causal attention, with $\rho_{ijk} = 1/(j+1)$, is not flat. The implementation reports path-weighted moments of $\log \rho_{ijk}^{(h)}$ during training and computes spectra of $\boldsymbol{\rho}_{ijk} T\widetilde{O}$ from checkpoints. Spectral summaries of this kind vanish at every unipotent holonomy (Proposition 34), so they are not complete curvature measures.

8.4 Latent link generators

The implemented links use one learned generator per layer, the same on every edge. Promoting the generators to latent variables with a normalized prior places their dynamics inside a variational free energy rather than in a postulated mechanics. This subsection states the model, its relation to the implemented chart, and its stationarity condition. It is not implemented, and nothing in it has been trained.

Index layers by $\ell = 0, \dots, L$ and write the state of token i at layer ℓ in reference coordinates as ζ_i^ℓ. Each causal edge from (j,ℓ) to $(i,\ell+1)$, $j \le i$, carries a generator $X_{ij}^\ell \in \mathfrak{gl}(K)$, and the local link is $U_i e^{X_{ij}^\ell} U_j^{-1}$. Two-step routes from (k,ℓ) to $(i,\ell+2)$ are compared with the route that first stays at k through the loops

$$W_{ijk}^\ell = \big(e^{X_{ik}^{\ell+1}} e^{X_{kk}^\ell}\big)^{-1} e^{X_{ij}^{\ell+1}} e^{X_{jk}^\ell}, \qquad k < j \le i, \tag{103}$$

which are time-like for $j = i$ and space-like for $j < i$. With the head projectors Π_h of Proposition 11, a transport noise $\bar{Q} \succ 0$, causal routing priors $\pi^{\ell,h}$, and token-type priors $p(\zeta_i^0)$, the latent-link model is

$$\begin{aligned} p(o,\zeta,J,X) &= p(X) \prod_i p(\zeta_i^0) \prod_{\ell=0}^{L-1} \prod_i \Big[\prod_h \pi_{i,J_i^{\ell,h}}^{\ell,h}\Big] \mathcal{N}\big(\zeta_i^{\ell+1}; m_i^\ell, \bar{Q}\big) \prod_i p(o_i \mid U_i \zeta_i^L), \\ m_i^\ell &= \sum_h e^{X_{iJ_i^{\ell,h}}^\ell} \Pi_h \zeta_{J_i^{\ell,h}}^\ell, \\ -\log p(X) &= \sum_{\ell,i,j \le i} \frac{\|X_{ij}^\ell - \bar{X}^\ell\|_F^2}{2\sigma_X^2} + \lambda_W \sum_{\ell,k<j\le i} w_{ijk}^\ell \Psi\big(W_{ijk}^\ell\big) + \text{const}, \end{aligned} \tag{104}$$

with anchor $\bar{X}^\ell$, width $\sigma_X > 0$, coupling $\lambda_W \ge 0$, loop weights $w_{ijk}^\ell = \bar{\pi}_{ij}^{\ell+1} \bar{\pi}_{jk}^\ell$ from the head-averaged generative routing prior, and the Frobenius defect $\Psi(W) = \frac{1}{2}\|W - I\|_F^2$. The prior is normalizable because the loop term is nonnegative, every other factor is a normalized conditional of a directed model, and no generative factor reads a recognition law.

Remark 33 (Static limit) *If $\bar{X}^\ell = OT$ for every layer and $\sigma_X \to 0$, every generator is pinned to the shared generator, and the mean of m_i^ℓ under routing weights β is $\sum_h \sum_j \beta_{ij}^{(h)} e^{OT} \Pi_h \xi_j = e^{OT} \sum_j T^{-1} \boldsymbol{\beta}_{ij} T \xi_j = \sum_j \widetilde{O} \boldsymbol{\beta}_{ij} T \xi_j$ by Eq. (41), which is the exponential-chart message of Proposition 10. The implemented exponential chart is therefore the zero-width limit of the latent model, and dynamic links are fluctuations around it.*

The curvature term must vanish exactly at flat configurations. On a compact group a class function, such as Wilson's plaquette density, can do this; on $\mathrm{GL}(K)$ with $K \ge 2$ no continuous class function can, because $\mathrm{GL}(K)$ then contains unipotent elements other than I.

Proposition 34 (Curvature measures on $\mathrm{GL}(K)$) *1. If $f : \mathrm{GL}(K,\mathbb{R}) \to \mathbb{R}$ is continuous and constant on conjugacy classes, then $f(u) = f(I)$ for every unipotent u.*

2. $\Psi(W) \geq 0$ *with equality exactly at* $W = I$*, and on* $\mathrm{O}(K)$*,* $\Psi(W) = K - \operatorname{Tr} W$*, Wilson's plaquette density.* $\Psi(W^{\ell}_{ijk})$ *is invariant under the passive site action. Under a reparametrization* $U_k \mapsto U_k h_k^{-1}$*,* $e^{X^{\ell}_{ab}} \mapsto h_a e^{X^{\ell}_{ab}} h_b^{-1}$*, which leaves every local link unchanged,* $W^{\ell}_{ijk} \mapsto h_k W^{\ell}_{ijk} h_k^{-1}$*, and* $\Psi(h_k W h_k^{-1}) = \Psi(W)$ *for every* W *exactly when* h_k *is a nonzero multiple of an orthogonal matrix.*

3. *For small generators,* $\Psi(W^{\ell}_{ijk}) = \frac{1}{2}\|X^{\ell+1}_{ij} + X^{\ell}_{jk} - X^{\ell+1}_{ik} - X^{\ell}_{kk}\|_F^2 + O(\|X\|^3)$*. For a time-like loop the argument is the depth difference of a cross-token generator corrected by the time-like generators at its ends, a discrete electric field.*

Proof (1) Write $u = RJR^{-1}$ with $J = I + N$ in real Jordan form, N nilpotent on the superdiagonal, and let $D_t = \operatorname{diag}(1, t^{-1}, \dots, t^{-(K-1)})$. Then $(D_t J D_t^{-1})_{p,p+1} = t J_{p,p+1}$, so $D_t J D_t^{-1} \to I$ as $t \to 0$, and $f(u) = f(D_t R^{-1} u R D_t^{-1}) \to f(I)$. (2) The zero set is a property of the norm, and for orthogonal W, $\|W - I\|_F^2 = 2(K - \operatorname{Tr} W)$. The loops are reference data, which gives passive invariance. Under the reparametrization each two-step route becomes $h_i(\cdots)h_k^{-1}$, so $W \mapsto h_k W h_k^{-1}$. If $h_k = cR$ with R orthogonal, $h_k(W - I)h_k^{-1} = R(W - I)R^{\top}$. Conversely, for nonzero u, v and small $t \neq 0$ the matrix $W = I + tuv^{\top}$ is invertible and $\|h_k(W - I)h_k^{-1}\|_F = |t||h_k u||h_k^{-\top} v|$, so invariance forces $|h_k u|/|u| = |v|/|h_k^{-\top} v|$ for all nonzero u, v; both sides then equal one constant $c > 0$, and h_k/c is orthogonal. (3) $W = (I - X^{\ell+1}_{ik} - X^{\ell}_{kk})(I + X^{\ell+1}_{ij} + X^{\ell}_{jk}) + O(\|X\|^2)$. ∎

Item 1 applies to every eigenvalue statistic, including the spectral summaries of the implemented triangle diagnostic, which all vanish on the shear $I + E_{12} \neq I$, whereas $\Psi(I + E_{12}) = \frac{1}{2}$. The Frobenius defect uses one structure that class functions lack, the reference metric, and that metric reduces the residual invariance of the curvature measure from $\mathrm{GL}(K)$ to the orthogonal group times scalars.

Proposition 35 (Link field equation) *Take the mean-field recognition law*

$$q(\zeta, J) = \prod_{\ell,i} \mathcal{N}(\zeta^{\ell}_i; \xi^{\ell}_i, C^{\ell}_i) \prod_{\ell,i,h} \beta^{\ell,h}_i(J^{\ell,h}_i)$$

for the states and labels and a point mass for the links, put $\bar{A} = \bar{Q}^{-1}$*, and let* $\mathbf{L}(X, E) = \int_0^1 e^{sX} E e^{(1-s)X} ds$ *be the Fréchet derivative of the matrix exponential. The free energy of the model in Eq.* (104) *is stationary in* X^{ℓ}_{ij} *exactly when*

$$\frac{\partial S_X}{\partial X^{\ell}_{ij}} = \mathbf{L}\big(X^{\ell\top}_{ij}, \mathcal{J}^{\ell}_{ij}\big), \qquad \mathcal{J}^{\ell}_{ij} = \sum_h \beta^{\ell,h}_{ij} \bar{A} \mathbb{E}_q\big[(\zeta^{\ell+1}_i - m^{\ell}_i)\zeta^{\ell\top}_j \;\big|\; J^{\ell,h}_i = j\big]\Pi^{\top}_h, \tag{105}$$

where S_X *is the negative logarithm of the link prior up to a constant. For one head the current is* $\mathcal{J}^{\ell}_{ij} = \beta^{\ell}_{ij}\bar{A}[(\xi^{\ell+1}_i - e^{X^{\ell}_{ij}}\xi^{\ell}_j)\xi^{\ell\top}_j - e^{X^{\ell}_{ij}} C^{\ell}_j]$.

Proof Only the transport factor of receiver i at layer ℓ and the link prior depend on X^{ℓ}_{ij}, and under the recognition law the transport term is $\frac{1}{2}\mathbb{E}_q\|\zeta^{\ell+1}_i - m^{\ell}_i\|^2_{\bar{A}}$ up to X-free terms. The mean m^{ℓ}_i contains $e^{X^{\ell}_{ij}}\Pi_h\zeta^{\ell}_j$ exactly on the event $J^{\ell,h}_i = j$, so the derivative of the transport term in the group element is $-\mathcal{J}^{\ell}_{ij}$. The adjoint of $E \mapsto \mathbf{L}(X, E)$ in the Frobenius inner

product is $E \mapsto \mathbf{L}(X^\top, E)$ (Higham, 2008), which gives Eq. (105). For one head, mean-field independence of $\zeta_i^{\ell+1}$ and ζ_j^ℓ gives $\mathbb{E}[(\zeta_i^{\ell+1} - e^X \zeta_j^\ell)\zeta_j^{\ell\top}] = \xi_i^{\ell+1}\xi_j^{\ell\top} - e^X(C_j^\ell + \xi_j^\ell \xi_j^{\ell\top})$. ■

The current is a precision-weighted prediction error times the source state, minus a term from the source's uncertainty. It has the form of the delta rule that updates fast-weight memories (Schmidhuber, 1992), the same rule that appears when linear attention is read as a fast-weight programmer (Schlag et al., 2021); here it is derived from a free energy and filtered through the derivative of the exponential chart. The link prior is necessary for a well-posed per-context problem. Without it, for one head with $\beta_{ij}^\ell > 0$ and $K \geq 2$, the generator enters only through the strictly convex quadratic $\mathcal{T}(G) = \frac{1}{2}\beta_{ij}^\ell \mathbb{E}_q \|\zeta_i^{\ell+1} - G\zeta_j^\ell\|_{\bar{A}}^2$ evaluated at $G = e^{X_{ij}^\ell}$. Its unique stationary point $G^\star = \xi_i^{\ell+1}\xi_j^{\ell\top}(C_j^\ell + \xi_j^\ell \xi_j^{\ell\top})^{-1}$ has rank at most one and lies outside the exponential chart, so by Proposition 35 a generator can be stationary only where $\mathbf{L}(X^\top, \cdot)$ is singular, at a critical point of the exponential map. Whether the infimum over the chart is approached at $G^\star$ depends on the sign of $\xi_j^{\ell\top}(C_j^\ell)^{-1}\xi_i^{\ell+1}$ (Proposition 36 in Appendix G).

For one head, linearized around a flat anchor for small links and sources, the field equation becomes a screened, massive discrete Maxwell equation of Euclidean signature on the depth-by-token complex, in which the anchor is a bare mass, the loop term is a curl-curl operator, and the time-like loops supply the electric term, as time-like plaquettes do in lattice gauge theory (Kogut and Susskind, 1975; Creutz, 1983). Inference in the links is first-order relaxation with the Fisher information of the transported law as friction, and inertia appears only for the most probable path jointly over depth. Directed causal matter induces no action on the links, because every factor of Eq. (104) other than the prior is a normalized conditional. Integrating out undirected Gaussian matter instead induces one half of the log-determinant of the matter precision, whose hopping expansion for orthogonal links and isotropic couplings begins with Wilson's triangle term, in the manner of induced gauge theories (Kazakov and Migdal, 1993); the loop prior of Eq. (104) has the form that such matter would induce. Appendix G proves these linearized, dynamical, and induced-action statements (Propositions 37 to 41). A first implementation would keep $\bar{X}^\ell = OT$ as the learned anchor and infer per-context deviations by a few Fisher-preconditioned steps or by one linearized solve; the all-edge cost $O(LN^2K^3)$ per step points to restricted supports, low-rank deviations, or amortization.

9 Language-Model Realization and Evaluation

Two records are reported. The first evaluates the causal residual architecture of Sections 2.3 and 2.4 at $K = 256$ with four layers; it predates the chart selector, applies the stored content maps directly, which is the affine chart, uses the residual $r = 1$, and passes the covariance through. The second compares the affine, first-order, and exponential charts, and the channel-likelihood receiver, in matched two-layer runs at $K = 20$ (Section 9.3). The saved configuration and evaluation protocol of each run define its empirical claim. The theoretical results apply to full Gaussian states; the stored covariance representation of each experiment is stated below.

9.1 Forward computation

The software selectors for the path evaluated at $K = 256$ are `projected_filtering` and `predictive_content`. The former names projected Gaussian routing; the latter selects direct value/output aggregation. This combination performs no iterative belief update. The route requires `gradient_mode=filtering` and inclusion of attention entropy for admission; neither setting is read by the predictive map, which performs no inner descent, so all source, score, and content dependencies remain available to outer cross-entropy differentiation.

Algorithm 1 specifies the forward and training operations. For the reported experiment, $K = 256$, $H = B = 8$, $d_b = r_h = s_h = 32$, $L = 4$, and $\tau_h = \sqrt{32}$. The implementation fixes $H = B$ and $r_h = s_h = d_b$ and computes $\tau_h = \kappa_\beta \sqrt{d_b}$ with $\kappa_\beta = 1$, so $\tau_h = \sqrt{r_h}$ holds by construction; independent head counts and ranks are not implemented. The numerical KL cost is bounded to $[0, 8K]$ before routing. The categorical optimization remains exact for this bounded cost; its identification with the unmodified Gaussian KL requires an inactive bound. Capping was checked on one 128-token window at each of 200 logged training steps and 10 evaluations, with no capped score, and on sampled windows the largest causal cost is about 300, well below $8K = 2048$. The bounded cost is nonnegative, so the evaluated routing is also the exact label posterior of the binary-record joint of Theorem 20 for that cost. The theoretical comparison in Section 5 is stated for uncapped costs.

9.2 Architecture and training

The WikiText-103 (Merity et al., 2017) experiment instantiates Section 2.3 with $K = 256$, eight 32-dimensional gauge blocks, four layers, and seed 6. Each layer has independent comparison and content maps. The configuration uses `projected_filtering` with full input scope, `predictive_content` readout, full output scope, and the additive attention residual. The group parameterization remains block-diagonal and vertex-factorized, with learned token and positional frames. Each block also contains affine LayerNorm and a coordinate GELU MLP with hidden width 2048, expansion factor eight, zero dropout, and covariance passthrough. These are the combined methods used in the recorded run; the saved results do not isolate the contribution of each choice.

The stored beliefs are diagonal Gaussians, but each role projection retains its full 32×32 reduced covariance. Diagonal storage is a partial gauge fixing: within each frame block every positive-definite covariance is congruent to a diagonal one and the frame absorbs the congruence, so with block-diagonal frames the realized reference covariances C_i are exactly the block-diagonal positive-definite matrices, and cross-block reference covariance is not representable. The operative router is Gaussian KL. Although the launch configuration contains `attention_score=squared_distance`, `mean_only=False` selects the Gaussian-KL path, as the saved resolved-operator metadata confirms. The causal ALiBi prior includes the current position and adds the headwise logit bias $-2^{-h}(i-j)$ for $j \le i$, $h = 1, \dots, 8$. The routing temperature is $\sqrt{32}$. Predictive content uses one readout per layer; the configured inference learning rates and belief trust bounds do not scale this update. The covariance and frame are held fixed through the stack, while outer cross-entropy training updates the learned tables, frames, role maps, MLPs, normalization, and decoder parameters through the forward graph. The decoder term $\kappa_{\text{uni}} \log p_{\text{uni}}$ uses the fixed coefficient $\kappa_{\text{uni}} = 1$ and

Algorithm 1 Causal Gaussian attention and supervised predictive learning

Require: Input tokens $x_0, \dots, x_{N-1}$ and, during training, targets $y_i = x_{i+1}$ for $0 \le i < N$, vocabulary Gaussian and frame tables, layer maps $S^{(\ell)}_{Q,h}, S^{(\ell)}_{K,h}, T^{(\ell)}_h, O^{(\ell)}_h$, a chart, and decoder parameters.

Initialize $(\mu_i^{(0)}, \Sigma_i^{(0)})$ from the vocabulary prior indexed by x_i; realize the ordered token–position frame $U_i = \exp(\phi_{x_i}) \exp(Y_i)$.

For each layer, evaluate the effective receiving maps $\widetilde{O}^{(\ell)}_h$ of the chart, Eq. (43); in the affine chart of the $K = 256$ run, $\widetilde{O}^{(\ell)}_h = O^{(\ell)}_h$.

Set $\log \pi^{(h)}_{ij} = -\alpha_h(i-j)$ up to a row constant for $j \le i$ and $\pi^{(h)}_{ij} = 0$ otherwise, with positive headwise slopes α_h and temperatures τ_h.

for $\ell = 0, \dots, L-1$ **do**

  Form $\xi_i = U_i^{-1}\mu_i^{(\ell)}$ and $C_i = U_i^{-1}\Sigma_i^{(\ell)}U_i^{-\top}$.

  for $h = 1, \dots, H$ **do**

    Form the query and key Gaussian pushforwards under $S^{(\ell)}_{Q,h}$ and $S^{(\ell)}_{K,h}$; denote their KL divergence by $D^{(\ell,h)}_{ij}$.

    Apply the numerical cost bound stated in the experimental methods to obtain $\bar{D}^{(\ell,h)}_{ij}$, then set $\beta^{(\ell,h)}_{ij} = \mathrm{softmax}_j\big(\log \pi^{(h)}_{ij} - \bar{D}^{(\ell,h)}_{ij}/\tau_h\big)$.

    Compute $z_i^{(\ell,h)} = \sum_j \beta^{(\ell,h)}_{ij} T^{(\ell)}_h \xi_j$.

  end for

  Compute $m_i^{(\ell)} = U_i \sum_h \widetilde{O}^{(\ell)}_h z_i^{(\ell,h)}$ and $\mu_i^{\mathrm{att},(\ell)} = r\mu_i^{(\ell)} + m_i^{(\ell)}$, with $r = 1$ in the $K = 256$ run.

  Set $u_i^{(\ell)} = \mathrm{LayerNorm}_{\gamma,b}(\mu_i^{\mathrm{att},(\ell)})$.

  Set $\mu_i^{(\ell+1)} = u_i^{(\ell)} + W_2^{(\ell)}\,\mathrm{GELU}(W_1^{(\ell)}u_i^{(\ell)} + b_1^{(\ell)}) + b_2^{(\ell)}$, and set $\Sigma_i^{(\ell+1)} = \Sigma_i^{(\ell)}$ under passthrough or the moment-matched covariance of Theorem 20 under moment matching.

end for

Compute $p(y_i \mid x_{\le i}) = \mathrm{softmax}\big(W_{\mathrm{out}}\mu_i^{(L)} + b_{\mathrm{out}} + \kappa_{\mathrm{uni}} \log p_{\mathrm{uni}}\big)_{y_i}$.

if training **then**

  Backpropagate the mean next-token cross-entropy through all layers, routing weights, content maps, frames, and initial vocabulary parameters; update trainable parameters with AdamW (Loshchilov and Hutter, 2019).

end if

the add-one-smoothed unigram frequency of the training stream; because b_{out} is learned without decay, this term acts as an initialization offset for b_{out}.

The stored covariance table holds log-variances initialized at $\log 15$, so the initial variance is 15. Decoupled weight decay 0.04 on this table carried the variances to 1.045 ± 0.054 per coordinate at the final step, with 84.6% of entries within 10% of unit variance; the run's covariance-diversity check recorded a coefficient of variation of $\mathrm{Tr}\,\Sigma$ of 0.047 against its threshold of 0.10. Because covariances and frames are fixed through the stack, $C_i = U_i^{-1}\Sigma_{p,x_i}U_i^{-\top}$ depends only on token type and position and is close to $1.04(U_i^\top U_i)^{-1}$, with a median relative Frobenius gap of 7.6% on sampled token-position pairs: the covariance

channel of the router is carried by the frames. On sampled windows of 128 tokens, replacing every routing covariance C_i by the identity raises cross-entropy by 0.325 nats on average over 8 windows (range 0.11 to 0.57), whereas replacing only the stored variances by the identity raises it by 0.024 ± 0.024 nats over 24 windows. The decay also acted as a hidden temperature for the Mahalanobis term, which a common rescaling of all covariances by c rescales by $1/c$; over training it sharpened that term about 14-fold. The experiment therefore does not test learned per-token uncertainty.

The run records 456,654,162 parameters, including 1,048,576 relational-map parameters and 4,203,520 MLP parameters. The token-frame table has $50{,}257 \times 8{,}192 = 411{,}705{,}344$ parameters, approximately 90.16% of the total. Per-type tables (token frames, prior means and log-variances, and decoder weights and biases) hold 98.6% of the parameters, the positional frames hold 1,048,576, and the context-processing parameters (interface maps, MLPs, and the shared normalization) number 5,252,608. This is capacity accounting; sparse table access does not make the count equivalent to dense-transformer FLOPs. The run completed 240,000 accepted updates and 491,520,000 training-token presentations over about 116.8 million unique training tokens, or about 4.21 passes. The saved environment identifies an NVIDIA GeForce RTX 5090, PyTorch `2.10.0.dev20251210+cu128`, and bfloat16 automatic mixed precision, which applies to the MLP matrix products; frame exponentials, Gaussian scores, content aggregation, layer normalization ($\epsilon = 10^{-6}$), decoding, and cross-entropy run in float32, and GELU is the exact error-function form.

Evaluation uses context length 128 and stride 32, excluding overlapping positions from repeated scoring. The test result covers 280,541 next-token targets from 280,542 tokens; validation covers 245,020 targets. The cache filename identifies tiktoken/GPT-2 tokenization with vocabulary size 50,257, but no verified tokenizer manifest is available. Perplexities refer to this saved protocol and are not directly comparable to conventional word-level WikiText-103 results. Table 4 lists the recorded perplexities.

Evaluation	Step	Perplexity
Best recorded validation	216,000	29.6088
Final validation	240,000	30.0100
Test, final weights	240,000	31.0389
Test, best-validation weights	216,000	pending
Training-split sample, evaluation protocol	240,000	pending

Table 4: Single-seed Gaussian attention language-model experiment. Test evaluation used the final weights, not the checkpoint with the lowest recorded validation perplexity. No variability estimate or causal ablation is supplied by this run.

The test cross-entropy is 3.43524149 nats per scored token. Validation cross-entropy reached its minimum at step 216,000 and rose by 0.0135 nats by the final step; the test split has not yet been evaluated at the best-validation checkpoint. The result is stored in the run directory `31.04_wikitext-103_K256_4-layer_noHM`, with configuration and resolved methods in `config.json`, evaluation in `test_results.json`, and source/data identity in `provenance.json`. The provenance records a dirty source tree and a change in source

identity during the run. The starting Git revision is `4e137ec88f45` and the final recorded revision is `965718d88cf7`; these revisions alone cannot identify the executed dirty-source bytes. The result therefore has incomplete source reproducibility.

A saved intervention probe on the first validation batch, 16 overlapping windows that score 608 consecutive targets, reports cross-entropy 3.0431 for the trained route (against 3.4015 on the full validation split), 4.1923 when the routing energy is removed, and 5.1182 when the positional prior is flattened on causal support. These frozen-model batch interventions support reliance on both content-dependent routing and the positional prior in that batch. They are not corpus-wide test results, retraining ablations, or evidence that any individual projection, the residual, or the MLP is necessary. These interventions measure prediction error, not variational descent.

In the trained model the explicit attention residual is numerically inert. On 24 sampled windows of 128 tokens, the per-token ratio $\|m_i\|/\|\mu_i\|$ has median 2.1×10^4, 5.9×10^3, 5.3×10^3, and 1.8×10^3 at layers 0 to 3, with minimum 693. The residual changes the normalized state by at most 1.8×10^{-3} in relative terms, and deleting it at every layer changes cross-entropy by less than 4×10^{-4} nats on every window. The relational maps train without weight decay and the layer normalization is invariant to the scale of its input, so the largest singular values of $O_h T_h$ grew from 1 at initialization to between 2.6×10^3 and 2.6×10^4 across heads and layers. At the final weights each layer therefore acts as $u_i = \mathrm{LN}(m_i)$ up to a relative 2×10^{-3}. The receiving token reaches the next layer through its frame and its routing self-weight, whose mean lies between 0.07 and 0.15 across layers and windows, rather than through the additive state. The role of the residual earlier in training was not measured.

Frozen intervention on every validation target	CE	ΔCE	95% interval
None (final weights)	pending	0	pending
Routing costs removed; positional prior retained	pending	pending	pending
Positional prior flattened on the causal support	pending	pending	pending
Residual term removed at every layer, $\mu_i + m_i \mapsto m_i$	pending	pending	pending
Stored covariances $\Sigma_{p,v}$ replaced by the identity	pending	pending	pending
All routing covariances C_i replaced by the identity	pending	pending	pending
Token frames replaced by the identity in every role	pending	pending	pending
Positional frames replaced by the identity in every role	pending	pending	pending
Frames removed from the input moments only (F1)	pending	pending	pending
Frames removed from the routing covariance only (F2)	pending	pending	pending
Frames removed from the normalization and MLP conjugation only (F3)	pending	pending	pending
Frames removed from the decoder only (F4)	pending	pending	pending

Table 5: Reserved frozen interventions on the complete validation split. Each row modifies one input of the trained predictor and records the change in cross-entropy, with intervals from an article-level bootstrap. The role-wise frame rows follow Figure 2. These are properties of one trained model; they are not retraining ablations and do not establish that any component is necessary.

Saved per-layer probes from one 128-token test window report minimum raw comparison-map singular values from 6.64 to 12.32, maximum projected-covariance condition numbers from approximately 88.7 to 550.5, all attained by query covariances, and zero recorded covariance repairs or capped scores. Both describe the numerical representative reached by the optimizer rather than the learned routing. The redundancy $(S_{Q,h}, S_{K,h}) \mapsto (AS_{Q,h}, AS_{K,h})$ of Theorem 14 changes them while leaving every cost unchanged, and the raw maps, which carry no weight decay, grew by factors of about 40 to 54 in norm along this direction during training. The basis-independent quantities are the generalized eigenvalues of the pairs $(V_{Q,i}, V_{K,j})$ and the relative gains, the square roots of the eigenvalues of $(S_{K,h}S_{K,h}^\top)^{-1}S_{Q,h}S_{Q,h}^\top$, which lie between 0.11 and 5.35. The mathematical full-row-rank condition does not itself guarantee conditioning throughout learned optimization.

PLACEHOLDER M4

Panel A: for each layer and head, the variance of each routing-logit component and its covariance with the total logit: whitened cross term $\mathsf{q}_i^\top \mathsf{k}_j/\tau_h$, key-norm term, source-dependent trace and log-determinant terms of Eq. (49), and positional prior.

Panel B: after whitening by the pooled projected key covariance, the distribution of ε_i from Eq. (56) and of the rowwise ℓ_1 distance between Gaussian and dot-product routing, with the bound of Eq. (57).

Panel C: dispersion of the projected key covariances $V_{K,j}$ across sources, and the share of that dispersion induced by the frames rather than by the stored covariances.

TO COLLECT: validation-set score components or projected moments for all layers and heads, with sampling coverage.

Figure 3: Reserved diagnostics for the conditions of Proposition 17 in the trained model. Panel A attributes the routing logits to their Gaussian components; Panels B and C measure the distance of the trained model from the constant-norm and shared-covariance hypotheses. The figure characterizes the realized operator. Whether covariance-aware routing improves prediction requires the retraining comparisons of Table 8.

The result supports feasibility of the combined architecture under the stated protocol.

9.3 Link charts and the channel receiver at small width

Four further runs compare the three charts of Eq. (43) and the channel receiver of Section 7 under one configuration. Each has $K = 20$, two heads of width 10 with block-diagonal $\mathrm{GL}(10) \times \mathrm{GL}(10)$ frames, two layers, residual coefficient $r = 0$, moment-matched covariance with floor $\epsilon_Q = 10^{-4}$, local-diagonal storage, affine LayerNorm followed by a coordinate GELU MLP of expansion eight, and the linear decoder. Positions enter through rotary rotations of the routing frame with base 100, not applied to values; there is no learned positional frame, and the source prior is uniform on the causal support. Training used seed 6, the same data order in every arm, 15,000 AdamW updates of batch 64 at context 128,

about 1.05 passes over the training tokens, and bfloat16 automatic mixed precision. Every arm starts from the same effective content maps, since the block-projector initialization $T = I$ with $O = I$, 0, and 0 in the affine, first-order, and exponential charts gives $\widetilde{O} = I$ in each. Evaluation follows the protocol of Section 9.2, with context 128, stride 32, and 280,541 scored test targets. Each model has 13,133,478 parameters, of which 16,401 lie outside the per-type tables and 3,200 in the comparison and content maps.

Arm	Content update	Test CE	Test PPL	Validation PPL	Time (s)
Affine, $\widetilde{O} = O$	predictive mean	4.80216	121.77	121.41	1344
First order, $\widetilde{O} = T^{-1} + O$	predictive mean	4.80390	121.98	122.38	1372
Exponential, $\widetilde{O} = e^{O}T^{-1}$	predictive mean	4.80016	121.53	121.69	1464
Exponential, $\widetilde{O} = e^{O}T^{-1}$	channel receiver	4.82773	124.93	124.98	1677

Table 6: Matched single-seed runs at $K = 20$ with two layers. Cross-entropy (CE) is in nats per scored token, perplexity (PPL) is its exponential, validation values are at the final step, which was also the best recorded validation step in every arm, and time is wall-clock training time on one RTX 5090. The receiver arm also holds the covariance at its encoding value instead of moment matching it.

The three predictive arms differ by at most 0.0037 nats per token on the test split and 0.0080 nats on the validation split, and their order differs between the splits: the exponential arm has the lowest test cross-entropy and the affine arm the lowest validation cross-entropy. Each arm is a single seed. The spread lies below the seed-to-seed variability previously estimated for $K = 20$ configurations of this model, a standard deviation of about 1.5 perplexity points near perplexity 120, or about 2.1 points for the difference of two single-seed runs; that estimate was not measured on these arms. The equality of held-out loss is the outcome predicted when optimization reaches comparable solutions, because the three charts represent the same tiled mean maps up to the real-logarithm domain (Section 3.6). The comparison therefore probes optimization geometry and domain, not expressivity. [**Data pending:** paired replicate seeds of all four arms, with seed-to-seed standard deviations measured on these configurations.]

The trained links differ in the way the charts predict (Table 7). The computed determinant of the layer-0 transfer of the first-order arm was negative at the snapshots of steps 1000 to 3000 and positive from step 4500 on, so the path of that link in parameter space crossed the singular set at least twice, which an affine parametrization permits. The exponential arms keep a positive determinant by construction, and their smallest singular value stayed above 0.11 after step 1000, against 8.7×10^{-3} and 8.5×10^{-3} at layer 0 of the affine and first-order arms. The trained generators have spectral norms between 1.9 and 3.0, where the first-order polynomial of the generator differs from its exponential by 33 to 58 percent in spectral norm, so the first-order arm is a different model rather than an approximation of the exponential arm. These singular values and condition numbers are properties of four trained models at $K = 20$; the structural statements they illustrate are Propositions 2 and 6 and the truncation argument of Section 3.6, which hold at every width.

Arm	Layer	Sign changes	$\min \sigma_{\min}(M)$	$\kappa(M)$	$\|X\|_2$	Gap
Affine	0	0	8.7×10^{-3}	677	–	–
Affine	1	0	0.110	51	–	–
First order	0	2	8.5×10^{-3}	79	–	–
First order	1	0	0.087	44	–	–
Exponential	0	0	0.144	34	2.67	0.58
Exponential	1	0	0.231	12	1.91	0.33
Exponential, receiver	0	0	0.115	42	3.00	0.49
Exponential, receiver	1	0	0.122	12	2.50	0.55

Table 7: Geometry of the trained content links of the runs in Table 6. $M = \widetilde{O}T$ is the stacked $K \times K$ transfer, the reference operator of an edge on which both heads select the same source with unit weight. Sign changes of $\det M$ are counted over twelve saved snapshots from step 0 to step 15,000, the minimum singular value is taken over the snapshots from step 1000 on, and the condition number $\kappa(M)$ is at the final step. For the exponential arms $X = OT$ is the final trained generator, and the gap is $\|e^X - I - X\|_2 / \|e^X\|_2$, the relative distance of e^X from its first-order polynomial. Values were computed in float64 from the saved factors.

The receiver arm replaces the predictive mean by the closed-form channel-likelihood update of Proposition 22, with tied measurements $\widetilde{O}_h^\top$ in the exponential chart, unit damping, channel floor $Q_h^{\mathrm{ch}} = I$, routing weight $\lambda_\beta = 1$, and the profiled per-coordinate coupling with $b_0 = c_0 = 1$. Its covariance is held at the encoding value and its layer prior is the incoming state, so by Section 7.3 each layer applies the Kalman update of Eq. (83), and the profiled coupling stays at its starting value c_0/b_0. The test cross-entropy of this arm is 0.0276 nats above that of the exponential predictive arm. The two arms differ in the content update and in the covariance contract, so the difference does not isolate the receiver, and as a single-seed difference it is about 1.6 times the standard deviation of a two-run difference estimated above.

These runs are small. They have two layers and two heads at $K = 20$, about one pass over the training tokens, and a positional and residual configuration that differs from the $K = 256$ run. Each run saved an archive of the executed source bytes, but the source trees were dirty, and two runs recorded a change of Git identity during training. No statement about larger models, other seeds, or longer training follows from them.

10 Discussion

10.1 Contribution and limits

The mathematical contribution is a specified full-Gaussian routing and content construction whose communication is carried by group-valued links, together with the factorization of those links into the selective channels of attention. Reference coordinates permit full-input role measurements, exact reduced Gaussian KL comparisons, and content channels with an explicit passive covariance law. Separate comparison and content maps are familiar from standard attention; their roles here are defined inside the framed Gaussian model. The

Model arm	Seeds	Parameters	Test CE	Test PPL	GPU-hours
Retraining arms at the evaluated configuration and token budget					
Evaluated: learned frames, decayed stored covariance	pending	pending	pending	pending	pending
Learned frames, stored covariance without decay	pending	pending	pending	pending	pending
Identity frames, decayed stored covariance	pending	pending	pending	pending	pending
Identity frames, stored covariance without decay	pending	pending	pending	pending	pending
Identity frames, per-type additive table of equal size	pending	pending	pending	pending	pending
Attention residual removed	pending	pending	pending	pending	pending
First-order content chart	pending	pending	pending	pending	pending
Exponential content chart	pending	pending	pending	pending	pending
Exponential chart with the channel receiver	pending	pending	pending	pending	pending
LayerNorm and MLP in reference coordinates	pending	pending	pending	pending	pending
Intertwiner-constrained nonlinear block	pending	pending	pending	pending	pending
MLP removed	pending	pending	pending	pending	pending
MLP replaced by Gaussian template routing	pending	pending	pending	pending	pending
Conventional transformers: same tokenizer, context, stride, and budget					
Matched context-processing parameters	pending	pending	pending	pending	pending
Matched measured compute	pending	pending	pending	pending	pending
Per-type table of the frame table's size	pending	pending	pending	pending	pending
Word-level protocol, every target scored					
Gaussian attention	pending	pending	pending	pending	pending
Conventional transformer, matched	pending	pending	pending	pending	pending

Table 8: Reserved main results under one declared protocol. Each retraining arm changes one component of the evaluated realization; the two reference-coordinate and intertwiner arms make the complete stack frame-covariant. Report the mean and standard deviation over at least three replicates that vary both the model seed and the data-order seed, paired across arms, with per-seed values in Table 11. Give parameters as context-processing (non-embedding) and total counts, since per-type tables hold most of the storage. Word-level rows permit comparison with published WikiText-103 results and are not compared with the BPE rows. Cells remain pending until each evaluation is complete.

PLACEHOLDER M3

Panel A: held-out cross-entropy versus training-token presentations for every arm of Table 8, one curve per seed.

Panel B: the same curves against measured device-hours.

Panel C: held-out minus training cross-entropy, with the training term evaluated on a fixed training-split sample under the evaluation protocol.

TO COLLECT: per-seed validation curves at a common cadence, token counts, passes over the unique training tokens, synchronized wall-clock time, and the checkpoint-selection rule.

Figure 4: Reserved learning curves for the arms of Table 8. Panels A and B separate data efficiency from compute efficiency. Panel C records the generalization gap on one protocol, which bears on interpretation because per-type tables hold most of the parameters.

independent-content joint establishes a consistent interpretation while leaving the content maps underdetermined. It therefore cannot, by itself, explain their empirical success or claim priority for Gaussian-KL or gauge-equivariant attention.

The exponential construction fixes the group domain, the endpoint law, and the order structure of the routed message, and it makes inverse-edge holonomy available on causal graphs. With one shared generator per layer it changes the parametrization, center, and domain of the content maps rather than the family of tiled mean maps, so it is not expected by itself to change held-out loss, and the matched runs at $K = 20$ agree with that expectation within their single-seed resolution. What the construction does change is measurable in the trained links: the exponential chart cannot cross the singular set, and in those runs its transfers stayed better conditioned than those of the affine and first-order charts. Whether the group geometry improves optimization at the target scale, or becomes expressive only once the generators vary with the edge, is not settled by these results.

The covariance of Theorem 14 holds for any operator that reads the states through reference moments, so it does not by itself constrain the learned maps; the coordinate normalization, MLP, and decoder are where the learned frames act (Section 4). Covariance passthrough is not propagation of predictive uncertainty; Eq. (69) gives the moment-matched alternative. The source-selection objective is exact for its declared costs, and routing and content have exact probabilistic readings. The channel receiver supplies an evidential alternative whose update is the exact power posterior of declared likelihoods, but it is a frozen-routing inference within one layer, and no common-free-energy descent statement is established for layer composition.

The $K = 256$ run supplies no matched baseline or variability estimate, and the $K = 20$ chart runs are single-seed comparisons at small width. Neither can establish parameter efficiency, sample efficiency, benefit from learned frames or from a chart, or causal necessity of covariance, the residual, or the MLP. The frame table dominates parameter count, and the short-context, tokenizer-specific evaluation must be kept explicit in comparisons. Dirty

and changing source provenance limits reconstruction of the executed model; a Git SHA alone does not recover its bytes.

10.2 Relation to a complete variational model

The general gauge-theoretic variational free energy (VFE) construction supplies a broader probabilistic setting in which normalized agent and observation kernels define one population joint and one exact evidence bound. Its latent-source construction yields categorical Gibbs routing under explicit likelihood and recognition restrictions; Appendix E states this connection. The predictor already has a normalized autoregressive likelihood. For the attention sublayer, a normalized latent model exists in which routing is the exact label posterior and the content-plus-residual update is the exact predictive mean (Theorem 20), and the linear decoder is a Gibbs readout (Section 6.3). The latent-link model of Section 8.4 supplies a normalized link sector for such a joint, with the implemented chart as its zero-width limit. Unresolved are the normalization and MLP maps, the collapse of the predictive mixture to a point state between layers, and the relation of cross-entropy training to that model's evidence. This distinction preserves the exact attention result without attributing the whole architecture to the Free Energy Principle.

10.3 Selective communication and information coding

The sending and receiving interfaces describe selected use of a framed state. They impose no claim that a learned direction is a measured cognitive faculty. A connection with Shannon-style coding remains an open question. The Gaussian channel of Proposition 27 provides one possible channel law; a coding formulation additionally requires a message model, an input or resource constraint, and a specified information measure. Its relation to the predictive mean operator remains to be established. Subspace dimension and low matrix rank alone do not supply a channel capacity.

10.4 Experiments needed to separate the contributions

The next comparison should freeze source bytes, tokenizer and data manifests, checkpoint selection, and evaluation masks. A conventional projected-attention baseline should share tokenizer, context, depth, MLP, training tokens, and a stated parameter and compute budget. Paired retraining comparisons can then isolate covariance-aware routing from a matched mean-only or dot-product route, and learned frames from an identity-frame model with its parameterization change reported. The mean-only and dot-product route used for that control was removed from the reference implementation after the reported run and must be restored in a tagged release; a capacity control with an additive per-type table of equal size is implemented. Residual and MLP necessity require their own controlled comparisons. Multiple paired seeds are needed to estimate variability; the existing frozen-batch probes do not replace retraining experiments.

Block-diagonal frames and cross-block communication are distinct design choices. Full-input S_Q, S_K, T and full-output O already communicate across frame blocks when the optional cross-block head-mixing layer is disabled, as here. A future transport-group comparison should hold these communication maps fixed. Likewise, projected-key norm measure-

ments can test the conditional dot-product reduction without assuming that LayerNorm enforces it. These are bounded tests of the attention mechanism.

For the links, the next comparisons are paired multi-seed runs of the three charts and of the channel receiver at the configuration of the $K = 256$ run, with the receiver compared against a predictive arm that shares its covariance contract. The trained triangle spectra of Proposition 32, the generator norms, and the singular values of the transfers should be recorded across training, since they are the quantities on which the charts differ. A group-constrained or Fisher-preconditioned update of the generator, and per-edge latent generators started from the shared anchor of Remark 33, are the extensions in which the group geometry could change the represented maps rather than only their parametrization.

11 Conclusion

Gaussian-KL routing between framed Gaussian states is the exact minimizer of an entropy-regularized source-selection objective and, at every positive temperature, the exact posterior of a source label under a normalized binary compatibility record; at unit temperature it is also the coordinate minimizer of a local Gaussian mixture KL. In dual coordinates it is a dot product between query expectation parameters and key natural parameters. With shared projected key covariance it equals whitened dot-product routing with a key-norm penalty that neither LayerNorm nor high-dimensional concentration removes; with source-dependent covariance it adds a per-source metric and pair biases. Content aggregation with its residual is the exact predictive mean of a normalized mixture-of-channels kernel, and read as evidence the same channels give a receiver whose exact minimizer is a Gaussian power posterior in closed form. The attention operator factors through frame-invariant reference moments, so its covariance constrains no learned map, and the frames affect computation only through the state parametrization and the frame-dependent normalization, MLP, and decoder.

In this construction the communication links are group valued. The link $\Theta_{ij} = U_i e^{X_{ij}} U_j^{-1}$ is invertible for every generator and obeys the endpoint law of a lattice link variable, and for the low-rank generator built from the value and output maps it splits exactly into the flat reference transport and a selective channel whose first-order part is the projected matrix link, a covariant composite of the bilinear kind distinguished by Wetterich (2014, Sec. I). With tiled heads and routing outside the exponent the routed message resums into the receiving maps $e^{OT} T^{-1}$, whose orders are a head-routed flat transport, the standard value path, and cross-head mixing. The affine, first-order, and exponential charts on the same stored maps represent the same tiled mean maps up to the real-logarithm domain, and only the exponential chart stays in $\mathrm{GL}^+(K)$. The reference transport is pure gauge and the causal support contains no closed walk through distinct positions, but invertible links supply inverse-edge triangle holonomies, and a shared generator has the same nonflat holonomy on every triangle. Lattice-gauge degrees of freedom require generators that vary with the edge together with an action for them; the latent-link model supplies a normalized prior of that kind, with the implemented chart as its zero-width limit, and it is not implemented here.

The four-layer model records final-weight test perplexity 31.04 under its saved protocol. This is a feasibility result with one seed and incomplete source reproducibility, in which

the attention residual is numerically inert and the stored variances collapsed to near unity. In matched single-seed runs at $K = 20$ the three charts reach test cross-entropies within 0.004 nats per token of one another, consistent with their common function class, while the exponential chart keeps the trained transfers in the identity component and better conditioned. A controlled baseline, paired seeds, and component comparisons are still needed to establish an empirical advantage. A complete generative interpretation would require an explicit joint law, observation model, and variational update that recover the whole predictor. [**Data pending:** replace with the final matched comparison of Table 8 and the frozen interventions of Table 5.]

Code availability

The reference implementation is available at `https://github.com/cdenn016/V3_Transformer`. The mean-only and dot-product control route proposed in Section 10.4 was removed from that repository after the reported run, and the current configuration loader rejects the archived configuration; a tagged release containing the evaluated route, the control route, and a loader for the archived configuration is required. [**Data pending:** tag, commit, and archive identifier of that release.] Section 9.2 identifies the saved configuration, evaluation record, and source-provenance limits of the reported experiment. A public repository does not by itself recover the modified source bytes used in that run. The content-link charts and the channel receiver are implemented in VFE 3.0 (`https://github.com/cdenn016/VFE_3.0`), selected by `content_link_chart` and by `attention_mode=projected_gradient` with `relational_receiver_law=channel_likelihood` and `relational_receiver_update=closed_form`; each run of Section 9.3 saved an archive of its executed source. The float64 identity checks of Appendix H and the measurement script for Table 7 are distributed with the manuscript source.

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Appendix A. The Smooth-Bundle Construction

This appendix gives the construction behind the frames of Section 2.1. Figure 5 gives the geometric picture.

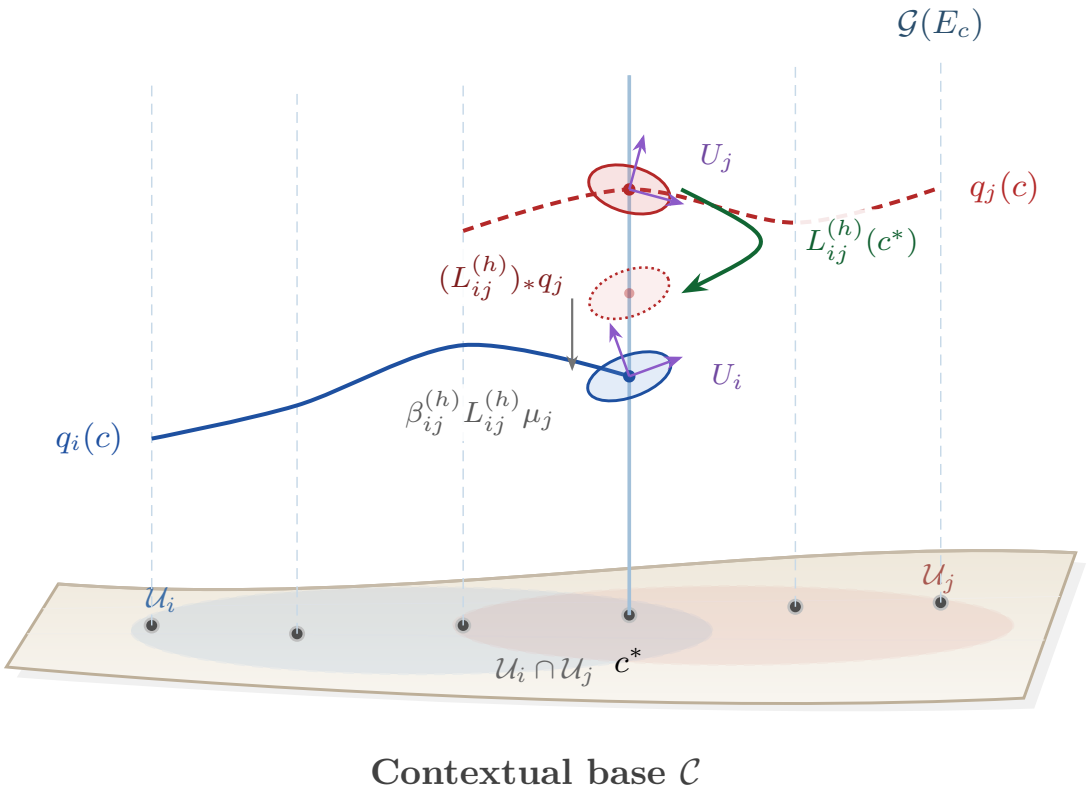


Figure 5: **Local Gaussian sections and directed communication links.** The curved surface represents the contextual base of the principal bundle; the curves are sections of its associated Gaussian-state bundle over overlapping neighborhoods. At a shared context c^*, the local frames accompany the state glyphs. For one communication head indexed by h, the green arrow applies $L_{ij}^{(h)} = U_i \widetilde{O}_h T_h U_j^{-1}$ to the sender. The dotted contour depicts its Gaussian pushforward; the gray arrow identifies the weighted mean contribution to the receiver. A full-rank link is illustrated; a lower-rank link may produce a singular pushforward. The receiver's state need not equal that pushforward. Routing uses the separate comparison channel in Figure 1. The finite token construction requires no smooth contextual base.

The geometric construction begins with a principal bundle of reference frames over a space of contexts. Let $\mathcal{C}$ be a smooth contextual base and take $G = \mathrm{GL}(K, \mathbb{R})$. A principal bundle

$$\pi : \mathcal{P} \longrightarrow \mathcal{C} \tag{106}$$

carries a free right action $n \mapsto n \cdot g$ that preserves π and is transitive on each fiber. Locally, $\mathcal{P}|_{\mathcal{U}} \simeq \mathcal{U} \times G$, with the group acting on the second factor. Thus a principal fiber consists of frame choices; choosing one frame identifies the fiber with G. The probabilistic states are attached to this principal structure through associated bundles (Nakahara, 2003; Frankel, 2011).

Let $\mathsf{V} = \mathbb{R}^K$ carry the defining representation of G, and let

$$\mathcal{B} = \{\mathcal{N}(\mu, \Sigma) : \mu \in \mathsf{V},\ \Sigma \succ 0\}, \qquad \rho(g)\mathcal{N}(\mu, \Sigma) = \mathcal{N}(g\mu, g\Sigma g^\top). \tag{107}$$

The vector and Gaussian statistical bundles are

$$E = \mathcal{P} \times_G \mathsf{V}, \qquad \mathcal{G}(E) = \mathcal{P} \times_\rho \mathcal{B}. \tag{108}$$

For the statistical bundle, the quotient convention is

$$(n \cdot g, b) \sim (n, \rho(g)b). \tag{109}$$

Equivalently, (n, b) and $(n \cdot g, \rho(g^{-1})b)$ represent the same associated point. The analogous relation uses $v \mapsto gv$ for the vector bundle. The principal bundle carries gauge freedom, the vector bundle carries latent values, and the associated statistical bundle carries distributions over those values.

An agent over a contextual region $\mathcal{U}_i$ selects a local principal section u_i and maintains belief and prior sections:

$$\begin{aligned} u_i : \mathcal{U}_i \longrightarrow \mathcal{P}, \qquad & \pi \circ u_i = \mathrm{id}_{\mathcal{U}_i}, \\ q_i, p_i \in \Gamma\big(\mathcal{G}(E)|_{\mathcal{U}_i}\big). & \end{aligned} \tag{110}$$

Here Γ denotes smooth sections, and the belief and prior have coordinates (μ_i, Σ_i) and $(\mu_{p,i}, \Sigma_{p,i})$ in the section u_i. Communication additionally uses one reference trivialization σ_0 of $\mathcal{P}$ on its working domain. The frame field is the relative element U_i defined by $u_i = \sigma_0 \cdot U_i^{-1}$, so that $\xi_i = U_i^{-1}\mu_i$ are the σ_0-coordinates of the same vector. Identifying the fiber with its coordinates in the section u_i, written E_c, the frame is the linear isomorphism $U_i(c) : \mathsf{V} \to E_c$ that converts σ_0-coordinates into u_i-coordinates. On overlaps, agents can maintain different Gaussian sections at the same context. They are not required to be local representatives of one common belief. Figure 5 illustrates these associated statistical sections and their local frames. The attention construction uses this belief/prior sector; it does not require an additional bundle of generative-model parameters.

For token attention, the contextual dependence is evaluated at a fixed context c_0, while $i \in \mathcal{I} = \{0, \ldots, N-1\}$ labels the token agents. Keeping a separately labeled copy of the fiber for each token is the pullback along the constant map $\iota : \mathcal{I} \to \mathcal{C}$, $\iota(i) = c_0$:

$$\mathcal{P}_{\mathcal{I}} = \iota^*\mathcal{P}, \qquad E_{\mathcal{I}} = \iota^* E. \tag{111}$$

Write E_i for the vector fiber at token i. This notation distinguishes token incidence from the contextual base of the general construction. Because ι is constant, $\iota^*\mathcal{P} = \mathcal{I} \times \mathcal{P}_{c_0}$ canonically. The token construction therefore sees no connection, curvature, or topology of $\mathcal{P}$, and its gauge group is G^N, on which the contextual gauge group acts only through the diagonal. Placing tokens at distinct contexts $\iota(i) = c_i$ and realizing links by parallel transport along declared base curves would make the contextual connection enter the attention operator. The section of each token agent and the reference trivialization give $U_i : \mathsf{V} \to E_i$, and its statistical fiber is

$$\mathcal{G}(E_i) = \{\mathcal{N}(\mu, \Sigma) : \mu \in E_i,\ \Sigma \succ 0\}. \tag{112}$$

A general contextual principal bundle need not admit a global section. The factorized attention formulas use the reference trivialization σ_0 on their working domain; on the finite token pullback this trivialization is canonical. The communication graph is additional data; the matrix links on it are determined by the frames, the shared interfaces, and the routing weights. A link $L_{ij} : E_j \to E_i$ can be singular, so it is not in general a principal-bundle transition function or a group-valued parallel transport. Sections 2 and 8 construct these links from comparison and content interfaces while retaining the underlying principal-bundle structure.

Appendix B. Irreducible Frame Blocks and Communication Heads

The representation carried by a state and the organization of its communication heads are separate structures. To make their relationship precise, fix a decomposition $\mathsf{V} = \bigoplus_{b=1}^{B} \mathsf{V}_b$, with $\mathsf{V}_b \simeq \mathbb{R}^{d_b}$ and $K = \sum_b d_b$, and restrict the frame group to

$$G = \prod_{b=1}^{B} \mathrm{GL}(d_b, \mathbb{R}), \qquad \rho(g_1, \dots, g_B) = \bigoplus_{b=1}^{B} g_b, \qquad U_i = \bigoplus_{b=1}^{B} U_i^{(b)}. \tag{113}$$

Here B counts independently transforming frame blocks, while H counts communication heads. Each V_b is an irreducible real representation of G: a nonzero invariant subspace contains the orbit of any one of its nonzero vectors, and the bth general linear factor maps that vector to every other nonzero vector in V_b. Thus no proper nonzero invariant subspace exists. These representations remain inequivalent when their dimensions coincide, because the product factors act independently. Repeating the same group element on several blocks would instead give equivalent copies of one representation. Equation (113) is an explicit specialization of the full $\mathrm{GL}(K)$ action, not a theorem that an arbitrary attention architecture has this decomposition.

In this appendix O_h denotes the applied receiving map, written $\widetilde{O}_h$ in the main text; in the affine chart the two coincide. Let $\mathrm{pr}_b : \mathsf{V} \to \mathsf{V}_b$ and $\iota_b : \mathsf{V}_b \to \mathsf{V}$ be the coordinate projection and inclusion, and put $E_i^{(b)} = U_i \mathsf{V}_b$. A head h is local to a block $b = b(h)$ when its interface maps factor as

$$S_{Q,h} = \overline{S}_{Q,h} \,\mathrm{pr}_b, \quad S_{K,h} = \overline{S}_{K,h} \,\mathrm{pr}_b, \quad T_h = \overline{T}_h \,\mathrm{pr}_b, \quad O_h = \iota_b \overline{O}_h. \tag{114}$$

The barred maps act on the selected block; the comparison maps retain the full-row-rank requirement. The resulting $L_{ij}^{(h)} = U_i \iota_b \overline{O}_h \overline{T}_h \,\mathrm{pr}_b \, U_j^{-1}$ communicates only between the corresponding site subspaces $E_j^{(b)}$ and $E_i^{(b)}$. Taking $H = B$ and a one-to-one assignment gives a precise sense in which heads can be organized by irreducible blocks. The irreducible object is the vector carrier $E_i^{(b)}$, however, not the Gaussian density or the entire attention operation. The common reference channels remain fixed under the passive site action of Eq. (4). A proper selected row space inside the carrier is not invariant under the full defining $\mathrm{GL}(d_b)$ action. In the full-input/output construction, S_Q, S_K, T, O need not satisfy Eq. (114); a head can access several blocks. Its endpoint covariance follows from the transforming frames, without requiring the fixed reference maps to be group intertwiners.

Independent Gaussian blocks require an additional statistical assumption. If every local covariance is block diagonal, $\Sigma_i = \bigoplus_b \Sigma_i^{(b)}$, then the full Gaussian laws factor and block-diagonal reference transport gives

$$D_{\mathrm{KL}}\big(q_i \| (\Omega_{ij})_* q_j\big) = \sum_{b=1}^{B} D_{\mathrm{KL}}\big(q_i^{(b)} \| (\Omega_{ij}^{(b)})_* q_j^{(b)}\big), \quad \Omega_{ij}^{(b)} = U_i^{(b)} (U_j^{(b)})^{-1}. \tag{115}$$

Here $q_i^{(b)}$ is the block marginal. The identity follows by writing the logarithm of the product-density ratio as a sum and integrating each normalized factor (van Erven and Harremøes, 2014, Thm. 28). Block-diagonal frames alone do not imply it. For example, $q = \mathcal{N}(0, \left(\begin{smallmatrix} 1 & c \\ c & 1 \end{smallmatrix}\right))$, $p = \mathcal{N}(0, I_2)$, and $0 < |c| < 1$ have identical one-dimensional marginals but $D_{\mathrm{KL}}(q\|p) = -\frac{1}{2}\log(1-c^2) > 0$. Furthermore, separate routing distributions for the blocks are a modeling choice: softmax of the summed KL chooses one shared source and does not equal independent headwise source selection.

The frame generators have a related but different decomposition. At a block, $\mathfrak{gl}(d_b) = \mathfrak{sl}(d_b) \oplus \mathbb{R} I_{d_b}$ separates traceless generators from the central dilation generator; it is not a decomposition of the state space into two irreducible heads. At the ambient level,

$$\mathfrak{gl}(K) = \left(\bigoplus_b \mathfrak{gl}(d_b)\right) \oplus_{\mathrm{vec}} \mathfrak{m}, \qquad \mathfrak{m} = \bigoplus_{b \neq c} \mathrm{Hom}(\mathsf{V}_c, \mathsf{V}_b), \qquad \dim \mathfrak{m} = K^2 - \sum_b d_b^2. \tag{116}$$

The subscript emphasizes a vector-space splitting. The off-block complement is preserved by block-diagonal conjugation but is generally not a Lie subalgebra: for two scalar blocks, $[E_{12}, E_{21}] = E_{11} - E_{22}$ is block diagonal. Restoring these generators enlarges the frame group; it does not add the only possible cross-block communication, since full-width interface maps already provide such communication.

Representation-based attention has established precedents. The SE(3)-Transformer uses irreducible rotation types with invariant attention and equivariant values; its multi-head construction partitions multiplicity channels while retaining several representation types in each head (Fuchs et al., 2020, Secs. 2.3 and 3.2, App. D.3). Gauge Equivariant Transformer combines local-frame-equivariant multi-head attention with regular fields and their irreducible decomposition (He et al., 2021, Secs. 4.1–4.2). The subject here is the particular combination of full Gaussian comparison laws, separately specified sending/receiving interfaces, and the matrix links $U_i O_h T_h U_j^{-1}$. The frame blocks specify the state representation, while the comparison and content interfaces specify how that state is used for communication.

Appendix C. Positional and Masking Priors

With a nonuniform prior, Eq. (66) adds $\log \pi_{ij}$ to the logits with unit coefficient while the cost is divided by τ, so temperature never rescales a positional preference. Masks and positional biases therefore enter as priors over sources. This identification, and the entropy-regularized or transport form of the softmax that underlies it, is established in the attention literature (Niculae and Blondel, 2017; Litman, 2025; Bianchessi et al., 2026; Litman and Guo, 2026). On the causal support $S_i = \{0, \dots, i\}$, the uniform prior $\pi_{ij} = \mathbf{1}[j \leq i]/(i+1)$

gives the support and row normalization of GPT-family attention (Radford et al., 2019). A window of radius w, $\pi_{ij} \propto \mathbf{1}[i - w \le j \le i]$, gives the sliding-window support of Longformer and Mistral (Beltagy et al., 2020; Jiang et al., 2023). The prior $\pi_{ij} \propto \exp[-\alpha_h(i-j)]$ gives the linear biases of attention with linear biases (ALiBi) (Press et al., 2022), and $\log \pi_{ij} = f_h(i-j) + \text{const}$ gives learned relative biases, parameterized by sinusoids in the global positional bias of Transformer-XL and shared within logarithmic offset buckets in T5 (Dai et al., 2019; Raffel et al., 2020). Press et al. (2022) state the geometric slope schedule $\alpha_h = 2^{-8h/H}$; their reference implementation uses it when H is a power of two and otherwise appends alternate slopes of the next power-of-two schedule. The construction fixes only the form in which a prior enters. It does not select a prior, except that the autoregressive factorization requires $\pi_{ik} = 0$ for $k > i$ at every layer, because each state depends on its own token.

Appendix D. An Optional Prior-Bank Vocabulary Readout

Let q_i be the Gaussian state after the chosen forward computation and p_v a normalized class template on the same latent space, with finite $D_{\mathrm{KL}}(q_i \| p_v)$. Declaring the templates in reference coordinates and comparing them with $\widetilde{q}_i = (U_i^{-1})_* q_i$ makes the readout invariant under the passive action; comparing local coordinates directly defines a different, non-covariant readout. No convergence of q_i is assumed. Specify a categorical readout $\hat{p}_i$ by minimizing

$$\mathcal{J}_{\mathrm{decode}}(\hat{p}_i) = \sum_v \hat{p}_i(v) D_{\mathrm{KL}}(q_i \| p_v) + \tau_{\mathrm{dec}} D_{\mathrm{KL}}(\hat{p}_i \| \omega), \qquad \tau_{\mathrm{dec}} > 0, \tag{117}$$

where ω is a class prior. The unique optimum on its support is

$$\hat{p}_i(v) \propto \omega_v \exp\left[-D_{\mathrm{KL}}(q_i \| p_v)/\tau_{\mathrm{dec}}\right]. \tag{118}$$

This is a specified Gibbs readout. At unit temperature it is also the categorical mean-field coordinate update for a fixed q_i under the normalized mixture $p(v, x) = \omega_v p_v(x)$, as in the general expected-log update of Blei et al. (2017, §2.4, Eqs. 17–18). It is not generically the posterior-predictive average of a classifier.

The score identity is exact:

$$\mathbb{E}_{q_i}[\log p_v(X)] = -D_{\mathrm{KL}}(q_i \| p_v) - H(q_i). \tag{119}$$

The entropy is independent of v and cancels in the categorical softmax. For Gaussian $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ and $p_v = \mathcal{N}(\mu_{p,v}, \Sigma_{p,v})$, the expectation is

$$-\frac{1}{2}\left[\mathrm{Tr}(\Sigma_{p,v}^{-1}\Sigma_i) + (\mu_i - \mu_{p,v})^\top \Sigma_{p,v}^{-1}(\mu_i - \mu_{p,v}) + \log\det \Sigma_{p,v} + K\log(2\pi)\right]. \tag{120}$$

These are Gaussian cross-entropy identities (Bishop, 2006).

For the normalized point classifier

$$p(v \mid x) = \frac{\omega_v p_v(x)}{\sum_w \omega_w p_w(x)}, \tag{121}$$

posterior prediction instead averages probabilities, $\bar{p}_i(v) = \mathbb{E}_{q_i}[p(v \mid X)]$. This generally differs from normalizing $\exp(\mathbb{E}_{q_i} \log p(v \mid X))$. For example, with equal class priors, $q_i = \mathcal{N}(0, 4)$ and templates $p_0 = \mathcal{N}(0, 1)$, $p_1 = \mathcal{N}(2, 1)$, the unit-temperature Gibbs readout is $\hat{p}_i(1) = 1/(1 + e^2) \approx 0.11920$. The predictive average is

$$\bar{p}_i(1) = \mathbb{E}_{X \sim \mathcal{N}(0,4)} \left[\frac{1}{1 + \exp(2 - 2X)} \right] \approx 0.32383. \tag{122}$$

The expected-log construction therefore does not replace the predictive integral. Both readouts are variational: among joints whose latent marginal is fixed to q_i, the unrestricted minimizer of the KL to $\omega_v p_v(x)$ has class marginal $\bar{p}_i$, and the minimizer within the product family is the Gibbs readout. The same distinction separates the Gibbs routing row from the posterior-predictive source probability of Section 6.2.

Uniform ω gives logits $-D_{\mathrm{KL}}(q_i \| p_v)/\tau_{\mathrm{dec}}$. Projecting a full-covariance belief before decoding makes this identity exact for the projected distribution only. Choosing a temperature or learning an additional scalar is a modeling choice. Cross-entropy training of this readout does not become optimization of the full belief free energy merely because its scores are divergences. The linear decoder used in the experiment of Section 9 is itself the unit-temperature Gibbs readout for the shared-covariance templates $p_v = \mathcal{N}(\Sigma_* w_v, \Sigma_*)$, with w_v the vth row of W_{out} and class prior $\log \omega_v = b_{\mathrm{out},v} + \kappa_{\mathrm{uni}} \log p_{\mathrm{uni}}(v) + \frac{1}{2} w_v^\top \Sigma_* w_v$ up to a constant, for any $\Sigma_* \succ 0$; the belief covariance cancels from every logit difference. The token-bank readout differs by giving each class its own template covariance.

Appendix E. Relation to a Normalized Latent-Source Model

A broader gauge-theoretic variational free energy (VFE) theory begins with a normalized generative law and a separately specified population recognition law. Geometry alone supplies neither. Let z collect the declared agent and auxiliary latent variables (the latent configuration, written y in the broader theory), let X denote fixed structural data, and let each record kernel have a normalized density ℓ_a. A normalized latent law p_θ^Z, constructed for example by finite directed kernel composition, gives

$$p_\theta(z, o \mid X) = p_\theta^Z(z \mid X) \prod_a \ell_a(o_a \mid z_{\partial a}, X). \tag{123}$$

Integrating the records first leaves the unit mass of p_θ^Z. Record scopes may overlap because each record is a conditional output of the same latent configuration. The factors are fixed while recognition varies. Latent variables may themselves take values in a space of probability laws; this does not permit a generative factor to read the recognition distribution being optimized.

Fix a regular observation slice with $0 < p_\theta(o \mid X) < \infty$. Let Q have density q, satisfy $Q \ll P_\theta(\cdot \mid o, X)$, and have finite expectations of $|\log q|$ and $|\log p_\theta(z, o \mid X)|$. Defining $\mathcal{L}_\theta(Q) = \mathbb{E}_Q[\log p_\theta(z, o \mid X) - \log q(z)]$ gives

$$\log p_\theta(o \mid X) = \mathcal{L}_\theta(Q) + D_{\mathrm{KL}}\big(Q \| P_\theta(\cdot \mid o, X)\big). \tag{124}$$

Indeed, Bayes' formula makes the log density ratio equal to $\log q - \log p_\theta(z, o \mid X) + \log p_\theta(o \mid X)$; integrating proves the identity. The KL is nonnegative and vanishes exactly at equality

of the full laws. No factorization of Q is assumed. Outside the finite-density domain, the extended free energy is defined as $-\log p_\theta(o \mid X) + D_{\mathrm{KL}}(Q \| P_\theta(\cdot \mid o, X))$, avoiding undefined differences of infinite expectations. This is the standard evidence-bound construction (Blei et al., 2017, Sec. 2.2, Eq. 14); consistent passive pushforwards preserve its relative entropy.

The broader theory also supplies a route to categorical attention. Give each receiver-head pair (i,h) an independent source label $J_i^{(h)}$ with prior $\pi_i^{(h)}$ and a binary compatibility record $\chi_i^{(h)} \in \{0,1\}$, and take

$$\begin{aligned} p_\theta(z, j, \chi, o \mid X) = p_\theta^Z(z \mid X) \prod_{i,h} \pi_i^{(h)}\big(j_i^{(h)}\big)\, \mathrm{Bern}\Big(\chi_i^{(h)}; \exp\big[-D^{(h)}_{i j_i^{(h)}}(z)/\tau_h\big]\Big) \\ \times \prod_a \ell_a(o_a \mid z_{\partial a}, X), \end{aligned} \tag{125}$$

in which no other factor reads a label. Each record kernel is normalized for every $\tau_h > 0$ because $D_{ij}^{(h)} \geq 0$; for a cost that is only finite, replacing D by $D - \min_k D_{ik}$ multiplies the selected-record likelihood by a label-independent factor. Conditioning on $\chi = \mathbf{1}$ gives the exact conditional rows $P(J_i^{(h)} = j \mid z, \chi, o) = \mathrm{softmax}_j\big(\log \pi_{ij}^{(h)} - D_{ij}^{(h)}(z)/\tau_h\big)$, independent across (i,h). This is the optimality-variable device of control as inference (Levine, 2018). Under $Q(dz, dj) = Q_Z(dz) \prod_{i,h} \beta_i^{(h)}(dj_i^{(h)})$ with $\mathbb{E}_{Q_Z} |D_{ij}^{(h)}| < \infty$, the row-dependent free energy and its optimizer are

$$\mathcal{F}_i^{(h)}(\beta_i^{(h)}) = D_{\mathrm{KL}}(\beta_i^{(h)} \| \pi_i^{(h)}) + \tau_h^{-1} \sum_j \beta_{ij}^{(h)} \mathbb{E}_{Q_Z} D_{ij}^{(h)}, \tag{126}$$

$$\beta_{ij}^{(h)*} = \mathrm{softmax}_j\big(\log \pi_{ij}^{(h)} - \mathbb{E}_{Q_Z} D_{ij}^{(h)}/\tau_h\big). \tag{127}$$

Expansion of the factorized density gives the first expression up to a row-independent term; a normalization multiplier gives the second. The pointwise generative posterior instead uses $D_{ij}^{(h)}(z)$ inside softmax. Multiplying a row objective by τ_h preserves its optimizer, but does not license independently rescaling sectors of one global evidence lower bound (ELBO).

The fixed-context Gaussian mixture of Section 6.1 supplies a different local joint with the same unit-temperature row objective: at unit temperature its KL is the projected Gaussian comparison cost plus categorical relative entropy. Appending the same normalized content law in the model and recognition factors makes that content law cancel, as in Eq. (73). It therefore permits the chosen message means without deriving the learned content maps or residual. The general Gaussian realization likewise permits singular conditional mean maps with positive-definite receiving noise, but that normalization fact does not select the predictive attention architecture.

The autoregressive softmax predictor already defines a normalized probability law, and its outer cross-entropy is the corresponding negative log-likelihood; in evaluation each conditional is computed from a bounded window, which leaves every factor normalized. For the attention sublayer, a normalized latent model exists in which routing is the exact label posterior and the content-plus-residual update is the exact predictive mean (Theorem 20). What remains unresolved is a latent joint and recognition construction that also covers the normalization and MLP maps, the collapse of the predictive mixture to a point state between layers, and the relation of cross-entropy training to that model's evidence. A model

with no latent variables would make the evidence identity tautological without deriving those operations.

Appendix F. Additional Scope Results for Gaussian Predictive Architectures

The main text distinguishes geometric attention from a complete model of population inference and nonlinear state evolution. The following calculations state the population-covariance, Gaussian-closure, and causal-prediction conditions used in the main text. They concern the declared states and predictive maps.

F.1 Population covariance and link-residual costs

Let a joint population state have law $\mathcal{N}(\boldsymbol{\mu}, \Gamma)$, with covariance blocks $\Gamma_{ij} = \mathrm{Cov}(k_i, k_j)$ and $\Gamma_{ii} = \Sigma_i$. For a fixed link $L_{ij} : E_j \to E_i$, set $e_{ij} = k_i - L_{ij}k_j$ and $\bar{e}_{ij} = \mu_i - L_{ij}\mu_j$. Bilinearity of covariance gives

$$V_{ij} := \mathrm{Cov}(e_{ij}) = \Sigma_i + L_{ij}\Sigma_j L_{ij}^\top - \Gamma_{ij}L_{ij}^\top - L_{ij}\Gamma_{ji}. \tag{128}$$

This identity needs no inverse of L_{ij}. For any receiving covariance $R_i \succ 0$, expansion around the mean gives the expected quadratic cost

$$\mathbb{E}[e_{ij}^\top R_i^{-1} e_{ij}] = \bar{e}_{ij}^\top R_i^{-1}\bar{e}_{ij} + \mathrm{Tr}(R_i^{-1}V_{ij}). \tag{129}$$

The centered cross terms have zero expectation, leaving the trace term. These are Gaussian moment identities (Bishop, 2006); the two displayed moment formulas also hold for non-Gaussian laws with finite second moments.

A product approximation to the population sets the off-diagonal blocks Γ_{ij} to zero. Local state covariances alone do not justify that substitution. For a scalar example, take unit marginal variances, zero means, $L_{ij} = 1$, and cross covariance c, with $|c| < 1$. The expected squared residual is $2(1-c)$, while every individual Gaussian marginal is unchanged as c varies. Thus comparing local distributions and computing a correlated population residual are different operations.

A Gaussian record with mean $k_i - L_{ij}k_j$ and positive-definite noise can use Eq. (129) inside a normalized population model of Appendix E. Its expected log-likelihood cost is not generically the KL between the two projected Gaussian marginals in the main text. That KL has its own trace and determinant terms. Substituting a marginal divergence for the record cost changes the probabilistic model unless a separate identity establishes their equality.

F.2 Nonlinear mean maps and full Gaussian laws

An operation on a Gaussian mean does not automatically specify the pushforward of its entire law. To see this for LayerNorm (Ba et al., 2016), put $P_0 = I - \mathbf{1}\mathbf{1}^\top/K$ and define

$$\mathrm{LN}_\epsilon(x) = \frac{P_0 x}{\sqrt{\|P_0 x\|^2/K + \epsilon}}. \tag{130}$$

For $K = 2$, $\epsilon = 0$, and a nondegenerate Gaussian input, $x_1 = x_2$ has probability zero and

$$\mathrm{LN}_0(x) = \mathrm{sign}(x_1 - x_2)(1, -1). \tag{131}$$

The output law has two atoms. For $\epsilon > 0$ and arbitrary $K \geq 2$, the output remains in the proper zero-sum hyperplane, and it is bounded, $\|\mathrm{LN}_\epsilon(x)\|^2 < K$, so its law is not even a degenerate Gaussian. Fixed affine gains and biases map this support into an affine subspace of dimension at most $K-1$. Neither construction is a nondegenerate full Gaussian pushforward. Applying normalization only to μ_i, while retaining Σ_i for routing, remains a valid specified coordinate operation; it has a different statistical meaning.

F.3 Prefix-based causal prediction

A predictive conditional requires a state determined only by the available prefix and shared across all candidate next tokens. If one separately recomputes a binary state after inspecting each candidate target and assigns that target probability 3/4, each row is normalized but the two candidate scores sum to 3/2. They are not one conditional probability law. By contrast, a single prefix-based state followed by one categorical readout defines a normalized kernel, and the product of such kernels defines a finite sequence law by successive summation. The main architecture uses this direct causal construction. Outer backpropagation from training targets updates shared parameters; it does not insert those targets into the prediction-time state.

Appendix G. Proofs for Latent Link Generators

This appendix proves the statements of Section 8.4 that follow Proposition 35. The notation is that of Eq. (104) and Proposition 35. The links are point masses, the states and labels carry the mean-field recognition law of that proposition with $C_j^\ell \succ 0$, $\bar{A} = \bar{Q}^{-1}$, and $\mathbf{L}(X, E) = \int_0^1 e^{sX} E e^{(1-s)X} ds$ is the Fréchet derivative of the matrix exponential, which is linear in E and is the differential of the exponential map at X (Higham, 2008). Matrices carry the Frobenius inner product $\langle E, E' \rangle = \mathrm{Tr}(E^\top E')$ and its norm $\|E\|$, which is submultiplicative, and a family $X = (X_e)$ of generators indexed by the edges $e = (i \leftarrow j, \ell)$, $j \leq i$, carries $\langle X, Y \rangle = \sum_e \langle X_e, Y_e \rangle$. We write $\mathcal{F}(X)$ for the free energy of Proposition 35 as a function of the point-mass links at fixed recognition of the states and labels. With one head, $\Pi_1 = I_K$, the label J_i^ℓ selects one source, and its recognition law assigns probability β_{ij}^ℓ to the source j. Write

$$S_j^\ell = C_j^\ell + \xi_j^\ell \xi_j^{\ell\top} \succ 0 \tag{132}$$

for the second moment of ζ_j^ℓ under the recognition law. A chart link is a matrix e^X with $X \in \mathfrak{gl}(K)$. By Theorem 1 of Culver (1966), a real matrix is a chart link exactly when it is invertible and every Jordan block belonging to a negative eigenvalue occurs an even number of times; in particular, every invertible real matrix without negative eigenvalues is a chart link.

G.1 The unregularized link problem

Proposition 36 (The chart infimum without a link prior) *Let $S_X = 0$, take one head, $K \geq 2$, and $\beta_{ij}^{\ell} > 0$, write $\xi' = \xi_i^{\ell+1}$ and $S = S_j^{\ell}$, and let $\mathcal{T}(G) = \frac{1}{2}\beta_{ij}^{\ell}\mathbb{E}_q\|\zeta_i^{\ell+1} - G\zeta_j^{\ell}\|_{\bar{A}}^2$ for real $K \times K$ matrices G.*

1. *Up to a constant independent of X_{ij}^{ℓ}, the free energy depends on X_{ij}^{ℓ} only through $\mathcal{T}(e^{X_{ij}^{\ell}})$, and*

$$\mathcal{T}(G) - \mathcal{T}(G^{\star}) = \tfrac{1}{2}\beta_{ij}^{\ell}\big\|\bar{A}^{1/2}(G - G^{\star})S^{1/2}\big\|^2, \qquad G^{\star} = \xi'\xi_j^{\ell\top}S^{-1}. \tag{133}$$

   Hence $G^{\star}$ is the unique stationary point and the unique minimizer of $\mathcal{T}$; it has rank at most one and is not a chart link.

2. *A generator X is stationary for $X \mapsto \mathcal{T}(e^X)$ only where $\mathbf{L}(X^{\top}, \cdot)$ is singular, which is exactly where X is a critical point of the exponential map.*

3. *The trace $\tau = \operatorname{Tr} G^{\star}$ is*

$$\tau = \frac{\xi_j^{\ell\top}(C_j^{\ell})^{-1}\xi'}{1 + \xi_j^{\ell\top}(C_j^{\ell})^{-1}\xi_j^{\ell}}, \tag{134}$$

   so it has the sign of $\xi_j^{\ell\top}(C_j^{\ell})^{-1}\xi_i^{\ell+1}$.

4. *If $\tau \geq 0$, the infimum of $\mathcal{T}$ over chart links equals $\mathcal{T}(G^{\star})$, is not attained, and is approached only along sequences of links that converge to $G^{\star}$.*

5. *If $\tau < 0$, no sequence of chart links converges to $G^{\star}$, and the infimum of $\mathcal{T}$ over chart links lies strictly above $\mathcal{T}(G^{\star})$.*

Proof (1) With one head, $m_i^{\ell} = e^{X_{iJ}^{\ell}}\zeta_J^{\ell}$ with $J = J_i^{\ell}$, and $-\log\mathcal{N}(\zeta_i^{\ell+1}; m_i^{\ell}, \bar{Q}) = \frac{1}{2}\|\zeta_i^{\ell+1} - m_i^{\ell}\|_{\bar{A}}^2 + \frac{1}{2}\log\det(2\pi\bar{Q})$. Under the recognition law the label is independent of the states, so the expected transport term of receiver i at layer ℓ is $\sum_{j'}\beta_{ij'}^{\ell}\frac{1}{2}\mathbb{E}_q\|\zeta_i^{\ell+1} - e^{X_{ij'}^{\ell}}\zeta_{j'}^{\ell}\|_{\bar{A}}^2$ up to a constant. Its summand $j' = j$ is $\mathcal{T}(e^{X_{ij}^{\ell}})$, and with $S_X = 0$ no other term of the free energy contains X_{ij}^{ℓ}. The variables $\zeta_i^{\ell+1}$ and ζ_j^{ℓ} belong to different layers and are independent under the recognition law, so $\mathbb{E}_q[\zeta_j^{\ell}\zeta_i^{\ell+1\top}] = \xi_j^{\ell}\xi'^{\top} = SG^{\star\top}$ and $\mathbb{E}_q[\zeta_j^{\ell}\zeta_j^{\ell\top}] = S$. Expanding the square,

$$\tfrac{2}{\beta_{ij}^{\ell}}\mathcal{T}(G) = \mathbb{E}_q\|\zeta_i^{\ell+1}\|_{\bar{A}}^2 - 2\operatorname{Tr}\big(\bar{A}GSG^{\star\top}\big) + \operatorname{Tr}\big(\bar{A}GSG^{\top}\big),$$

and completing the square in G gives Eq. (133), because $\operatorname{Tr}(\bar{A}(G - G^{\star})S(G - G^{\star})^{\top}) = \|\bar{A}^{1/2}(G - G^{\star})S^{1/2}\|^2$. Since $\bar{A}$ and S are positive definite, the right side of Eq. (133) vanishes only at $G = G^{\star}$, and it is a positive-definite quadratic form in $G - G^{\star}$. The matrix $G^{\star} = \xi'(S^{-1}\xi_j^{\ell})^{\top}$ has rank at most one, hence is singular for $K \geq 2$, whereas $\det e^X = e^{\operatorname{Tr} X} > 0$.

(2) The gradient of $\mathcal{T}$ at G is $\beta_{ij}^{\ell}\bar{A}(GS - \xi'\xi_j^{\ell\top}) = \beta_{ij}^{\ell}\bar{A}(G - G^{\star})S$, which is the negative of the one-head current $\mathcal{J}_{ij}^{\ell}$ of Proposition 35 at G. With $S_X = 0$ that proposition makes X stationary exactly when $\mathbf{L}(X^{\top}, \mathcal{J}_{ij}^{\ell}) = 0$. Because $e^X \neq G^{\star}$ by item 1 and $\bar{A}$ and S are invertible, $\mathcal{J}_{ij}^{\ell} \neq 0$, so $\mathbf{L}(X^{\top}, \cdot)$ has a nonzero kernel. The substitution $s \mapsto 1 - s$ gives

$\mathbf{L}(X,E)^\top = \mathbf{L}(X^\top, E^\top)$, so $\mathbf{L}(X^\top,\cdot)$ is singular exactly when $\mathbf{L}(X,\cdot)$ is, that is, exactly when the differential of the exponential map at X is singular.

(3) By the Sherman–Morrison formula, $S^{-1}\xi_j^\ell = (C_j^\ell)^{-1}\xi_j^\ell/(1+\xi_j^{\ell\top}(C_j^\ell)^{-1}\xi_j^\ell)$, and $\tau = \mathrm{Tr}(\xi'\xi_j^{\ell\top}S^{-1}) = (S^{-1}\xi_j^\ell)^\top\xi'$. The denominator is positive.

(4) and (5) The rank-one matrix $G^\star = \xi' w^\top$, $w = S^{-1}\xi_j^\ell$, has characteristic polynomial $z^{K-1}(z-\tau)$. By Eq. (133), $\mathcal{T}(G)-\mathcal{T}(G^\star) \geq \frac{1}{2}\beta_{ij}^\ell\lambda_{\min}(\bar{A})\lambda_{\min}(S)\|G-G^\star\|^2$. If $\tau \geq 0$, then for $\varepsilon > 0$ the matrix $G^\star + \varepsilon I$ has the eigenvalues $\tau+\varepsilon$ and ε, all positive, so it is a chart link, and $\mathcal{T}(G^\star+\varepsilon I) \to \mathcal{T}(G^\star)$ as $\varepsilon \to 0$. No chart link equals $G^\star$, so the infimum $\mathcal{T}(G^\star)$ is not attained, and by the lower bound every sequence of links whose energies tend to $\mathcal{T}(G^\star)$ converges to $G^\star$. If $\tau < 0$, then τ is a simple eigenvalue of $G^\star$, and the other eigenvalue, 0, lies at distance $|\tau|$. By continuity of the roots of the characteristic polynomial there is $r > 0$ such that every real matrix G with $\|G-G^\star\| < r$ has exactly one eigenvalue, counted with multiplicity, in the disk $|z-\tau| < |\tau|/2$. That disk is symmetric under complex conjugation and the characteristic polynomial of G is real, so this eigenvalue is real and negative. Its algebraic multiplicity is one, so it has a single Jordan block, which occurs once, and by the criterion above G is not a chart link. No chart link lies within r of $G^\star$, and the lower bound gives $\mathcal{T}(e^X) \geq \mathcal{T}(G^\star) + \frac{1}{2}\beta_{ij}^\ell\lambda_{\min}(\bar{A})\lambda_{\min}(S)r^2$ for every X. ∎

G.2 The linearized link equation

Proposition 37 (Linearized link equation) *Take one head and the flat anchor $\bar{X}^\ell = 0$ for every layer. For $0 \leq \ell \leq L-2$ and $k < j \leq i$, define the loop map $\mathbf{d}$ and the loop-weight operator $\mathbf{w}$ by*

$$(\mathbf{d}X)_{ijk}^\ell = X_{ij}^{\ell+1} + X_{jk}^\ell - X_{ik}^{\ell+1} - X_{kk}^\ell, \qquad (\mathbf{w}\mathbf{d}X)_{ijk}^\ell = w_{ijk}^\ell(\mathbf{d}X)_{ijk}^\ell, \tag{135}$$

let $\mathbf{d}^\top$ be the adjoint of $\mathbf{d}$ for the Frobenius inner products on edges and loops, write $\beta_e = \beta_{ij}^\ell$ and $\mathcal{J}_e^0 = \beta_e\bar{A}[(\xi_i^{\ell+1}-\xi_j^\ell)\xi_j^{\ell\top} - C_j^\ell]$ for the one-head current of Proposition 35 at zero generators on the edge $e = (i \leftarrow j, \ell)$, and define the operator

$$(\mathbf{K}X)_e = \sigma_X^{-2}X_e + \lambda_W(\mathbf{d}^\top\mathbf{w}\mathbf{d}X)_e + \beta_e\bar{A}X_eS_j^\ell. \tag{136}$$

1. *$\mathbf{K}$ is self-adjoint and $\langle X, \mathbf{K}X\rangle \geq \sigma_X^{-2}\|X\|^2$. The linear equation $\mathbf{K}X = \mathcal{J}^0$ therefore has a unique solution X_{lin}, and $\|X_{\mathrm{lin}}\| \leq \sigma_X^2\|\mathcal{J}^0\|$.*

2. *Write the stationarity condition of Proposition 35 as $\Phi(X) = 0$, with $\Phi_e(X) = \partial S_X/\partial X_e - \mathbf{L}(X_e^\top, \mathcal{J}_e(X))$ and $\mathcal{J}_e(X)$ the one-head current at X. There is a constant c, depending on λ_W, the loop weights, $\bar{A}$, the weights β_e, and the matrices S_j^ℓ, such that for $\|X\| \leq 1$*

$$\Phi(X) = \mathbf{K}X - \mathcal{J}^0 + R(X), \qquad \|R(X)\| \leq c\big(\|X\|^2 + \|X\|\|\mathcal{J}^0\|\big). \tag{137}$$

To first order in the generators and the sources, the field equation is therefore $\mathbf{K}X = \mathcal{J}^0$, that is,

$$\sigma_X^{-2}X_e + \lambda_W(\mathbf{d}^\top\mathbf{w}\mathbf{d}X)_e + \beta_e\bar{A}X_eS_j^\ell = \beta_e\bar{A}\big[(\xi_i^{\ell+1}-\xi_j^\ell)\xi_j^{\ell\top} - C_j^\ell\big] \quad \text{for every edge } e. \tag{138}$$

3. *Every stationary link field $X^\star$ with $\|X^\star\| \le 1$ satisfies $\|X^\star - X_{\mathrm{lin}}\| \le c\sigma_X^2(\|X^\star\|^2 + \|X^\star\|\|\mathcal{J}^0\|)$. If in addition $\|X^\star\| + \|\mathcal{J}^0\| \le 1/(2c\sigma_X^2)$, then $\|X^\star\| \le 2\sigma_X^2\|\mathcal{J}^0\|$ and $\|X^\star - X_{\mathrm{lin}}\| \le 2c\sigma_X^4(2\sigma_X^2+1)\|\mathcal{J}^0\|^2$, which is of second order in the sources.*

Proof (1) For matrices Y and X and symmetric $\bar{A}$ and S, $\langle Y, \bar{A}XS\rangle = \mathrm{Tr}(Y^\top \bar{A}XS) = \langle \bar{A}YS, X\rangle$ and $\langle X, \bar{A}XS\rangle = \|\bar{A}^{1/2}XS^{1/2}\|^2 \ge 0$, so each matter block is self-adjoint and nonnegative. The loop weights are nonnegative and $\lambda_W \ge 0$, so $\lambda_W \mathbf{d}^\top\mathbf{w}\mathbf{d}$ is self-adjoint with $\langle X, \lambda_W \mathbf{d}^\top\mathbf{w}\mathbf{d}X\rangle = \lambda_W \sum w_{ijk}^\ell \|(\mathbf{d}X)_{ijk}^\ell\|^2 \ge 0$. Adding σ_X^{-2} times the identity gives the bound, so $\mathbf{K}$ is invertible with $\|\mathbf{K}^{-1}\| \le \sigma_X^2$.

(2) With the flat anchor, $S_X(X) = \sum_e \|X_e\|^2/(2\sigma_X^2) + \lambda_W g_0(X)$ with $g_0(X) = \sum w_{ijk}^\ell \Psi(W_{ijk}^\ell)$ and $W_{ijk}^\ell = e^{-X_{kk}^\ell} e^{-X_{ik}^{\ell+1}} e^{X_{ij}^{\ell+1}} e^{X_{jk}^\ell}$, an entire function of X. Item 3 of Proposition 34 gives $g_0(X) = \frac{1}{2}\langle \mathbf{d}X, \mathbf{w}\mathbf{d}X\rangle + O(\|X\|^3)$, so the function $g = g_0 - \frac{1}{2}\langle \mathbf{d}X, \mathbf{w}\mathbf{d}X\rangle$ has vanishing value, gradient, and Hessian at $X = 0$, and Taylor's theorem bounds $\|\nabla g(X)\|$ by $\frac{1}{2}\sup_{\|Y\|\le 1}\|D^3 g(Y)\|\|X\|^2$ for $\|X\| \le 1$. Hence $\partial S_X/\partial X_e = \sigma_X^{-2}X_e + \lambda_W(\mathbf{d}^\top\mathbf{w}\mathbf{d}X)_e + \lambda_W \nabla_e g(X)$. For the matter side, the one-head current at $G = e^{X_e}$ is $\mathcal{J}_e(X) = \beta_e \bar{A}(\xi_i^{\ell+1}\xi_j^{\ell\top} - GS_j^\ell) = \mathcal{J}_e^0 - \beta_e\bar{A}(e^{X_e} - I)S_j^\ell$. Write $\|\cdot\|_2$ for the operator norm, so that $\|\cdot\|_2 \le \|\cdot\|$ and $\|Y_1Y_2\| \le \min(\|Y_1\|_2\|Y_2\|, \|Y_1\|\|Y_2\|_2)$. For $\|Y\| \le 1$ and $0 \le t \le 1$, $\|e^Y - I - Y\| \le \sum_{n\ge 2}\|Y\|^n/n! \le \|Y\|^2$, $\|e^{tY} - I\|_2 \le e^{t\|Y\|} - 1 \le 2t\|Y\|$, and $\|e^{tY}\|_2 \le e$, so

$$\big\|\mathbf{L}(Y,E) - E\big\| \le \int_0^1 \big(\|e^{sY} - I\|_2\|E\|\|e^{(1-s)Y}\|_2 + \|E\|\|e^{(1-s)Y} - I\|_2\big)ds \le (e+1)\|Y\|\|E\|.$$

Therefore

$$\mathbf{L}(X_e^\top, \mathcal{J}_e(X)) = \mathcal{J}_e^0 - \beta_e\bar{A}X_eS_j^\ell - \beta_e\bar{A}(e^{X_e} - I - X_e)S_j^\ell + \big[\mathbf{L}(X_e^\top, \mathcal{J}_e(X)) - \mathcal{J}_e(X)\big],$$

where the third term has norm at most $\beta_e\|\bar{A}\|\|S_j^\ell\|\|X_e\|^2$ and the bracket has norm at most $(e+1)\|X_e\|(\|\mathcal{J}_e^0\| + 2\beta_e\|\bar{A}\|\|S_j^\ell\|\|X_e\|)$. Subtracting from $\partial S_X/\partial X_e$ gives Eq. (137) edge by edge, and summing the squares over the edges, with $\sum_e a_e^4 \le (\sum_e a_e^2)^2$ and $\sum_e a_e^2 b_e^2 \le (\sum_e a_e^2)(\sum_e b_e^2)$, gives the bound with one constant c. Dropping R, which is of second order in X and $\mathcal{J}^0$ jointly, gives Eq. (138).

(3) At a stationary point $\Phi(X^\star) = 0$, so $\mathbf{K}(X^\star - X_{\mathrm{lin}}) = -R(X^\star)$ and $\|X^\star - X_{\mathrm{lin}}\| \le \sigma_X^2\|R(X^\star)\|$ by item 1. Under the additional condition, $\|X^\star\| \le \|X_{\mathrm{lin}}\| + c\sigma_X^2\|X^\star\|(\|X^\star\| + \|\mathcal{J}^0\|) \le \sigma_X^2\|\mathcal{J}^0\| + \frac{1}{2}\|X^\star\|$, which gives $\|X^\star\| \le 2\sigma_X^2\|\mathcal{J}^0\|$, and substituting this bound into the first estimate gives the second. ∎

In the language of lattice gauge theory, Eq. (138) reads as follows; the reading names the terms of Proposition 37 and adds nothing to it. The anchor term $\sigma_X^{-2}X_e$ is a bare mass. The map $\mathbf{d}$ sums the generators around each elementary loop of Eq. (103) with its orientation, the discrete curl of the algebra-valued edge field, so $\lambda_W\mathbf{d}^\top\mathbf{w}\mathbf{d}$ is a weighted curl-curl operator. For a time-like loop, $j = i$, the loop sum is $(\mathbf{d}X)_{iik}^\ell = -[(X_{ik}^{\ell+1} - X_{ik}^\ell) - (X_{ii}^{\ell+1} - X_{kk}^\ell)]$, the depth difference of a cross-token generator corrected by the time-like generators at its ends, so these loops contribute the square of a discrete electric field to $\langle X, \mathbf{K}X\rangle$, as time-like plaquettes give the electric term of lattice gauge theory (Kogut and Susskind, 1975;

Creutz, 1983); the space-like loops give the magnetic part. The matter term $\beta_e \bar{A} X_e S_j^\ell$ is an additional, state-dependent mass that screens the field. Because $\mathbf{K}$ is positive definite, depth enters $\langle X, \mathbf{K}X\rangle$ with the same sign as the token directions, so the equation is elliptic, of Euclidean signature: a static, screened, massive discrete Maxwell (Proca) equation on the depth-by-token complex rather than a wave equation.

G.3 Inference dynamics and the most probable path

Proposition 38 (Fisher information and first-order relaxation) *For a link field X, let $\mathcal{I}_X$ be the Fisher information, with respect to X, of the transport kernels $\mathcal{N}(\zeta_i^{\ell+1}; m_i^\ell, \bar{Q})$ of Eq.* (104)*, averaged over the recognition law of the states and labels,*

$$\mathcal{I}_X[E, E'] = \sum_{\ell,i} \mathbb{E}_q\big\langle Dm_i^\ell[E], \bar{A} Dm_i^\ell[E']\big\rangle, \qquad Dm_i^\ell[E] = \sum_h \mathbf{L}\big(X^\ell_{iJ_i^{\ell,h}}, E^\ell_{iJ_i^{\ell,h}}\big)\Pi_h \zeta^\ell_{J_i^{\ell,h}}, \tag{139}$$

where $Dm_i^\ell[E]$ is the derivative of m_i^ℓ in the direction $E = (E_e)$.

1. *$\mathcal{I}_X$ is the Gauss–Newton form of the transport part of $\mathcal{F}$. For one head it is block diagonal over the edges, with blocks*

$$\mathcal{I}_e[E, E'] = \beta_{ij}^\ell \operatorname{Tr}\big(\bar{A}\mathbf{L}(X_e, E) S_j^\ell \mathbf{L}(X_e, E')^\top\big), \qquad e = (i \leftarrow j, \ell), \tag{140}$$

   and $\mathcal{I}_e$ is positive definite exactly when $\beta_{ij}^\ell > 0$ and $\mathbf{L}(X_e, \cdot)$ is invertible.

2. *On the open set where $\mathcal{I}_X$ is invertible, the recognition flow $\dot{X} = -\mathcal{I}_X^{-1}\nabla\mathcal{F}(X)$ of the point-mass links is of first order. Its velocity is a function of the configuration alone and vanishes exactly at the critical points of $\mathcal{F}$, and it obeys the force balance $\mathcal{I}_X[\dot{X}, \cdot] = -d\mathcal{F}$, in which $\mathcal{I}_X$ is a friction tensor and no inertial term appears. Along every solution $\frac{d}{dt}\mathcal{F} = -\mathcal{I}_X[\dot{X}, \dot{X}]$, so $\mathcal{F}$ strictly decreases along every nonconstant solution and no nonconstant solution is periodic. At a critical point with invertible Hessian H, the linearized field $-\mathcal{I}_X^{-1}H$ has real eigenvalues, so no linearized mode oscillates.*

Proof (1) For a Gaussian kernel $\mathcal{N}(\zeta'; m_\theta, \bar{Q})$ with fixed covariance, the score in the direction E is $\langle Dm[E], \bar{A}(\zeta' - m_\theta)\rangle$, and its covariance under the kernel is $\langle Dm[E], \bar{A}\bar{Q}\bar{A}Dm[E']\rangle = \langle Dm[E], \bar{A}Dm[E']\rangle$; the covariance block of the Gaussian Fisher metric does not contribute because $\bar{Q}$ does not depend on X. Averaging over the conditioning states and labels gives Eq. (139). The transport part of $\mathcal{F}$ is $\sum_{\ell,i} \frac{1}{2}\mathbb{E}_q\|\zeta_i^{\ell+1} - m_i^\ell\|^2_{\bar{A}}$ up to X-free terms, as in the proof of Proposition 36, and its second derivative is $\mathcal{I}_X[E, E'] - \sum_{\ell,i} \mathbb{E}_q\langle \zeta_i^{\ell+1} - m_i^\ell, \bar{A} D^2 m_i^\ell[E, E']\rangle$; the Gauss–Newton form omits the second term. For one head, $Dm_i^\ell[E] = \mathbf{L}(X_{iJ}^\ell, E_{iJ}^\ell)\zeta_J^\ell$ with $J = J_i^\ell$, the events $J = j$ are disjoint with probabilities β_{ij}^ℓ and independent of the states, and $\mathbb{E}_q[\zeta_j^\ell \zeta_j^{\ell\top}] = S_j^\ell$, which gives the block form. Finally $\mathcal{I}_e[E, E] = \beta_{ij}^\ell \|\bar{A}^{1/2}\mathbf{L}(X_e, E)(S_j^\ell)^{1/2}\|^2$ vanishes for some $E \neq 0$ exactly when $\beta_{ij}^\ell = 0$ or $\mathbf{L}(X_e, \cdot)$ has a nonzero kernel.

(2) The free energy is smooth in the point-mass links, and so is $\mathcal{I}_X$, so on the open set where $\mathcal{I}_X$ is invertible the field $-\mathcal{I}_X^{-1}\nabla\mathcal{F}$ is smooth and its integral curves exist and are unique. The flow is a first-order equation in X, so its velocity is a function of X,

and it vanishes exactly where $\nabla\mathcal{F} = 0$. Applying $\mathcal{I}_X$ gives the force balance. Along a solution, $\frac{d}{dt}\mathcal{F} = \langle \nabla\mathcal{F}, \dot{X}\rangle = -\mathcal{I}_X[\dot{X}, \dot{X}] \le 0$, with equality only at a critical point. A solution that reaches a critical point is constant by uniqueness, so a nonconstant solution has nonvanishing velocity, $\mathcal{F}$ decreases strictly along it, and it cannot return to an earlier point. At a critical point with invertible Hessian H, $\mathcal{I}_X^{-1}H$ is similar to the symmetric matrix $\mathcal{I}_X^{-1/2}H\mathcal{I}_X^{-1/2}$, whose eigenvalues are real. ■

Proposition 39 (Inertia of the most probable path over depth) *Let $m^\ell > 0$ for $0 \le \ell \le L-1$, let V_ℓ, $0 \le \ell \le L$, be differentiable functions on a finite-dimensional inner-product space, and let a path $X^0, \dots, X^L$ with fixed ends minimize*

$$\sum_{\ell=0}^{L-1} \frac{m^\ell}{2}\left\|X^{\ell+1} - X^\ell\right\|^2 + \sum_{\ell=0}^{L} V_\ell(X^\ell). \tag{141}$$

Then, for $0 < \ell < L$,

$$m^\ell\big(X^\ell - X^{\ell+1}\big) + m^{\ell-1}\big(X^\ell - X^{\ell-1}\big) + \nabla V_\ell(X^\ell) = 0. \tag{142}$$

For $m^\ell = m$ and $V_\ell = V$ this is $m(2X^\ell - X^{\ell+1} - X^{\ell-1}) + \nabla V(X^\ell) = 0$, the central-difference form of $m\ddot{X} = \nabla V$, Newton's equation with mass m in the potential $-V$.

Proof For $0 < \ell < L$ the point X^ℓ is an unconstrained variable of the differentiable objective (141), so its gradient in X^ℓ vanishes at a minimizer, and that gradient is the left side of Eq. (142). ■

The increment term of Eq. (141) is, up to a constant, the negative logarithm of a Gaussian increment density with precision m^ℓ. In the latent-link model the time-like loops supply such a term. If the time-like generators $X_{ii}^{\ell+1}$ and X_{kk}^ℓ are held at zero, then $W_{iik}^\ell = e^{-X_{ik}^{\ell+1}} e^{X_{ik}^\ell}$, and item 3 of Proposition 34 gives

$$\lambda_W w_{iik}^\ell \Psi(W_{iik}^\ell) = \tfrac{1}{2}\lambda_W w_{iik}^\ell \left\|X_{ik}^{\ell+1} - X_{ik}^\ell\right\|^2 + O(\|X\|^3),$$

an increment term for the cross-token generator of the edge $(i \leftarrow k)$ with mass $m^\ell = \lambda_W w_{iik}^\ell = \lambda_W \bar{\pi}_{ii}^{\ell+1}\bar{\pi}_{ik}^\ell$ at quadratic order. Precision is in this sense the inertial mass of the most probable path, and the sign is the Euclidean one: Eq. (142) describes motion in the inverted potential $-V_\ell$. The inertia belongs to the joint optimization over depth, whereas the recognition flow of Proposition 38 is of first order.

G.4 Directed and undirected matter

Proposition 40 (Directed matter induces no link action) *In the model of Eq.* (104)*, the conditional law of the observations, states, and labels given the links has total mass one for every X. The link marginal of the model is therefore the prior $p(X)$, and the links reach the posterior only through the evidence $p(o \mid X)$.*

Proof Every factor of Eq. (104) other than $p(X)$ is a normalized conditional of a directed acyclic model: $p(o_i \mid U_i \zeta_i^L)$ integrates to one over o_i, $\mathcal{N}(\zeta_i^{\ell+1}; m_i^\ell, \bar{Q})$ integrates to one over $\zeta_i^{\ell+1}$ for every value of the states and labels of layer ℓ and of the links, the routing prior sums to one over the labels, and $p(\zeta_i^0)$ integrates to one. Integrating first over the observations and then over the states and labels of layers $L, L-1, \dots, 0$, which Tonelli's theorem permits for nonnegative integrands, leaves 1. The posterior of the links is then proportional to $p(X)p(o \mid X)$. ∎

For undirected matter, let a finite set V of sites carry states $\zeta_v \in \mathbb{R}^K$, let a set E of directed edges $e = (a \leftarrow b)$ between distinct sites, at most one for each pair of sites, carry reference links $G_e = e^{X_e}$, and declare the pairwise consistency energy

$$\mathcal{E}(\zeta, X) = \tfrac{1}{2} \sum_{v \in V} \zeta_v^\top \Lambda_0 \zeta_v + \tfrac{1}{2} \sum_{e=(a \leftarrow b) \in E} \big\| \zeta_a - G_e \zeta_b \big\|_{\Lambda_E}^2, \qquad \Lambda_0, \Lambda_E \succ 0, \tag{143}$$

the energy-based reading of the alignment terms. As a quadratic form in $\zeta = (\zeta_v)_{v \in V}$, $\mathcal{E} = \frac{1}{2} \zeta^\top \mathbf{P}(X) \zeta$ with the matter precision $\mathbf{P}(X) = \mathbf{D}(X) - \mathbf{H}(X)$, whose diagonal blocks are

$$\mathbf{D}_v(X) = \Lambda_0 + \sum_{e=(v \leftarrow b) \in E} \Lambda_E + \sum_{e=(a \leftarrow v) \in E} G_e^\top \Lambda_E G_e, \tag{144}$$

and whose off-diagonal blocks are $\mathbf{H}_{ab} = \Lambda_E G_e$ and $\mathbf{H}_{ba} = G_e^\top \Lambda_E$ for $e = (a \leftarrow b) \in E$, and zero for sites not joined by an edge.

Proposition 41 (Induced action of undirected Gaussian matter) *1. Let $\tilde{p}$ be a probability density on the generators, and let the joint density of states and generators be proportional to $\tilde{p}(X) e^{-\mathcal{E}(\zeta, X)}$, normalized as a whole and not conditionally on X. This joint is normalizable, and its generator marginal is proportional to $\tilde{p}(X) e^{-S_{\mathrm{ind}}(X)}$ with*

$$S_{\mathrm{ind}}(X) = \tfrac{1}{2} \log\det \mathbf{P}(X) \geq \tfrac{1}{2} |V| \log\det \Lambda_0. \tag{145}$$

2. If the spectral radius of $\mathbf{D}^{-1}\mathbf{H}$ is below one,

$$S_{\mathrm{ind}}(X) = \tfrac{1}{2} \log\det \mathbf{D}(X) - \tfrac{1}{2} \sum_{n \geq 1} \frac{1}{n} \operatorname{Tr}\big[(\mathbf{D}^{-1}\mathbf{H})^n\big], \tag{146}$$

and $\operatorname{Tr}[(\mathbf{D}^{-1}\mathbf{H})^n]$ is the sum, over the closed walks $(v_0, v_1, \dots, v_n = v_0)$ of length n whose steps join sites adjacent in E, of $\operatorname{Tr}(\mathbf{D}_{v_0}^{-1}\mathbf{H}_{v_0 v_1}\mathbf{D}_{v_1}^{-1}\mathbf{H}_{v_1 v_2} \cdots \mathbf{D}_{v_{n-1}}^{-1}\mathbf{H}_{v_{n-1} v_0})$, in which each step carries the link of its edge or the transpose of that link.

3. Let $\Lambda_0 = \lambda_0 I$ and $\Lambda_E = \lambda_E I$, let every link be orthogonal, and let the sites be token positions with the edges oriented causally, $e = (a \leftarrow b)$ with $b < a$. Then $\mathbf{D}_v = d_v I$ with $d_v = \lambda_0 + \deg(v) \lambda_E$ independent of the links, the spectral radius of $\mathbf{D}^{-1}\mathbf{H}$ is below one, the terms $n = 1$ and $n = 2$ of Eq. (146) do not depend on the links, and the term $n = 3$ is, up to a link-independent constant,

$$\sum_{k<j<i} \frac{\lambda_E^3}{d_i d_j d_k} \big(K - \operatorname{Tr} F_{ijk}\big) = \sum_{k<j<i} \frac{\lambda_E^3}{d_i d_j d_k} \Psi\big(G_{ik}^{-1} G_{ij} G_{jk}\big), \tag{147}$$

summed over the triples whose three edges lie in E, with F_{ijk} the triangle holonomy of Definition 29. On a single triangle, $V = \{i, j, k\}$, the term $n = 4$ is also link independent, and S_{ind} equals a constant plus the single summand of Eq. (147) *plus a remainder of absolute value at most $\frac{3K}{10}\rho^5/(1-\rho)$, where $\rho = 2\lambda_E/(\lambda_0 + 2\lambda_E) < 1$.*

Proof (1) Each pairwise term of Eq. (143) is nonnegative, so $\mathbf{P}(X) \succeq I_{|V|} \otimes \Lambda_0 \succ 0$, and since $\log\det$ is monotone on positive-definite matrices the bound in Eq. (145) follows. The Gaussian integral gives $\int e^{-\mathcal{E}(\zeta,X)} d\zeta = (2\pi)^{|V|K/2} \det \mathbf{P}(X)^{-1/2}$, which lies in $(0, (2\pi)^{|V|K/2}(\det \Lambda_0)^{-|V|/2}]$. Integrating against the probability density $\tilde{p}$ therefore gives a finite positive normalizer, and the generator marginal is proportional to $\tilde{p}(X) \det \mathbf{P}(X)^{-1/2} = \tilde{p}(X) e^{-S_{\mathrm{ind}}(X)}$. Had the states been given a law normalized conditionally on X, this factor would cancel, as in Proposition 40.

(2) $\mathbf{D}(X) \succeq I_{|V|} \otimes \Lambda_0$ is invertible, and $\log\det(\mathbf{D} - \mathbf{H}) = \log\det \mathbf{D} + \log\det(I - \mathbf{D}^{-1}\mathbf{H})$. Under the spectral condition, $\log\det(I - M) = \operatorname{Tr}\log(I - M) = -\sum_{n\geq 1} \operatorname{Tr}(M^n)/n$ with $M = \mathbf{D}^{-1}\mathbf{H}$. Expanding the block product $\operatorname{Tr}(M^n)$ gives the sum over sequences of sites, and $\mathbf{H}_{vu} = 0$ unless v and u are joined by an edge, which leaves the closed walks.

(3) With orthogonal links, $G_e^\top \Lambda_E G_e = \lambda_E I$, so $\mathbf{D}_v = d_v I$. For the norm $\max_v \|x_v\|$ on block vectors, the induced norm of M is at most $\max_v \sum_u \|M_{vu}\|_2 = \max_v \deg(v)\lambda_E/d_v < 1$, and the spectral radius is bounded by any induced norm. There are no edges from a site to itself, so $\mathbf{H}_{vv} = 0$ and the term $n = 1$ vanishes. A closed walk of length two is $v \to u \to v$, with trace $\lambda_E^2 \operatorname{Tr}(G_e G_e^\top)/(d_v d_u) = \lambda_E^2 K/(d_v d_u)$. A closed walk of length three visits three pairwise adjacent sites $k < j < i$. The walk $i \to j \to k \to i$ carries $\mathbf{H}_{ij}\mathbf{H}_{jk}\mathbf{H}_{ki} = \lambda_E^3 G_{ij} G_{jk} G_{ik}^\top$, whose trace is $\lambda_E^3 \operatorname{Tr}(G_{ik}^{-1} G_{ij} G_{jk}) = \lambda_E^3 \operatorname{Tr} F_{ijk}$, because $G_{ik}^\top = G_{ik}^{-1}$ and F_{ijk} is conjugate to $G_{ik}^{-1} G_{ij} G_{jk}$. The reversed walk carries the transpose and has the same trace, and cyclicity of the trace gives the same value for each of the three starting sites, so each triangle contributes $-\frac{1}{2} \cdot \frac{1}{3} \cdot 6\lambda_E^3 \operatorname{Tr} F_{ijk}/(d_i d_j d_k)$ to Eq. (146), which is the summand of Eq. (147) minus the constant $\lambda_E^3 K/(d_i d_j d_k)$. For orthogonal W, $\Psi(W) = K - \operatorname{Tr} W$ by item 2 of Proposition 34. On a single triangle, label the sites cyclically; each step of a closed walk then moves one place forward or backward, and the number of forward steps minus the number of backward steps is divisible by three. A walk of length four therefore has two steps in each direction, so it contains a step followed by its reversal, which contributes $\lambda_E^2 G_e G_e^\top/d = \lambda_E^2 I/d$ or $\lambda_E^2 G_e^\top G_e/d = \lambda_E^2 I/d$ with $d = \lambda_0 + 2\lambda_E$; removing that pair leaves a closed walk of length two, so the term $n = 4$ is link independent. For $n \geq 5$ there are at most $3 \cdot 2^n$ closed walks of length n, and each has trace of absolute value at most $K(\lambda_E/d)^n$, because a product of orthogonal matrices has trace at most K in absolute value. The remainder is therefore at most $\frac{1}{2}\sum_{n\geq 5} \frac{1}{n} 3K\rho^n \leq \frac{3K}{10}\rho^5/(1-\rho)$. ∎

On orthogonal links, the leading link-dependent term of the induced action is thus a positive multiple of the Frobenius defect of each triangle holonomy, with the cube of the hopping ratio λ_E/d as coupling, which is the form of the loop term of the prior in Eq. (104). For links outside the orthogonal group the blocks $\mathbf{D}_v$ and the walks of length two depend on the links, so the induced action then also contains edge-local terms that involve no loop.

Appendix H. Numerical Identity Checks

The identities of Sections 3, 7, and 8.3 were evaluated in float64 at small sizes with random inputs by the script `glk_lie_links_checks/checks.py`, whose output accompanies it. The written derivations carry the mathematical claims; the checks test the formulas, and their sizes, $K \le 7$ with at most three heads and four sources, carry no information about trained models or large networks. All 58 checks passed.

The low-rank exponential of Lemma 4 holds to a relative error of 8.5×10^{-16}, the augmented-exponential evaluation of φ_1 agrees with its series to 2.6×10^{-15}, and the split of Proposition 5 holds to 5.5×10^{-16}, with the identity on $\ker T_h$, the image condition, and the determinant identity at the level of rounding. The first-order defect $(\Theta_{ij}(s) - \Omega_{ij})/s - L_{ij}^{(h)}$ is 8.5×10^{-3}, 8.5×10^{-4}, and 8.5×10^{-5} at $s = 10^{-2}$, 10^{-3}, and 10^{-4}. The passive law and the adjoint action of a common reference change hold to 8×10^{-16}, and the numerical matrix logarithm of $\mathrm{diag}(-1, -2)$ is complex, as item 4 of Proposition 2 requires. On the principal-logarithm domain the conversion of Proposition 6 gives $TO = Z$ to 2.4×10^{-15} and $\exp(OT) = I + O_{\mathrm{lin}}T$ to 1.6×10^{-16}, and the embedding Υ is a determinant-preserving homomorphism satisfying items 2 and 3 to 3×10^{-15}. The complement split of Lemma 7 holds to 2.7×10^{-14} at $K = 7$ with $S = 4$, and exact enumeration over head-label tuples reproduces Eq. (36) with a receiver-carried complement to 3.4×10^{-16}. In the tiled case the three forms of the resummed receiving map, the effective transfer $T\widetilde{O} = e^{TO}$, the head-sum identity, the projector relations of Proposition 11, and the order series of Eq. (40) hold to 2×10^{-15}; the second-order truncation vanishes exactly at the stated matrix, and $\sigma_{\min}(e^X) \ge e^{-\|X\|_2}$ held on 200 random generators. The pinned latent-link transport reproduces the exponential-chart message of Remark 33 to 2.4×10^{-15}.

For the holonomy, the triangle spectra of Proposition 32 match those of $\boldsymbol{\rho}_{ijk}TO$ and $\boldsymbol{\rho}_{ijk}e^{TO}$ to 1.3×10^{-14} with $TO \ne I$, the head-sum loop with $TO = I$ matches Table 3 to 2×10^{-15}, and the flat identification has trivial holonomy. The errors of the first- and second-order truncations of Eq. (101) scale as the square and the cube of the generator scale. The flat routing pattern $c = (1, 0.5, 0.2, 0.05, 0.01)$ has positive rows summing to one and zero cocycle defect. All eigenvalue statistics vanish on the shear $I + E_{12}$ while its Frobenius defect is $\frac{1}{2}$, the Frobenius defect is invariant under a multiple of an orthogonal matrix, and a non-conformal diagonal reparametrization changes it.

For the receiver, 200 random perturbations do not lower the energy of Eq. (76) below its value at the closed form of Proposition 22, central differences of the energy vanish there to 1.8×10^{-9}, the closed form equals the Gaussian power posterior to 9×10^{-16}, and $F - \lambda_\alpha D_{\mathrm{KL}}(q \| p_{\mathrm{pow}})$ is constant in q to 5×10^{-15}. The unobserved marginal of item 3 of Corollary 23 equals the prior marginal to 1.1×10^{-15}, the predictive limit of Proposition 24 holds to 1.3×10^{-9} at prior precision 10^{-10}, the natural-parameter interpolation of Proposition 25 agrees with a fourth-order Runge–Kutta integration of the Fisher flow to 6×10^{-15}, and the frozen energy decreases along the damped full-family update. The profiled per-coordinate majorize-minimize step never increased the profiled energy over 300 random instances of four iterations each, the episode-start step equals the Kalman form of Eq. (83), the damped diagonal counterexample reproduces 1.500, 2.503, and 1.000, and the local closed form is the frame image of the reference closed form. A Monte Carlo estimate

from 4×10^5 draws matches the moment-matched covariance of Eq. (69) to a relative error of 2.3×10^{-3} at $r = 0$ and 2.0×10^{-3} at $r = 1$.

Appendix I. Experimental Record and Diagnostics

This appendix reserves the experimental record of the final runs. Each table and figure below is a placeholder to be replaced by recorded values or artwork.

Reporting field	Final-run value or artifact identifier
Source and configuration	pending: source archive and hash, configuration schema, resolved operator selectors
Data and tokenizer	pending: dataset and split revision, tokenizer and version, vocabulary, concatenation policy, token-cache hashes, unique training tokens, passes
Architecture	pending: K, blocks, heads, ranks r_h, s_h, depth, normalization placement, residual, MLP width, decoder
Initialization	pending: prior means and covariances, frame generators, interface maps, as executed
Optimization	pending: learning rate and weight decay for every parameter table, schedule, warmup, clipping groups, post-update frame projection
Numerical realization	pending: precision per operation, cost bound, reduced-covariance factorization and failure policy, matrix-exponential policy
Training budget	pending: token presentations, updates, batch and accumulation, device-hours
Evaluation	pending: checkpoint-selection rule, context, stride, scored targets, masking

Table 9: Reproducibility contract for the final reported runs. Populate each field from the executing source archive and its saved metadata, and separate effective settings from accepted but inactive selectors.

Parameter role or resource	Evaluated run	Final run
Per-type tables (token frames, prior means and covariances, decoder)	pending	pending
Positional frame generators	pending	pending
Context-processing parameters (interface maps, MLP, normalization)	pending	pending
Parameters read per token	pending	pending
Optimizer state and peak device memory	pending	pending
Training throughput and device-hours	pending	pending
Inference latency at declared batch and context	pending	pending

Table 10: Reserved storage and cost record. Sparse access to per-type tables reduces the parameters read per token but not the stored parameters; report both, together with measured throughput under a declared precision.

Model arm	Seeds	Selected step	Validation CE	Test CE	Test PPL
pending	pending	pending	pending	pending	pending

Table 11: Reserved per-seed results for every arm of the main results table. List the model seed and the data-order seed, and apply the checkpoint-selection rule identically across arms.

Model and source	Tokenization	Context	Word-normalized test PPL
Evaluated realization	GPT-2 BPE	pending	pending
Published reference points	pending	pending	pending

Table 12: Reserved calibration of the headline result. Word-normalized perplexity is the exponential of the summed test negative log-likelihood divided by the number of word-level test tokens, after an audit confirming that both token streams encode the same text. Published rows use different protocols and are contextual references, not matched comparisons.

PLACEHOLDER S1

Panel A: per layer, quantiles across tokens of the ratio $\|m_i\|/\|\mu_i\|$ of message to residual state, over training.

Panel B: common scale and scale-free invariants of the comparison maps (relative gains and principal angles of $S_{Q,h}, S_{K,h}$), and the largest singular value of $O_h T_h$, over training.

Panel C: mean and spread of the stored log-variances over training, with the curve implied by weight decay alone.

Panel D: counts, with denominators, of cost-bound activations, factorization fallbacks, frame-generator projections, and skipped updates.

TO COLLECT: per-table norms and invariants at every evaluation, and instrumented event counters.

Figure 6: Reserved training-dynamics and numerical diagnostics. The routing costs are unchanged when a common invertible map acts on $S_{Q,h}$ and $S_{K,h}$, and the messages are unchanged under $(T_h, O_h) \mapsto (BT_h, O_h B^{-1})$. Gradient flow conserves the corresponding quantities $S_{Q,h}S_{Q,h}^\top + S_{K,h}S_{K,h}^\top$ and $T_h T_h^\top - O_h^\top O_h$, whereas adaptive optimizers without weight decay move along these directions; Panel B therefore separates motion along the symmetries from changes of the realized operator.

PLACEHOLDER S2

Panel A: per layer, the $H \times B$ matrix of Frobenius norms of each head's maps $S_{Q,h}, S_{K,h}, T_h, O_h$ restricted to each frame block.

Panel B: attention entropy per layer and head, with the minimum-entropy rows and the self-weights $\beta_{ii}^{(h)}$.

TO COLLECT: trained interface maps and validation-set routing weights.

Figure 7: Reserved head and frame-block diagnostics. Panel A measures whether communication heads coincide with the irreducible frame blocks of Appendix B or access several blocks through full-width maps.

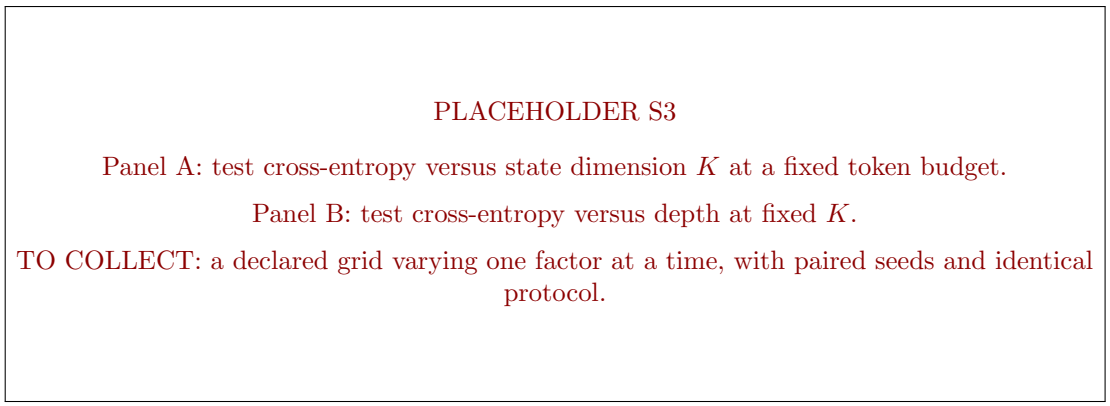


Figure 8: Reserved width and depth sweep. Existing preliminary runs differ in training duration, context, stride, learning rates, and covariance state, so they do not define this controlled comparison.

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